Hydrogeological Dynamics, Behavioural Clustering and Management Intervention Analysis at Newborough Warren Coastal Sand Dune Aquifer, Wales

A state-space modelling and empirical analysis of groundwater behaviour, forest management, dune scraping, hydrological boundaries and critical rainfall forecasting

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Abstract

Coastal sand dune aquifers are ecologically sensitive systems whose groundwater regimes determine the persistence of legally protected dune slack habitats. Newborough Warren, a 1,300-hectare Special Area of Conservation (SAC) on the Isle of Anglesey, Wales, has undergone sustained water table decline over recent decades, threatening the humid dune slack communities for which the site is designated. This study presents a reproducible, data-driven analytical pipeline applied to a network of 69 dipwells monitored over 2005–2026, integrating hierarchical clustering, Pearson affinity analysis, state-space modelling, intercept boundary auditing, comparative model benchmarking, critical rainfall threshold forecasting, an expanded clear-felling BACI experiment, dune scraping analysis, cluster-level water balance decomposition, and water table fluctuation specific yield estimation.

Agglomerative hierarchical clustering (Ward's linkage, k = 6, silhouette coefficient = 0.41) identified a primary geological binary within the dune field: the till-and-estuary-constrained Eastern Block and the deep-sand Western Block. The Lumped Catchment Storage Coefficient (LCSC = $100/\beta_1$) translates this geological contrast into a hydrogeological behavioural spectrum. The Eastern Block (C1: LCSC = 63.9%, C2: LCSC = 59.7%) behaves as a shallow pan — flashy, with a large water table response per mm of rainfall but limited total storage volume. The Western Block (C3: LCSC = 83.3%) behaves as a deep sponge — attenuated, with a modest water table response per mm of rainfall but large total storage capacity. The Forest cluster (C4: LCSC = 104.5%) sits apart from this geological axis: its high LCSC reflects land cover rather than substrate, the Corsican pine plantation intercepting approximately 24% of rainfall before it reaches the aquifer. Together the three clusters define a spectrum from shallow pan through deep sponge to canopy sponge that underpins all subsequent analysis.

Pearson affinity analysis confirmed cluster membership across all 69 reference wells and extended the hydrogeological classification to 21 additional legacy and experimental wells. Twenty-three reference wells were flagged as Multi-Cluster Affinity members occupying structural transition zones between geological blocks, identifying the locations of greatest restoration potential per unit management effort.

A dual-model intercept audit mapped unmeasured boundary fluxes across all 69 reference wells. Positive intercepts concentrated at the northern forest margin and lake-buffer sector identify zones of lateral recharge descending from the bedrock ridge, with CEH14 showing the largest boundary subsidy in the network (α = +0.222 m/month). Whilst a state-space model with an intercept improved single-step diagnostic fit at boundary-affected wells, it degraded iterative forecasting stability, confirming that an intercept-free formulation must be retained for long-term simulation.

The SSM substantially outperformed the traditional linear model in iterative forecasting across all 69 wells through its explicit parameterisation of aquifer storage decay ($-\beta_3$): median Nash–Sutcliffe Efficiency improvement +0.586; 68 of 69 wells showing positive NSE with SSM versus 33 of 69 with TLM. Spatial mapping of SSM performance gain physically delineates subsurface structural boundaries between deep sand reserves, perched slacks and boundary discharge zones without invasive coring.

Cluster-level mechanistic equations were derived to produce critical rainfall threshold (P_flood) equations — operational management tools giving the minimum monthly winter rainfall required to raise the water table to the slack floor from any given antecedent depth. The Western Block requires approximately 15–25% more monthly winter rainfall than the Eastern Block to close an equivalent water table deficit, reflecting its lower recharge sensitivity. Peak flood transfer functions confirm that antecedent summer minimum water table depth is the dominant control on winter flooding severity in the Western Block (R^2 = 0.684, $p < 0.001$); winter rainfall amount alone is not a significant predictor ($p = 0.818$) — a finding with direct implications for restoration planning under changing climate seasonality.

An expanded clear-felling BACI experiment across 16 wells, incorporating a cumulative water balance covariate to partition intervention signals from climate variability (ANCOVA-BACI, R^2 = 0.729), found that the December 2017 clearfell imposed a climate-corrected displacement of −0.093 m on the core impact zone, on top of a −0.137 m displacement attributable to the prior 2015 dune scraping, giving a combined climate-corrected hydrological cost of −0.230 m. Three independent robustness analyses — SSM residual normalisation, synthetic control, and cluster transition testing — found no detectable clearfell-specific signal above the noise of climate variability and the scraping confound; the ANCOVA displacement should therefore be treated as an upper bound on any clearfell effect rather than a causal attribution. Post-felling storage decay coefficients ($-\beta_3$) increased at 13 of 14 paired wells, including all regional controls several hundred metres from the felling boundary; because increases are not confined to the intervention footprint, they are attributed to SSM fitting artefacts of the transition between equilibria rather than genuine drainage acceleration. The clearfell demonstrably increased recharge sensitivity (β_1 increased at 12 of 13 paired wells) and eliminated the canopy's sacrificial evaporative shielding: β_2 increased at all 13 paired wells, deepening summer water table minima proportionally more than any winter improvement. By February 2026 the climate-corrected cumulative BACI displacement showed no sign of stabilisation, and the climate sensitivity of the impact zone was reduced by 79% following canopy removal.

Dune scraping produced spatially variable outcomes that depend critically on position within the drainage system. CEH36 and CEH18 share a common control well (CEH4) and occupy a drainage sequence descending from the plantation boundary toward the coast: CEH36 (scraped April 2015) lies closest to the felling zone, CEH18 (scraped October 2023) further

downgradient, and CEH4 (unmanipulated throughout) closest to the coast. At CEH36, a climate-corrected net benefit of +0.143 m was recorded during the pure scraping era. The Tier 1 CUSUM at CEH4 shows progressive deterioration relative to the regional mean (−0.096 m) over the same period, consistent with coastal drainage dominating its water balance independently of any scraping signal. Following the December 2017 clearfell the CEH36 benefit was partially eroded (post-felling net benefit −0.047 m). CEH18 showed a measurable improvement following its October 2023 scraping (era mean 0.950 m vs post-felling 1.223 m). CEH21, paired against the coastal control CEH22 at a separate location on the western margin, showed limited response to its October 2023 scraping. $-\beta_3$ increased at all five monitored wells including both unmanipulated controls; because this pattern extends well beyond any intervention footprint it reflects a network-wide climate-driven structural signal rather than a localised scraping response.

Network-wide summer minimum water table depths have declined at statistically significant rates across all structural domains (C1: −0.010 m yr^{-1}, C2: −0.012 m yr^{-1}, C3: −0.015 m yr^{-1}, C4: −0.017 m yr^{-1}), with winter maxima showing no significant trend in any cluster. The spatial coherence of this signal — consistent across clusters confirmed as hydrogeologically distinct, and unrelated to differences in canopy cover, substrate type or management history — points to a single system-wide forcing. At Newborough Warren, where the open warren has no drainage network and sea level rise would act to raise rather than lower the coastal water table base level, the only driver consistent with a summer-concentrated, network-wide declining trend is increasing atmospheric evaporative demand under a warming climate, compounded by increasing seasonal amplitude in rainfall. Over the monitoring period, winter rainfall shows a marginally significant increasing trend (+4.75 mm/yr, $p = 0.049$), but the highly transmissive sandy aquifer drains this winter surplus rapidly — as confirmed by the high $-\beta_3$ values across all clusters — leaving summer water table floors increasingly exposed to elevated evaporative demand. Individual drought summers (notably 2022, with only 747 mm annual rainfall and a +1.62°C temperature anomaly) illustrate the dual-stress conditions under which this seasonal redistribution is most ecologically damaging. This pattern is directly confirmed by the RAF Valley record, which shows a statistically significant warming trend in summer maximum temperatures of +0.013°C yr^{-1} over the full 95-year record ($p < 0.001$), with the post-2013 period running 0.87°C above the pre-2013 baseline. Earlier studies attributing water table depression primarily to forest evapotranspiration were conducted during the driest decade in the 95-year record and without climate partitioning — a conflation that the ANCOVA-BACI framework applied here is designed to prevent.

Mapped against the eco-hydrological thresholds established by Curreli et al. (2013), the current water table regime at Newborough Warren is already outside the range supporting wet dune slack communities across most of the monitored network. The Western Block (C3) summer minimum has fallen below the wet slack viability threshold in every monitored year and below the dry slack threshold in 81% of years, whilst winter maxima exceeded the wet slack flooding threshold in only 1 of 21 monitored years. The Eastern Block retains meaningful flooding capacity — C1 achieves wet slack winter flooding in 67% of years — but C2 is already marginal at 50%, and at the current rate of summer minimum decline C1 will approach the dry slack summer threshold by the early 2030s. The ecological window for effective intervention is therefore narrow and closing — the P_flood analysis confirms that Western Block summer minima have already deepened beyond the point at which any

realistic winter rainfall total can restore surface flooding, placing these slacks in a structurally compromised hydrological state that neither clearance nor scraping alone can reverse without addressing the underlying climate-driven summer drawdown.

The major finding of this study concerns the hydrological consequences of clearance for dune slack conservation. The near-identical summer minimum decline rates across canopy-free and forested clusters confirm that progressive forest evapotranspiration is not driving the network-wide declining trajectory, whilst post-felling increases in the atmospheric draw coefficient (β_2) confirm that canopy removal deepens summer water table drawdown — the variable that governs winter flooding probability. Where the objective is maintenance or restoration of humid dune slack communities, clearance as currently implemented is therefore counterproductive on hydrological grounds, regardless of its merits for other habitat management objectives. The same β-coefficient framework that quantifies the cost of clearance identifies silvicultural management of canopy density as the primary calibratable hydrological intervention: stand density rather than species composition is the key management variable, and a more open canopy structure at reduced stand density would recover the majority of the recharge deficit whilst retaining the shielding, buffering and drainage-attenuation functions that the current closed stand provides. Observational evidence from the 1993 clearfell and 1996 broadleaf restocking on the northern plantation margin — monitored through dipwell NW10 over an 18-year shared record — suggests, from a single dipwell, that broadleaf conversion may produce deeper summer minima than adjacent Corsican pine, though its position precludes firm attribution, conversion produces consistently deeper summer minima than adjacent Corsican pine at comparable elevation, with the mechanism reflecting higher recharge-season throughfall rather than amplified atmospheric draw (β_2 essentially identical between broadleaf and pine wells). An emerging deepening trend as the canopy closes (−28.8 mm/yr, $p = 0.042$, 2019–2025) indicates that the full hydrological cost of a mature closed broadleaf canopy may not yet be expressed in the current record.

Cluster-level water balance decomposition reveals that every structural domain operates under a persistent monthly deficit under mean climate conditions. The water balance residual — the flux required to close the monthly deficit under mean conditions — ranges from 314–351 mm/year across clusters, compared to local recharge from direct rainfall of only 99–173 mm/year. For the Forest cluster, ridge-derived lateral recharge is confirmed as the primary sustaining mechanism (C4 mean $\alpha = +0.097$ m/month; CEH14 $\alpha = +0.222$ m/month); the open dune clusters are sustained by a combination of topographic position and high recharge sensitivity — a finding that reframes the relative importance of within-site land cover management versus protection of the ridge recharge pathway. Water table fluctuation analysis confirms broadly uniform specific yield across the site ($S_y = 0.22$–0.26), with the Forest cluster value (0.227 after correction for canopy interception) indistinguishable from the open dune clusters, confirming that the plantation's distinctive hydrological signature is a surface boundary effect rather than a property of the sand.

The integrated analytical pipeline — combining network clustering, mechanistic state-space modelling, boundary subsidy mapping, intervention BACI analysis, and critical rainfall threshold forecasting — provides a transferable and reproducible framework for quantitative groundwater management in legally protected dune slack systems. The P_flood threshold equations and SSM framework have been implemented in the Newborough Warren

Hydrogeological Modelling Suite scenario viewer (Hollingham, 2026b), a companion browser-based tool that operationalises the P_flood forecasting equations, allows managers to explore current water table positions relative to flooding thresholds, and provides indicative sensitivity analysis under projected climate forcing. Scenario simulation of land cover change is included as an illustrative application of the SSM equilibrium framework; the limitations of this approach relative to a fully calibrated spatial model are discussed in Section 5.7. Slug tests and a ground-penetrating radar survey of aquifer thickness are identified as the priority field measurements that would enable a properly calibrated spatial model in future work.

1. Introduction

Sand dune systems are among the most ecologically dynamic and internationally important coastal habitats in northwest Europe (Dargie, 1995). Their characteristic dune slack communities — low-lying inter-dune hollows that flood seasonally where the water table intersects the surface — support a concentration of nationally rare plant species including creeping willow (*Salix repens*), and a diverse suite of wetland-associated invertebrates. The persistence of these communities is fundamentally dependent on a shallow, responsive groundwater table that rises into the slack floor each winter.

Newborough Warren on the Isle of Anglesey (National Grid Reference SH 406 636) is one of the largest and most ecologically important coastal dune systems in Wales. Designated as a Special Area of Conservation primarily for its humid dune slack and creeping willow communities, the site comprises approximately 1,300 hectares of late-glacial blown sand resting on weakly permeable glacial till (Stratford et al., 2007). The northern 700 hectares were afforested with Corsican pine (*Pinus nigra* var. *laricio*) between 1948 and 1965.

A well-documented trend of water table decline has been observed at the site over the past six decades (Ranwell, 1959; Jennings, 1990). Conservation management since 2018 has included an experimental clearfell of approximately 8 ha within the plantation, and topographical dune scraping at selected slack sites. These interventions provide natural experiments by which the hydrological consequences of management can be assessed empirically. Despite the importance of the site and the scale of management investment, no systematic network-wide quantitative analysis has previously been undertaken.

Restoration of dune slack hydrology is widely assumed to be achievable through topographical intervention provided sufficient winter rainfall occurs. However, if summer water table minima are declining at a rate that progressively increases the recharge deficit that must be overcome each winter, there exists an intervention window — a period during which topographical management can still realistically restore flooding conditions — beyond which no realistic winter rainfall total can compensate for the accumulated summer deficit. Quantifying the rate of closure of this window, and identifying the site-specific mechanisms that drive it, is therefore an urgent practical and scientific priority. This requires not only a long baseline record of groundwater behaviour but a mechanistic understanding of how recharge, evapotranspiration and drainage interact across the hydrogeologically distinct sub-units of the dune system.

This study presents such an analysis. The specific objectives are to: (1) classify the dipwell network into hydrogeologically coherent groups by hierarchical clustering and extend membership to the full monitoring network via Pearson affinity analysis; (2) characterise mechanistic groundwater signatures using a state-space model benchmarked against a traditional regression approach; (3) map and interpret unmeasured boundary subsidies; (4) quantify the effects of dune scraping using a two-tier hierarchical CUSUM-BACI framework that separates localised treatment signals from background system-wide deterioration; (5) conduct an expanded ANCOVA-BACI assessment of the December 2017 plantation clearance across a 16-well hierarchical experimental network, partitioning intervention signals from climate variability; and (6) derive operational critical rainfall threshold equations for conservation planning.

1.1. The Newborough Warren Analytical Architecture

To achieve a rigorous characterisation of the aquifer's hydrogeological dynamics, this study employs a nine-tiered analytical framework that scales from individual well diagnostics to network-wide predictive modelling. A further objective is to demonstrate that the analytical framework is replicable at comparable dune slack systems using only manual dipwell records and publicly available climate station data, without specialist survey equipment, telemetry or laboratory analysis.

- **Hydrogeological Clustering:** The 69-well reference network is partitioned into hydrogeologically distinct zones using agglomerative hierarchical clustering with Ward's minimum variance linkage (Section 3.2). A correlation-based dissimilarity metric isolates temporal hydrograph dynamics from absolute water table elevation, and the optimal cluster number (k = 6) is selected using a dual-metric approach combining Ward's merge distances and mean silhouette coefficient. The resulting six clusters provide the spatial framework for all subsequent analysis.
- **Extended Network Classification and Pattern Matching:** The primary hydrogeological classification is extended from the 69-well reference network to the full 90-well monitoring array using Pearson Affinity Analysis (Section 3.3). Unlike the mechanistic models, this phase utilizes pattern-matching on raw observed time series without model fitting. It identifies core cluster members and pinpoints Multi-Cluster Affinity (MCA) flags at structural transition zones where local topographic drivers are superseded by system-wide climatic forcing. The wells in the extended network are not used in the rest of the analysis.
- **Mechanistic Modelling and Storage Characterization:** At the cluster scale (n=6), the State-Space Model (SSM) is used to derive physical coefficients (β_1,β_2,β_3) directly from cluster-average time series (Section 3.4.1). These coefficients characterize the fundamental contrast between the "shallow pan" and "deep sponge" blocks, providing the basis for both the Lumped Catchment Storage Coefficients (LCSC) and the operational P_flood threshold equations used for management forecasting.
- **Predictive Validation and Model Benchmarking:** The predictive skill of the SSM is formally validated through matched comparison against a Traditional Linear Model (TLM) at every individual well (n=69) in the reference network (Section 3.4.2). This benchmarking uses iterative Nash-Sutcliffe Efficiency (NSE) and one-step R^2 to assess predictive skill across the network. Spatial mapping of the performance gains (ΔR^2, ΔNSE) demonstrates where the explicit parameterisation of storage decay is indispensable for accurate multi-step forecasting.
- **Boundary Flux and Intercept Auditing:** A Boundary Subsidy Audit (Section 3.4.3) is performed by comparing the mass-balanced Model A (α=0) with the free-intercept Model B across all 69 reference wells. Mapping the resulting α values allows for the identification of localized recharge sources and drainage sinks—such as the northern forest margin and lake-buffer sectors—that operate independently of meteoric forcing.
- **Intervention Analyses:** The hydrological consequences of two management interventions are quantified using complementary BACI frameworks (Sections 3.5.4–3.5.5):

  - *Dune Scraping:* A two-tier hierarchical CUSUM-BACI framework across five focal wells separates localised scraping signals from background system-wide deterioration. Tier 1 validates local control wells against a regional mean; Tier 2 differences each treatment well against its matched control. Era-specific storage decay coefficients ($-\beta_3$) are estimated independently for each management period to assess whether scraping has altered subsurface drainage mechanics.
  - *Plantation Clearfell:* An expanded ANCOVA-BACI experiment across 16 wells in a three-zone hierarchical design — core impact, transition/edge, and regional control — assesses the hydrological consequences of the December 2017 clearfell. A cumulative water balance covariate partitions the intervention signal from concurrent climate variability, and era-specific SSM coefficient analysis characterises the spatial footprint of drainage acceleration across the forest–dune boundary.
- **Robustness and Confound Assessment:** Independent robustness analyses are conducted for both intervention analyses to test whether the detected signals are genuine rather than artefacts of control well limitations or climate confounds (Sections 3.5.4–3.5.5). For the scraping analysis, the reliability of CEH4 as a stable long-term control is tested using two independent methods: a synthetic control constructed as an optimally-weighted combination of 11 donor wells calibrated on the pre-scraping period, and an SSM forward residual computed by calibrating the model on the pre-scraping period and running it forward iteratively. Convergence between the raw BACI estimate, the synthetic control gap and the SSM residual confirms whether the scraping signal at CEH36 is detectable independently of CEH4's own coastal erosion and felling pulse trajectory. For the clearfell analysis, three independent analyses test whether a clearfell-specific signal can be detected once the principal confounds — scraping baseline contamination, the anomalously wet 2015–16 pre-clearfell period, and the common post-2018 climate signal — are accounted for: SSM residual normalisation against the regional control mean; synthetic control analysis for each impact well; and cluster transition analysis testing whether SSM coefficients at the felled wells converge toward C3 Western Block values post-clearfell. A spatial transect analysis additionally tests whether post-felling step changes show a distance-dependent gradient consistent with a localised clearfell effect or a spatially uniform response indicative of a common climate signal.

- **Climate Trajectory and Threshold Exceedance Analysis:** Annual summer minima and winter maxima are extracted for C1–C3 and projected to 2040 against the eco-hydrological viability thresholds established by Curreli et al. (2013), quantifying the rate of closure of the ecological intervention window and the proportion of monitored years in which flooding thresholds are exceeded across the principal structural domains (Section 3.6.4).

- **Water Balance Decomposition:** Cluster-level SSM coefficients are used to decompose the mean monthly water balance into its component fluxes — recharge, atmospheric draw, gravity drainage and boundary subsidy — expressed both as head-space quantities (m/month) and as indicative volumetric fluxes (mm/year) using assumed specific yield (Section 3.7.2). This tier quantifies the structural dependence of the aquifer on ridge-derived lateral recharge across all four principal clusters, independently of land cover or management history.

- **Water Table Fluctuation Specific Yield Estimation:** Empirical upper bounds on specific yield are derived independently from the dipwell record using the water table fluctuation method (Healy and Cook, 2002), applying an interception correction for the Forest cluster following Freeman (2008). The resulting cluster-specific Sy estimates (Section 3.7.3) bracket the volumetric water balance from above, confirming that the assumed values used in the primary decomposition are conservative lower bounds, and confirm broadly uniform aquifer storage properties across the site irrespective of surface land cover.

2. Study Site

Newborough Warren occupies the south-western tip of the Isle of Anglesey in north-west Wales (Figure 1). The dune field is underlain by weakly permeable glacial till forming the effective base of the sand aquifer. A bedrock ridge runs south-west to north-east and forms the northern boundary of the study area, its western end entering the sea. A link to an interactive map of the long term monitoring network is in the supplementary materials section. The hydrological study area, defined by the extent of the DEM-derived flow network (Figure 1: modelled drainage lines are derived from topographic analysis of the digital elevation model, and represent theoretical flow paths rather than mapped surface watercourses), covers approximately 1,172 ha (~4.0 km × 3.0 km). The 90-dipwell monitoring network spans a ~720 ha core zone (~3.3 km × 2.2 km) within this, lying entirely on the south-eastern flank of the ridge. The dune system on the north-western flank of the ridge, draining toward the Malltraeth estuary, is hydrologically separate from the main Newborough Warren aquifer; monitoring wells located on that flank do not have sufficient record length for inclusion in the clustering and state-space analyses reported here..

Lateral recharge descending from both flanks of the ridge into the dune sands contributes to the unmeasured boundary subsidies detected in the state-space model, and is most pronounced at the forest and lake-buffer wells in the northern part of the study area, which lie closest to the ridge crest. The dune aquifer south-east of the ridge is divided into an eastern block, where shallower sand over till and estuarine deposits produces a more responsive, heterogeneous water table regime, and a western block, where deep sand accumulation over a broader platform permits significant groundwater storage and more buffered seasonal fluctuations regime (Stratford et al., 2007).

Recharge discharges naturally across the foreshore as tides recede, and to a lesser extent towards landward salt marsh deposits. A small lake, Llyn Rhos-ddu, acts as a near-fixed head body, receiving groundwater in winter and recharging the aquifer during drier months (Stratford *et al.*, 2007). The full monitoring network at the site comprises 117 dipwells (Figure 1), of which 69 have sufficient record length for inclusion in the core analyses.

The northern 700 hectares are managed plantation dominated by mature Corsican pine. A partial clearance of approximately 8.4 ha (just over 1% of the plantation) began in December 2017, removing trees across a defined felling zone at the plantation–open dune transition (Figure 2). Dune scraping has been undertaken at selected western slack locations: CEH36 in April 2015, and CEH18 and CEH21 in October 2023.

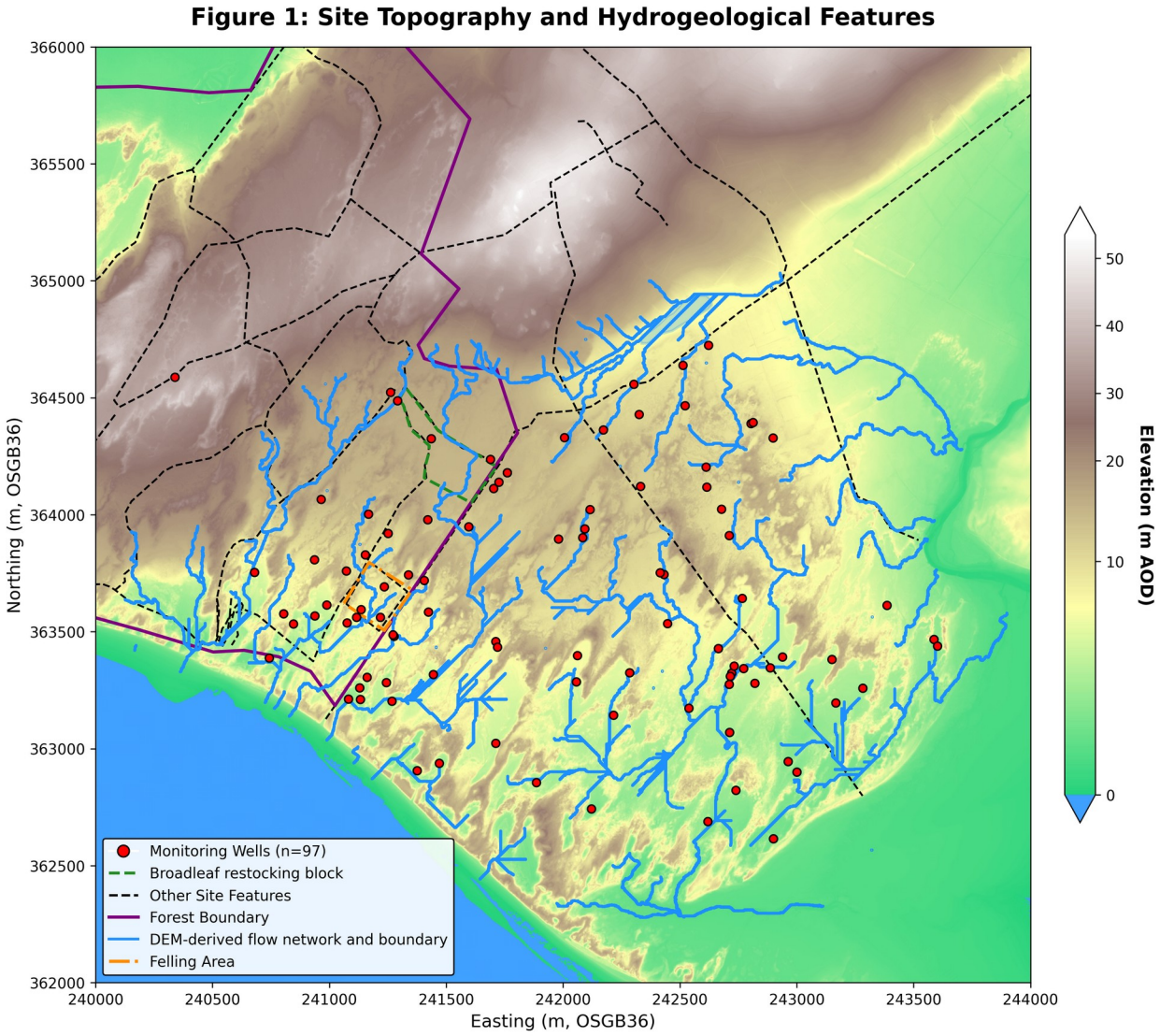


Figure 1. *Site topography and hydrogeological features. Digital elevation model of Newborough Warren showing the full 117-well monitoring network, rock ridge, forest boundary, DEM-derived flow network (modelled drainage lines from topographic convergence analysis; not mapped surface watercourses) and watershed boundary. (12_01_dem_site_overview.png) (LiDAR data: © NRW & OS; see Section 9 for full attribution).*

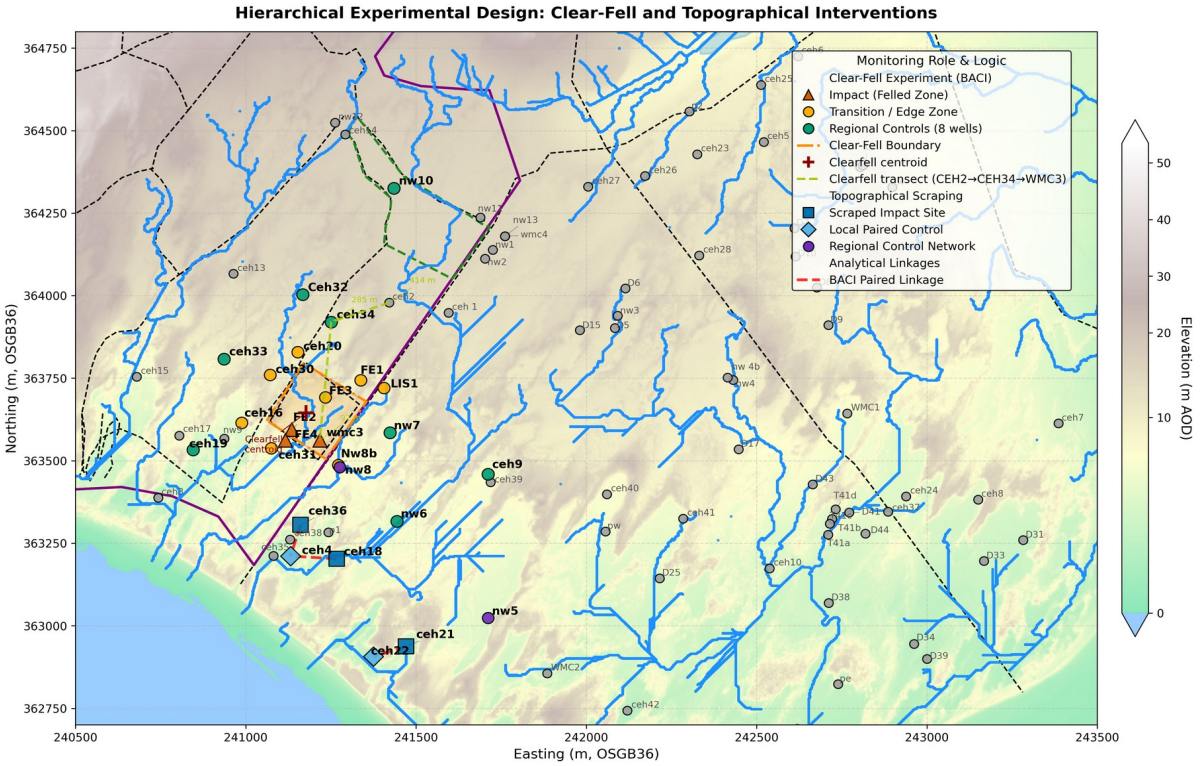


Figure 2. *Hierarchical experimental design. Spatial layout of the monitoring network showing core impact wells (FE2, FE4, WMC3), edge zone wells (FE1, FE3, CEH31, LIS1, CEH20, CEH30, CEH16, NW8b), regional controls (CEH32, CEH34, CEH33, NW10, CEH19), scraped sites (CEH36, CEH18, CEH21/22), felling boundary and BACI paired linkages. (13_01_experimental_setup_map.png) (LiDAR data: © NRW & OS; see Section 9 for full attribution).*

3. Methods

3.1. Data Sources and Preparation

All analyses were implemented as a sequential, numbered Python pipeline of 23 scripts. Each script reads from and writes to a shared outputs directory, and scripts are numbered to reflect execution order (00 through 21, including scripts 11b, 19a and 19b). The pipeline is documented in the accompanying README and all outputs are reproducible by running the published scripts in sequence. The companion browser-based modelling suite, which operationalises the P_flood forecasting equations and scenario tools derived from this pipeline, is also hosted at https://newbroman.github.io/Newborough-Hydrology_models/. Where script names are cited in the text below, they refer to files in the pipeline repository.

Groundwater Records

Groundwater level data were obtained from a network of 90 dipwells. Field measurements record depth to water below the top of each standpipe as a positive number in metres; water standing above the standpipe top is recorded as a negative value. Records span April 2005 to February 2026. Raw records were quality-controlled by removing values exceeding 4.0 m below standpipe top as physically implausible for this dune system, and aggregated to calendar monthly means; where multiple field measurements fell within a single month these were averaged to reduce sub-monthly variability. Gaps of up to three consecutive months were filled by time-based linear interpolation; longer gaps were retained as missing values. A reference network of 69 wells, each with more than 100 months of valid record extending to February 2026, formed the basis of the clustering and state-space modelling analyses. Wells not meeting these criteria were assigned to an extended network of 21 wells used in the Pearson affinity analysis to extend the hydrogeological classification beyond the reference network boundary.

Ground surface elevations for all dipwells were derived from the Natural Resources Wales LiDAR Digital Terrain Model (2 m resolution, captured March 2023; Natural Resources Wales, 2023), obtained from the DataMapWales repository (https://datamap.gov.wales/maps/lidar-viewer/). The accuracy of the DEM-derived elevations was assessed by comparison against independently surveyed DGPS ground elevations at 61 wells (R^2 = 0.976, RMSE = 0.477 m, mean bias = +0.264 m). The bulk of this RMSE is attributable to six outlier wells where discrepancies exceeded 0.5 m; five are located within the Corsican pine plantation, where satellite signal attenuation under dense evergreen canopy is a known source of DGPS positional error, and the sixth is an open-ground well whose DGPS-recorded elevation matches that of a plantation well 245 m distant, strongly suggesting a recording error. Excluding these six wells, the RMSE falls to 0.231 m with a mean bias of +0.203 m, consistent with the NRW stated DTM accuracy of ±0.15 m RMSE on open ground plus a modest systematic offset attributable to residual vegetation returns. The DEM is therefore adopted as the primary elevation source throughout this study.

Water table elevation in metres above Ordnance Datum (maOD) was derived using surveyed pipe-top elevations (Pipe_Top_Elev) and field-measured upstand heights (u, the height of the pipe top above ground surface), as maOD = Pipe_Top_Elev + raw_reading, where raw readings follow the negative depth-below-pipe-top convention. Upstand heights ranged from

−0.14 m (CEH28, a sunken casing) to +0.71 m (CEH2); for wells where upstand height was not directly recorded, values were assigned from field notes, all less than 0.10 m and within the stated vertical accuracy of the DTM. The maOD head surfaces are used in the spatial groundwater analysis (Section 3.8) and the broadleaf comparison (Section 4.6.8); the temporal clustering, SSM parameterisation, and BACI analyses use depth-below-pipe-top records throughout for consistency with the original monitoring convention.

For the cluster average hydrograph construction (Section 3.4.1), each well's depth-to-water series was corrected to a common ground-surface datum by subtracting its surveyed upstand height before averaging across cluster members, ensuring that between-well differences in pipe-top elevation do not bias the cluster mean. This correction was not applied to individual well SSM fits, where the pipe-top offset cancels in the first-difference operation, nor to the clustering and Pearson affinity analyses, where z-score standardisation renders absolute depth irrelevant. Network summary statistics and well characterisation figures are presented in Section 4.1.2 and Figure 4.

3.1.1 Climate Data

Monthly total precipitation and potential evapotranspiration were obtained from the RAF Valley Met Office climate station (SH 30691 75549), approximately 16 km north of the site. The station record extends from December 1930 to February 2026, with a mean annual rainfall of 890 mm over the well-record period (calculated from complete calendar years 2006–2025; the partial years at the start and end of the monitoring record are excluded from this summary statistic). A single missing monthly rainfall value in the historic record (June 1941) was retained as missing rather than substituted; it falls outside the well-record period and has no influence on any analysis.

Potential evapotranspiration was calculated using the Thornthwaite and Mather (1957) method with a day-length and month-length correction, applied to mean monthly air temperature derived as the mean of daily maximum and minimum. The method proceeds in three steps. First, the annual heat index I is computed as the sum of twelve monthly contributions:

$$I = \sum_{m=1}^{12} \left(\frac{T_m}{5} \right)^{1.514}$$

where T_m is mean monthly temperature clipped to zero for months at or below 0°C. Second, unadjusted monthly PET is calculated as:

$$PET_{unadj} = 16 \times \left(\frac{10\,T}{I} \right)^{alpha}$$

where: $alpha = 6.75 \times 10^{-7}\, I^3 - 7.71 \times 10^{-5}\, I^2 + 1.792 \times 10^{-2}\, I + 0.49239$

For months where mean temperature exceeds 26.5°C a high-temperature correction is applied $PET_{unadj} = -415.85 + 32.24\,T - 0.43\,T^2$; this threshold is not reached at RAF Valley under the historic record but is retained for formal completeness. Months with mean temperature at or below zero are assigned PET = 0. Third, the unadjusted value is scaled by a day-length and month-length correction factor:

$$PET = PET_{unadj} \times K \text{, where } K = \left(\frac{N}{12}\right) \times \left(\frac{d}{30}\right)$$

where N is the maximum possible sunshine duration for the month (hours), calculated from site latitude (53.25°N, RAF Valley) using standard solar geometry, and d is the number of days in the month. This correction accounts for the fact that the original Thornthwaite equation was calibrated for a standard 30-day month and 12-hour photoperiod.

Thornthwaite PET is known to overestimate evaporative demand in humid maritime climates relative to physically-based methods such as Penman-Monteith (Donohue et al., 2010), and it was selected here because it requires only mean air temperature, consistent with the low-cost replication objective described in Section 3.8. Although forest recharge at Newborough is estimated to be 100–200 mm yr^{-1} lower than open dune recharge (Stratford, 2006), these absolute differences do not affect the comparative conclusions of this study. Because the same PET series is applied uniformly across all clusters and time periods, systematic biases cancel in cluster-to-cluster and before-versus-after contrasts.

A cumulative water balance series was computed for the well-record period by summing monthly net balance (P − PET), with a mean-centring correction applied to remove the systematic long-term offset between precipitation and Thornthwaite PET inherent to this maritime climate; the corrected series oscillates around zero and directly reflects periods of cumulative surplus and deficit relative to the long-term mean. The long-run detrending mean was 19.46 mm $month^{-1}$. The correspondence between this cumulative balance series and the unsmoothed network mean water level was assessed by linear regression at zero lag (R^2 = 0.515), confirming that interannual water table variability is substantially explained by antecedent climate forcing and providing the empirical motivation for the state-space modelling framework described in Section 3.4.

Baseline climate figures and well network characterisation statistics (record length, mean water level, seasonal amplitude, and monthly box plots) were produced by 00_climate_summary.py, with data preparation, quality control, gap-filling and upstand correction implemented in 01_data_prep.py.

3.2. Hierarchical Clustering

To characterize the spatial structure of the aquifer and identify distinct zones of hydrological behaviour, agglomerative hierarchical clustering was applied to the continuous dipwell monitoring network.

3.2.1 Data Pre-processing and Dissimilarity Metrics

Because the absolute elevation of the water table varies significantly across the site due to topography, analyzing raw head data would artificially separate functionally identical slacks. To isolate the temporal dynamics—the "shape" of the hydrograph—each well's monthly time series was used directly in the correlation matrix; the correlation-based distance metric inherently removes mean and variance differences between series, making explicit z-score standardisation unnecessary. (Note that explicit z-score standardisation is applied at the cluster average construction stage of the subsequent Pearson affinity analysis, where it ensures that wells with different absolute ranges contribute equally to the cluster average time series.)

Pairwise dissimilarity between the standardized time series was then computed using a correlation-based distance metric $d_{i,j}$:

$$d_{i,j} = 1 - r_{i,j}$$

where $r_{i,j}$ represents the Pearson correlation coefficient calculated over contemporaneous, overlapping temporal records for wells i and j. This specific metric translates the statistical correlation directly into the physical behaviour of the aquifer:

- **Synchronized Slacks (r=1):** Where two dipwells rise and fall in perfect harmony, the distance metric becomes zero (d=0). The algorithm recognizes identical hydrogeological drivers and groups them tightly.
- **Unrelated Slacks (r≈0):** Where one well responds to flashy winter storms and another is heavily buffered, the correlation approaches zero, yielding a distance of one (d=1), effectively pushing them apart in the clustering space.
- **Opposing Slacks (r=−1):** Where wells exhibit inverse temporal behaviour, the distance maximizes at two (d=2).

By utilizing this correlation-based distance, the absolute topographic elevation of the sand dunes is stripped from the analysis. The algorithm is forced to group wells purely on the timing and shape of their temporal hydrographs, allowing deep aquifer wells to cluster together even if they are situated at vastly different physical elevations.

3.2.2 Agglomeration Strategy

The dissimilarity matrix was processed using agglomerative hierarchical clustering, employing Ward’s minimum variance linkage. At the initiation of the algorithm, each dipwell is treated as an individual cluster. The algorithm then iteratively merges the two clusters that result in the smallest possible increase in the total within-cluster variance, quantified mathematically as the Error Sum of Squares (ESS).

For any given cluster C containing n wells, the ESS is defined as the sum of the squared distances between each well's standardized time series (X_i) and the mean temporal signature (centroid, $\bar{x_C}$) of that cluster:

$$ESS_C = \sum_{i \in C} \left\| x_i - \bar{x_C} \right\|^2$$

When evaluating the potential merge of two clusters, A and B, the algorithm calculates the resulting increase in total variance (*ΔESS)*. This merge cost is computed algebraically as:

$$\Delta ESS(A, B) = ESS_{A \cup B} - (ESS_A + ESS_B)$$

Which simplifies to the objective function:

where n_A and n_B represent the number of wells in clusters A and B, respectively.

This specific algebraic formulation is hydrogeologically essential for mapping a continuous dune aquifer. Because the spatial groundwater gradient transitions smoothly from the bedrock ridge to the foreshore, single-linkage (nearest-neighbour) algorithms frequently suffer from "chaining"—sequentially merging adjacent wells until the entire site collapses into a single, uninterpretable cluster.

Ward's mathematical objective function actively resists this chaining effect. The $\frac{n_A n_B}{n_A + n_B}$ term acts as a strict mathematical penalty against merging disproportionately large clusters or appending single outliers to established groups. Instead, it forces the algorithm to prioritize merging smaller, tightly correlated groups first, ensuring that the hydrographic signatures within any newly formed group remain highly synchronized. By penalizing variance, Ward's linkage imposes discrete analytical boundaries onto a continuous physical system, delineating the continuous sand matrix into compact, functionally distinct hydrogeological catchment zones (e.g., the rapid-draining Western Block and the near-perennial Eastern Block).

3.2.3 Cluster Determination and Domain-Informed Selection

The optimal number of clusters (k) was evaluated iteratively for k=2 to k=10 using a dual-metric approach: the elbow method applied to Ward's linkage merge distances, and the mean silhouette coefficient (Yuan and Yang, 2019).

Mathematically, the silhouette coefficient peaked at k=2, reflecting a fundamental, high-level binary split between the shallow and deep aquifer responses across the site. However, selecting k=2 provides insufficient resolution to distinguish the complex geomorphological units that define conservation management at Newborough Warren. Therefore, a final solution of k=6 was deliberately selected.

Rather than representing a post-hoc adjustment, this selection reflects a fundamental principle of environmental data science: mathematical optimization must be balanced with physical interpretability. A six-cluster solution successfully isolates the distinct geomorphological units operating within the Warren—the lake-buffer zone, the eastern mature dune, the western mature dune, the plantation, the coastal margin, and the open-water

boundary. Furthermore, the model maintains a stable silhouette score of 0.41 at k=6 prior to experiencing severe mathematical fragmentation at higher k values (Wu et al., 2021; Zanotti et al., 2022).

While a silhouette score of 0.41 indicates moderate rather than absolute separation, this is entirely consistent with the physical reality of a contiguous, highly permeable sand aquifer where boundaries are inherently gradational rather than impermeable. Ultimately, the robust, contiguous spatial coherence of the resulting clusters (detailed in Section 4.2) provides the independent, physical validation that a purely statistical metric cannot capture in isolation.

Clustering, silhouette validation, and cluster assignment were implemented in **02_clustering.py**. *Cluster map was produced by* **04_cluster_visualisations.py**.

3.3. Checking Cluster Allocation with Pearson Affinity Analysis

While hierarchical clustering imposes discrete boundaries, the physical reality of a highly transmissive sand aquifer is one of continuous hydraulic gradients. To evaluate the rigidity of the Ward's linkage assignments and quantify the gradational nature of the aquifer, a secondary Pearson Affinity Analysis was conducted to measure the probabilistic membership of each well to all identified clusters (Rao and Srinivas, 2006).

3.3.1 Cluster Membership and Affinity Scoring

First, a cluster average hydrograph time series, $\bar{X}_k(t)$, was generated for each of the six identified clusters (k=1,…,6) by calculating the mean standardized response of all wells definitively assigned to that cluster by the Ward's algorithm.

The temporal affinity of every individual well i to each cluster centroid k was then quantified using the Pearson correlation coefficient ($r_{i,k}$) over contemporaneous overlapping records of length N:

$$r_{i,k} = \frac{\sum_{t=1}^{N} \left(x_i(t) - \bar{x}_i\right)\left(\bar{X}_k(t) - \bar{\bar{X}}_k\right)}{\sqrt{\sum_{t=1}^{N} \left(x_i(t) - \bar{x}_i\right)^2 \sum_{t=1}^{N} \left(\bar{X}_k(t) - \bar{\bar{X}}_k\right)^2}}$$

3.3.2 Classification Criteria

To categorize the spatial transition of the groundwater dynamics, let r_{1st} represent a well's highest correlation with any cluster centroid, and r_{2nd} represent its second-highest correlation. The affinity margin, $\Delta r = r_{1st} - r_{2nd}$, serves as a mathematical index of well hydrograph behaviour. Based on these metrics, each well in the 69-well reference network was classified into one of four structural categories. The same affinity procedure was subsequently applied to the extended network of wells with shorter or earlier records allowing the hydrogeological classification to be extended beyond the reference network boundary(Section 3.3.3).

- **Core ($\Delta r \geq 0.05$):** The well exhibits a dominant, unambiguous affinity to a single cluster, and its highest correlation matches its spatial assignment from the Ward's

linkage. These represent the definitive hydrological signature of their respective geomorphological zones.

- **Fuzzy (Δr<0.05):** The well exhibits highly competing affinities between its primary and secondary clusters. This mathematical ambivalence identifies physical ecotones —transitional zones where the groundwater regime shifts smoothly between two distinct structural blocks (e.g., the boundary between the Western Block and the Forest margin).
- **Spy:** The well's highest temporal affinity (r_{1st}) belongs to a different cluster than the one assigned by the Ward's spatial linkage. This divergence highlights localized hydrogeological anomalies where sub-surface stratigraphy overrides broad spatial proximity (Hannah et al., 2000).
- **Unclassified:** The temporal record is of insufficient length or overlap to generate statistically significant correlation coefficients.

Multi-Cluster Affinity (MCA) Finally, Multi-Cluster Affinity (MCA) flags were raised for any well where three or more distinct cluster centroids achieved a correlation of r>0.90. An MCA flag mathematically identifies highly generic, highly responsive boundary wells that lack a specialized local signature, acting instead as structural nodes governed by system-wide climatic forcing rather than local topography.

3.3.3 Extended Network Classification

To extend the hydrogeological classification beyond the 69-well reference network, the Pearson affinity procedure was applied to all wells with a minimum of 24 months of record that did not meet the reference network criteria. The reference network cluster averages derived in Section 3.3.2 were used as fixed classification templates; these were not recalculated from the extended wells. Prior to correlation, all wells — both reference and extended — were z-score standardised jointly as a single combined dataset, ensuring that extended wells with shorter or earlier records were standardised relative to the same temporal distribution as the reference network rather than independently.

Extended wells were classified using the same affinity margin threshold (Δr = 0.05) and MCA criteria (r > 0.90 with three or more centroids) as the reference network audit. Because extended wells carry no prior Ward's cluster assignment, the Spy classification is not applicable; extended wells are assigned as either Core or Fuzzy based solely on the strength of their best-match affinity.

Affinity classification of the 69-well reference network was implemented in **05_pearson_affinity.py***; extension to the full 117-well monitoring array was implemented in** **06_pearson_extended.py**.

3.4. Using Models to derive the Aquifer Architecture

Predicting how a dune aquifer responds to rainfall and evaporation over time requires a model that captures not just the immediate effect of each month's weather, but also the aquifer's memory of its own state. A simple approach — the Traditional Linear Model (TLM) used as a benchmark in Section 3.4.2 — treats the aquifer as a bucket that fills with

rainfall and empties at a fixed background rate, regardless of how full it is (von Asmuth et al., 2002). This works reasonably well when the model is continuously corrected with observed dipwell readings, but fails when run autonomously over longer periods because it has no mechanism to slow drainage as the water table drops or accelerate it when the water table is high (Hypolite et al., 2021; Young, 2011).The state-space model (SSM) captures this physical reality by adding a head-dependent drainage term (β_3) that scales drainage rate to the current water table depth, consistent with the functional relationship between groundwater head and lateral discharge demonstrated in shallow sandy aquifers (Gumuła-Kawęcka et al., 2021; Manis et al., 2022). This single addition transforms the model from a statistical curve-fit into a physically constrained representation of aquifer behaviour, allowing it to simulate water table dynamics autonomously over multi-year periods without accumulating systematic error (Knotters and van Walsum, 1997). The SSM is applied here in two variants, described below.

A parsimonious state-space model (SSM) was developed in two variants: Model A ($\alpha = 0$, strict mass balance) and Model B (free intercept α, used only in the boundary subsidy audit, Section 3.4.3. The baseline formulation is of the form:

$$h(t) = \alpha + \beta_1 \cdot P(t) - \beta_2 \cdot PET(t) - \beta_3 \cdot h(t-1) + \varepsilon(t)$$

where $h(t)$ is depth-to-water-table; $\boldsymbol{\beta_1}$ = *recharge sensitivity;* $P(t)$ monthly rainfall; $\boldsymbol{\beta_2}$ = *atmospheric draw;* $PET(t)$ monthly potential evapotranspiration; $\boldsymbol{\beta_3}$ = *storage decay coefficient;* $\boldsymbol{h_{(t-1)}}$ antecedent water table depth, governing the rate of storage decay; **α** a site-specific intercept representing unmeasured boundary fluxes, constrained to zero in the baseline mass-balanced formulation (Model A) and estimated freely only in the boundary subsidy audit; and $\boldsymbol{\varepsilon(t)}$ a residual error term capturing unexplained variance.

β_3 is defined and reported as a positive quantity throughout this study. The negative sign appears explicitly in the model equation as $-\beta_3 \cdot h(t-1)$ to indicate that storage decay acts as a loss term — a deepe/ higher antecedent water table generates greater drainage, tending to restore equilibrium. Where the notation $-\beta_3$ appears in the text, figures and tables, it refers to this loss term and should be read as the magnitude of the drainage decay coefficient. An increase in $-\beta_3$ therefore indicates accelerated drainage — the water table is losing water more rapidly per unit of antecedent depth — and a decrease indicates retarded drainage or greater storage retention. This convention is maintained consistently throughout all analytical sections, including the BACI analysis, scraping analysis and cluster mechanistic equations.

3.4.1 Modelling the Cluster water levels

The standard state-space model (Model A) was applied at the cluster level to derive mechanistic coefficients representative of each hydrogeological zone. Prior to averaging, each well's monthly depth-to-water time series was corrected to a common ground-surface datum (see Section 3.1.1). The corrected depth series were then z-score standardised — the mean subtracted and the result divided by the standard deviation — so that wells with different absolute water table depths and seasonal amplitudes could be compared on the basis of their temporal dynamics alone rather than their absolute values (Rao and Srinivas, 2006; Zanotti et al., 2022).

For each of the six clusters, a mean monthly time series was computed by averaging the standardised, datum-corrected records of all member wells, producing a single representative hydrograph of seasonal water table behaviour for that hydrogeological zone. This average hydrograph was then rescaled back to physical units using the cluster mean and standard deviation, yielding a representative water table time series in metres below ground surface referenced to a consistent ground-surface datum. The SSM was fitted to this average hydrograph using the full available record rather than the 100-month window applied to individual wells, yielding a single set of coefficients (β_1, β_2, $-\beta_3$) representative of the cluster as a whole. This approach reduces the influence of individual well idiosyncrasies and local topographic noise, providing a parsimonious mechanistic characterisation of each hydrogeological zone.

Crucially, this regression was fitted with the intercept strictly forced to zero. By prohibiting an intercept constant, the model enforces a strict physical mass balance: in the absence of external climatic forcing (P = 0, PET = 0), the aquifer cannot spontaneously generate or lose water beyond the passive drainage decay dictated by its own antecedent storage state ($-\beta_3 \cdot h_t-1$). The resulting coefficients represent fundamental aquifer properties: β_1 as recharge sensitivity, β_2 as the atmospheric evaporative draw, and β_3 as the internal storage and drainage decay coefficient (Knotters and Bierkens, 2000).

The storage decay coefficient $-\beta_3$ provides a parsimonious empirical representation of head-dependent lateral drainage that does not require explicit knowledge of hydraulic conductivity, aquifer geometry or hydraulic gradient — parameters that are either unmeasured or spatially variable across a 700 ha dune system where borehole logs are absent across the majority of the study area. It is functionally equivalent to a linearised Darcy flux under the assumption that the hydraulic gradient scales proportionally with water table height above the drainage base, which is a reasonable approximation for a shallow unconfined dune aquifer draining to a fixed boundary (Knotters and van Walsum, 1997).

The Lumped Catchment Storage Coefficient (LCSC) is defined as the reciprocal of the recharge sensitivity expressed as a percentage (LCSC = $100/\beta_1$). Expressing it as LCSC% converts the recharge sensitivity into physically intuitive units — the millimetres of rainfall required to raise the water table by 10 cm — and scales in the same direction as storage capacity, making it directly comparable to the empirical specific yield concept used by site managers. A higher LCSC percentage therefore indicates a more buffered, less responsive aquifer: a shallow aquifer over impermeable till produces a large water table rise from a small rainfall event, a high β_1, and a low LCSC, whilst a deep sand aquifer absorbs rain across a thick unsaturated zone before the water table rises appreciably, producing a low β_1 and a high LCSC.

3.4.2 State-Space vs. Traditional Linear Models for Diagnostic Validation of Aquifer Architecture

Model Configurations . Both models were independently fitted to the most recent 100 months of data to predict the monthly change in water table elevation (Δh_t) using the same climatic forcing data, differing only in their structural representation of the aquifer:

1. **Traditional Linear Model (TLM)** treats the aquifer as a passive linear store driven purely by atmospheric inputs, with a fitted intercept term capturing background drift and unmeasured forcing, but no mechanism representing the aquifer's memory of its own state (von Asmuth et al., 2002):

$$\Delta h_t = \beta_1 P_t - \beta_2 PET_t + c$$

2. **State-Space Model (SSM):** The physically constrained mass-balance framework defined in Section 3.4, fitted through the origin (c = 0), with antecedent head h(t−1) included as a feedback term that causes drainage to accelerate when the water table is high and slow when it is low

$$\Delta h_t = \beta_1 P_t - \beta_2 PET_t - \beta_3 h_{t-1}$$

Forecasting Modes: Diagnostic vs. Iterative

Model performance was evaluated under two distinct operational modes to separate baseline statistical fit from true forecasting stability:

- **One-Step Diagnostic Fit:** The model predicts the change in head for month **t** using the *observed* field measurement of h(t−1) from the prior month. This tests the fundamental validity of the regression coefficients without compounding error.
- **Iterative Forecasting Stability:** The model predicts the change in head for month **t** using its own simulated head from the previous timestep (h_sim), relying entirely on its own simulated state over the 100-month evaluation window without periodic data assimilation; this tests resistance to cumulative drift over multi-year autonomous simulation.

Performance Metrics. Diagnostic fit (one-step) was evaluated using the Coefficient of Determination (R^2). Iterative forecasting stability was evaluated using R^2, the Root Mean Square Error (RMSE), and the Nash-Sutcliffe Efficiency (NSE) (Nash and Sutcliffe, 1970).

The RMSE is mathematically defined as:

$$R^2 = \left(\frac{\sum \left(h_{obs} - \bar{h}_{obs}\right)\left(h_{\square} - \bar{h}_{\square}\right)}{\sqrt{\sum \left(h_{obs} - \bar{h}_{obs}\right)^2 \sum \left(h_{\square} - \bar{h}_{\square}\right)^2}} \right)^2$$

The NSE is mathematically defined as:

$$NSE = 1 - \frac{\sum \left(h_{obs,t} - h_{,t}\right)^2}{\sum \left(h_{obs,t} - \bar{h}_{obs}\right)^2}$$

The NSE is the most informative metric for continuous hydrological forecasting. An NSE=1.0 indicates perfect predictive agreement, an NSE=0 indicates the model is only as accurate as using the long-term observed mean, and an NSE<0 indicates that the model is actively worse than using the observed mean, typically due to unchecked cumulative mass-balance drift.

Spatial Diagnostics of Subsurface Architecture using SSM and TLM comparison. Plotting the comparative performance metrics, defined as the improvement in variance explanation:

$$\Delta R^2 = R^2_{SSM} - R^2_{TLM}$$

and the improvement in iterative forecasting stability:

$$\Delta NSE = NSE_{SSM} - NSE_{TLM}$$

serves as more than a statistical validation; it acts as a spatial diagnostic tool for the underlying geological architecture (Beven, 2012; Gupta et al., 2008). Similar multi-methodological approaches have been employed to constrain recharge estimates and reduce uncertainty in shallow sandy systems by evaluating the sensitivity of model boundaries to observed head variations. Because the fundamental mathematical difference between the two models is the inclusion of the head-dependent internal storage decay coefficient ($-\beta_3 \cdot h(t-1)$), the spatial distribution of model improvement physically delineates the subsurface structural boundaries of the aquifer:

- **High ΔR^2 and ΔNSE zones:** Regions where the SSM most substantially outperforms the TLM are those where drainage is most strongly head-dependent at the monthly timescale (Fetter, 2001; Robins et al., 2013). In these zones the proportional drainage rate varies considerably between high and low water table states, making the $-\beta_3$ term essential for accurate autonomous simulation. The Western Block's deep sand buffer moderates this dynamic, and canopy interception in the Forest cluster smooths it further, producing a gradient of decreasing SSM gain from east to west that mirrors the structural gradient identified by clustering.
- **Low ΔR^2 and ΔNSE zones:** Conversely, areas exhibiting small improvement between the two models reflect zones where drainage is less strongly head-dependent. A small SSM gain in these zones does not indicate model failure but confirms that the TLM's linear approximation is adequate where drainage dynamics are sufficiently buffered or smoothed (Gumuła-Kawęcka et al., 2021).
- **Tidal boundary (ΔNSE failure):** Negative ΔNSE at tidal boundary wells confirms the physical limits of the rainfall-PET-drainage framework at tidal margins and identifies locations where specialist boundary-condition modelling would be required.

By mapping the success or failure of the storage decay coefficient $-\beta_3$, this comparative methodology physically draws the structural boundary lines between deep sand reserves, perched slacks, and complex discharge zones without requiring extensive invasive coring.

SSM versus traditional linear model benchmarking across all 69 reference wells was implemented in **08_model_benchmarking.py**.

3.4.3 Aquifer Mapping using State-Space Models With and Without an Intercept

To distinguish between the aquifer's intrinsic storage capacity and external boundary influences, a dual-model diagnostic audit was applied to all 69 wells in the reference network (Burnham & Anderson, 2002). This step was essential to ensure that the LCSC values across the network — particularly at wells proximal to the northern rock ridge and tidal margins — were not being artificially distorted by the adjacent rock ridge or localised drainage features (Knotters & van Walsum, 1997; von Asmuth et al., 2002).

Theoretical Framework: Model A vs. Model B. The standard state-space model (Model A) assumes a closed system where all head changes result from immediate meteorological drivers (P, PET) and internal state-feedback ($-\beta_3$). However, in the presence of a steady-state boundary flux — such as lateral baseflow from the northern rock ridge — Model A may produce biased coefficient estimates, potentially misattributing constant pressure to the rainfall coefficient β_1 (Young, 2011). To test for these boundary influences, a second model variant (Model B) was introduced, incorporating a constant intercept (α) to represent unmeasured recharge or drainage:

Model A (Strict Mass Balance):

$$\Delta h_t = \beta_1 P_t - \beta_2 PET_t - \beta_3 h_{t-1} + \varepsilon$$

Model B (Unconstrained Intercept):

$$\Delta h_t = \beta_1 P_t - \beta_2 PET_t - \beta_3 h_{t-1} + \alpha + \varepsilon$$

Hydrogeological Decision Logic. By mapping the calculated intercept (α) from Model B across the site, the audit acts as a hydrogeological probe to identify the aquifer's boundary fluxes. The sign and significance of the intercept provide a spatial diagnostic of site boundaries:

- **Positive Intercept (+α) — Boundary Subsidy:** A statistically significant positive intercept indicates a constant inflow of water (e.g., ridge-front recharge) that maintains the localised water table at a level independent of meteoric forcing.
- **Negative Intercept (−α) — Localised Sink:** A significant negative intercept indicates constant drainage loss, typically associated with proximity to steep topographical gradients, deep tidal channels, or severe localised forest transpiration.
- **Zero or Non-Significant Intercept — Pure Meteoric System:** These wells operate as clean reference sites, reacting strictly to atmospheric forcing and their own internal structural drainage without external boundary influences.

The NSE penalty for the intercept is defined as NSE_B − NSE_A; a positive value indicates that the unconstrained intercept improves iterative forecasting stability at that well, while a negative value indicates degradation. Both variants were evaluated under iterative 100-month forecasting to assess long-term predictive stability (Beven, 2012). Model B improves iterative performance at wells with genuine steady-state boundary fluxes, most notably at tidal and lacustrine boundary wells where a constant external input is physically real.

However, introducing a constant intercept at wells without a confirmed boundary flux risks accumulating systematic drift over multi-year simulations. Consequently, Model A is retained as the unified baseline for all site-wide mapping and forecasting, while Model B intercept values are used solely as a spatial diagnostic of boundary flux locations.

Dual-model intercept audit and spatial mapping of boundary subsidies were implemented in **07_boundary_intercept.py**.

3.5. Anthropogenic Interventions

3.5.1 The Before-After Control-Impact (BACI) Design

To evaluate the hydrological outcomes of anthropogenic interventions at the study site, a Before-After Control-Impact (BACI) experimental design was used (Stewart-Oaten et al., 1986). This framework is designed to isolate the effect of a specific localized disturbance from broader natural variability, such as seasonal weather patterns or regional climatic drift.

The fundamental mechanism of the BACI approach involves contrasting an impact zone (where the intervention occurred) with a reference control zone (which remained undisturbed) across designated pre-intervention and post-intervention periods. Rather than simply measuring absolute changes in water levels, the model calculates the difference (Δ) between the impact and control sites. The actual effect of the intervention is subsequently quantified by measuring the shift in this difference after the event.

For highly complex environments where multiple interventions overlap, the standard BACI method was developed into a Hierarchical Nested Control design to ensure robust baseline validation. By using multiple control sites to defend against localized hydrogeological anomalies, this study employs a "Beyond BACI" approach (Underwood, 1992).

General Before-After Control-Impact (BACI) Calculation. The standard BACI effect size isolates the intervention signal from natural background variation. The fundamental equation calculates the difference in means (μ) between the Impact (I) and Control (C) sites across the Before (B) and After (A) periods:

$$BACI_{\text{effect}} = \left(\mu_{I,A} - \mu_{C,A}\right) - \left(\mu_{I,B} - \mu_{C,B}\right)$$

Where:

- $\mu_{I,A}$ is the mean water level at the impact site post-intervention.
- $\mu_{C,A}$ is the mean water level at the control site post-intervention.
- $\mu_{I,B}$ and $\mu_{C,B}$ represent the respective means prior to the intervention.

Cumulative Sum (CUSUM) Statistical Verification. To validate the precise temporal onset and permanence of hydrological shifts identified by the BACI framework, a Cumulative Sum (CUSUM) sequential analysis was integrated into the analytical framework (Page, 1954). While the BACI contrast quantifies the magnitude of an intervention's effect, the CUSUM statistic is used to identify the exact temporal structural break — or regime shift

— within the time series data. Era-specific estimation of the storage decay coefficient $-\beta_3$ is additionally reported across treatment and control wells; changes that are confined to the intervention footprint would be consistent with a genuine drainage response, whilst changes extending uniformly to unmanipulated control wells indicate a network-wide signal attributable to climate or SSM fitting artefacts rather than the intervention itself.

Mathematical Framework. The CUSUM method operates by calculating a running total of the deviations of individual data points from an established baseline mean. In the context of this study, the input variables (x_i) are the sequential differences between the impact and control wells (Δ), and the baseline (μ_baseline) is the mean of these differences during the pre-intervention period. The cumulative sum at time t is formalized as:

$$C_t = \sum_{i=1}^{t} \left(x_i - \mu_{\text{baseline}} \right)$$

Interpretation of the CUSUM Trajectory. The resulting CUSUM trajectory provides a clear visualization of system stability:

- **Baseline Stability:** A stable relationship between the impact and control sites yields random fluctuations around the baseline mean, producing a roughly horizontal CUSUM trajectory.
- **Structural Break Detection:** A sustained shift in the hydrological relationship — such as that caused by canopy removal or slack scraping — results in directional deviations that accumulate rapidly, manifesting as a distinct change in the gradient of the CUSUM line.
- **Temporal Verification:** The exact point of inflection, where the trajectory breaks from its horizontal trend, marks the initiation of the intervention's physical effect.

By aligning these inflection points with the documented dates of the December 2017 clearfell and the 2015 and 2023 scraping interventions, the analysis confirms that the observed divergence is a direct consequence of the site works rather than gradual environmental drift. To isolate the localized effects of topographical and silvicultural interventions from background climatic drift, all CUSUM trajectories were calculated on the BACI delta (impact minus control) rather than raw water table depth. This ensures that any identified structural breaks represent a true divergence from the undisturbed regional baseline.

Where an impact site has experienced more than one intervention over the monitoring period, or where a secondary disturbance confounds the primary signal, the record is subdivided into discrete management eras defined by the dates of known interventions. Each era is treated as a distinct analytical period with its own mean water level, inter-era BACI delta and $-\beta_3$ estimate. This allows the analysis to separate the signal of a specific intervention from preceding baseline conditions and from subsequent confounding events — for example, isolating the effect of dune scraping before the onset of a plantation clearance pulse that would otherwise inflate or mask the scraping response. Era boundaries are defined a priori from documented intervention dates rather than from the data, preventing circular reasoning in the identification of structural breaks.

3.5.2 Hierarchical Nested Control Application

This framework operates across two distinct analytical tiers:

- **Tier 1 (Background Drift Validation):** Local control wells are first evaluated against a broader regional mean. This step verifies that the designated control sites are behaving consistently with macro-environmental trends and are not subject to anomalous localized influences, such as unmapped coastal drawdown. This tier verifies that the local control (C_local) accurately tracks the broader regional mean (C_region). The temporal difference (Δ_drift) at any time t is calculated as:

  $$\Delta_{\text{drift},t} = C_{\text{local},t} - C_{\text{region},t}$$

  A structurally stable Δ_drift,t over the baseline timeline validates the local control for use in Tier 2.

- **Tier 2 (Intervention Signal Isolation):** Once validated, these local controls are paired directly with specific impact wells. This pairing subtracts the regional noise and secondary intervention pulses, isolating the pure signal of the targeted experiment. The Tier 2 BACI effect calculates the net benefit of the intervention by contrasting the impact well (I) against the validated local control (C_local) across multiple distinct eras (E):

  $$\text{Net Benefit}_{E1 \to E2} = \left(\mu_{I,E2} - \mu_{C_{\text{local}},E2}\right) - \left(\mu_{I,E1} - \mu_{C_{\text{local}},E1}\right)$$

  This formulation systematically removes regional climatic noise and secondary pulses, standardizing the output to isolate the pure scraping signal.

3.5.3 Change in the (-β_3) Storage Decay Coefficient

Both the clearfell and scraping interventions may cause changes to subsurface drainage mechanics, assessed through era-specific estimation of the storage decay coefficient $-\beta_3$. To reduce the confounding influence of inter-era variation in rainfall and evapotranspiration, a two-step procedure was applied. First, β_1 and β_2 were estimated by fitting the full SSM to the complete well record.

$$\Delta h = \beta_1 P_m - \beta_2 PET + \beta_3 h_{prev} + \epsilon$$

The climate forcing contribution was then removed from each monthly observation to yield a drainage residual series (D_residual), representing the component of water table change attributable to head-dependent drainage after removing the estimated rainfall and evapotranspiration contributions:

$$D_{residual} = \Delta h - \beta_1 P_m + \beta_2 PET$$

Second, for each management era separately, $-\beta_3$ was estimated by regressing this drainage residual against −h_prev with a constant term:

$$D_{residual} = -\beta_3 \cdot \left(-h_{prev}\right) + c + \epsilon$$

This procedure reduces the influence of differential climate forcing between eras, but does not eliminate all sources of artefact: where the water table undergoes a sustained directional shift between eras — as occurs following both clearfell and scraping — the h_prev term

itself tracks a non-stationary trajectory, and the model may interpret systematic head changes as an elevated drainage signal. Era-specific $-\beta_3$ estimates should therefore be interpreted cautiously, and spatial extent is used as the primary diagnostic: changes confined to the intervention footprint are consistent with a genuine drainage response, whilst changes extending uniformly to unmanipulated control wells point to a network-wide artefact rather than an intervention effect.

3.5.4 Dune Scraping Intervention Analysis

A two-tier hierarchical BACI framework was applied across five focal wells. Three wells served as scraping treatment sites — CEH36 (central Warren, scraped April 2015), CEH18 (western boundary, scraped October 2023) and CEH21 (coastal margin, scraped October 2023) — and two as local controls: CEH4 (central Warren, unmanipulated) and CEH22 (coastal margin, unmanipulated). A regional mean was computed as the arithmetic mean of six wells distributed across the unmanipulated network (CEH9, NW8, NW8b, NW5, NW6, NW7) to characterise background system-wide trends.

Because the three treatment wells span different management histories, each was decomposed into site-specific eras defined by intervention dates:

CEH36 and its paired control CEH4 were decomposed into three eras:

- *Baseline* — pre-April 2015
- *Pure Scraping* — April 2015 to November 2017
- *Felling Pulse* — December 2017 onwards, capturing the confounding influence of the plantation clearance

CEH18 and CEH21 were decomposed into three eras:

- *Baseline* — pre-December 2017
- *Felling Pulse* — December 2017 to October 2023
- *After Scraping* — October 2023 onwards

Three limitations of the control wells bear directly on interpretation and are stated here to avoid repetition in the results.

First, CEH4 is subject to two independent sources of instability: progressive ground surface lowering from coastal erosion, and sustained water table deterioration attributable to the post-2017 felling pulse confirmed by its Tier 1 CUSUM trajectory. Net benefit estimates at CEH36 during the Pure Scraping era should be treated as upper bounds; an additional upward bias may arise from contemporaneous foredune build-up at the site contributing to a rising water table independently of the scraping intervention. Net benefit estimates at CEH18 over the After Scraping era are similarly subject to caution given CEH4's post-felling deterioration. The robustness of the CEH36 scraping signal is tested independently in Section 4.5 using a synthetic control constructed from 11 donor wells calibrated on the pre-scraping period.

Second, CEH22 experienced accelerating deterioration throughout the monitoring period that, as the network-wide $-\beta_3$ analysis demonstrates, reflects a system-wide signal rather than a process localised to the coastal margin. CEH22 does not constitute a stable Tier 1 baseline; net benefits are therefore not reported for CEH21, and raw Tier 2 BACI shifts are reported for completeness only.

Tier 1 CUSUM analysis evaluated each local control well against the regional mean to verify that the controls were tracking system-wide background trends consistently. Tier 2 paired CUSUM analysis differenced each treatment well against its matched local control to isolate the scraping signal from regional climatic drift. Net benefit was computed as the BACI shift in the treatment-minus-control difference between consecutive eras as defined in Section 3.5.1. The SSM was re-fitted independently for each era at treatment and control wells and $-\beta_3$ compared across eras with 95% confidence intervals, interpreted as described in Section 3.5.3.

The two-tier hierarchical CUSUM-BACI scraping analysis was implemented in **09_scraping_intervention.py**.

3.5.5 Clearfell BACI experiment

A three-zone hierarchical BACI experiment was conducted across 16 wells to assess the hydrological consequences of the December 2017 clearfell. Wells were categorised by spatial proximity to the cleared area into three zones, with zone membership defined a priori from well locations relative to the felling boundary rather than from the data, and verified against distances from the clearfell centroid (E=241177, N=363645, BNG):

- core impact (FE2, FE4, WMC3 — 65–102 m from the clearfell centroid, within or immediately adjacent to the felled block),
- edge zone (FE1, FE3, CEH31, LIS1, CEH20, CEH30, CEH16, NW8b — 65–241 m, within the broader forest–dune boundary zone),
- regional control (CEH32, CEH34, CEH33, NW10, CEH19, CEH9, NW7, NW6 — outside the zone of direct influence).

CEH36, CEH18 and CEH21 are not included in the edge zone: CEH36 (339 m) was scraped in April 2015 prior to the clearfell and its water table record therefore reflects post-scraping rather than undisturbed pre-intervention conditions; CEH18 (451 m) and CEH21 (766 m) are too distant for edge classification. The control pool comprises eight wells — CEH32, CEH34, CEH33, NW10, CEH19, CEH9, NW7 and NW6 — all open warren or forest interior wells unaffected by either intervention.

The BACI displacement for each zone was calculated as the mean water level of that zone minus the mean of the regional control zone, evaluated across defined temporal eras. Because CEH36 was scraped in April 2015, 32 months before the clearfell, the pre-felling period is not hydrologically homogeneous — the network-level BACI displacement reflects post-scraping conditions from April 2015 onwards regardless of zone membership. The pre-felling period was therefore subdivided into two eras on the basis of the known scraping date: a pre-scraping era (start of record to March 2015) and a post-scraping pre-felling era (April 2015 to November 2017). The post-scraping pre-felling era was adopted as the clean baseline for calculating the felling step change and CUSUM. The pre-scraping era mean is reported separately to quantify the displacement attributable to the 2015 scraping event and to allow the combined hydrological cost of both interventions to be calculated.

To separate the intervention signals from background climate variability, an ANCOVA-BACI model (Underwood, 1992; Gotelli & Ellison, 2004) was fitted to the monthly BACI displacement series with the centred cumulative water balance anomaly (P − PET − 19.47 mm/month, cumulated from the start of the BACI record) as a continuous climate covariate.

The model included step terms for the 2015 scraping event and the December 2017 clearfell, and an interaction term between the climate covariate and the post-felling indicator to test whether climate sensitivity changed after felling. An additional step term for the October 2023 scraping event was tested but not retained ($p = 0.258$, $\Delta AIC = +0.66$). The climate-corrected CUSUM starts at zero at the clearfell date (December 2017) and accumulates post-felling deviations from the post-scraping pre-felling fitted mean; the pre-felling CUSUM is zero by construction.

Five independent robustness analyses were conducted to test whether a clearfell-specific signal could be detected once three principal confounds are accounted for: scraping baseline contamination at CEH36; the anomalously wet 2015–16 pre-clearfell period, which elevated water tables across the whole network before the clearfell and inflates the reference level against which post-clearfell conditions are measured; and the common post-2018 declining residual visible across all zones including controls, consistent with reduced lagged ridge recharge following the drier 2018–2021 period and indistinguishable from a clearfell-specific effect within the ANCOVA framework.

SSM residual normalisation The SSM was calibrated on the pre-scraping period for each well and run forward iteratively using observed climate forcing. Residuals were normalised against the regional control zone mean to remove the common climate signal, providing a climate-independent test of intervention-specific displacement.

Synthetic control analysis For each impact well with a sufficient pre-scraping record, a synthetic counterfactual was constructed as an optimally-weighted combination of 11 donor wells calibrated on the pre-scraping period. The gap between observed and synthetic provides an estimate of the intervention effect after removing the common climate signal, without relying on the ANCOVA parametric assumptions.

Cluster transition analysis. If clearfell converts C4 Forest behaviour to C3 Western Block behaviour, SSM coefficients of felled wells should converge toward C3 reference values post-clearfell. Rolling 48-month β_1 and β_3 estimates were tracked for the felled zone and compared against C3 and C4 reference trajectories.

A spatial transect analysis was conducted to test whether the post-felling response showed a spatial gradient consistent with a localised clearfell effect. Era mean depths were compared across a transect from the uncleared plantation interior (CEH2, 414 m from clearfell centroid) through the regional control (CEH34, 285 m) to the core impact zone (WMC3, 92 m) and edge wells (184–191 m). A genuine clearfell-specific effect would produce a gradient strongest at WMC3 and diminishing with distance; a spatially uniform response would indicate a common climate signal rather than a localised intervention effect.

SSM coefficient comparison. The SSM was fitted independently to the pre- and post-felling records at each of the 16 wells, and the storage decay coefficient β_3 compared between periods with 95% confidence intervals and significance testing as described in Section 3.5.3. FE4 and LIS1 had no pre-felling records, and FE3 had only six pre-felling observations; all three are represented by post-felling values only.

The clearfell BACI experiment was implemented in *10_clearfell_baci.py* (v10.5.4).

3.6. Threshold Forecasting

To translate the observed spatial dynamics into predictive management tools, operational forecasting equations were derived using a multi-tiered mathematical framework. This approach combines continuous mass-balance modelling with seasonal transfer functions to quantify the aquifer's climatic vulnerability.

3.6.1 Cluster-Level Mechanistic State-Space Equations

The cluster-level SSM coefficients derived in Section 3.4.1 were used to construct operational P_flood threshold equations for each of the four principal clusters. The model was fitted with the intercept strictly forced to zero, ensuring that the coefficients represent a strict physical mass balance and providing the basis for algebraic inversion in Section 3.6.3.

3.6.2 Seasonal Transfer Functions

While the SSM operates on a continuous monthly basis, conservation management typically requires inter-seasonal risk forecasting — specifically, predicting winter flood peaks and summer drought minima from antecedent conditions. To address this, autoregressive transfer functions were fitted at a hydrological-year resolution for the three primary structural domains: Eastern Block, Western Block, and Forest clusters.

Two equations are fitted for each domain. The winter transfer function predicts the peak winter water table (h_peak) from cumulative winter precipitation and the antecedent summer minimum:

$$h_{peak} = \beta_1 P_{winter} + \beta_2 h_{min} + c$$

The summer transfer function predicts the summer minimum water table (h_min) from cumulative summer precipitation and the antecedent winter peak:

$$h_{min} = \beta_1 P_{summer} + \beta_2 h_{maxwinter} + c$$

where P_winter is cumulative precipitation over October–March, P_summer is cumulative precipitation over April–September, h_min is the preceding summer minimum water table depth, h_max is the preceding winter maximum water table depth, and c is a fitted intercept. Data are organised by hydrological year commencing 1 October.

The antecedent state terms (h_min in the winter equation, h_max in the summer equation) explicitly formalise the memory of the dune aquifer system — the degree to which a winter flood peak is conditioned by the depth of the preceding summer drought, and vice versa. This parameterisation allows managers to construct simple conditional forecasts: given a known or projected winter rainfall total and the current summer water table position, the expected winter flood maximum can be estimated directly from the fitted equation without running the full SSM.

3.6.3 Algebraic Derivation of the Critical Rainfall Threshold (P_flood)

The most operationally significant metric for dune slack management is the critical rainfall threshold (P_flood)—the precise minimum winter rainfall depth required to raise the water table from its current antecedent position to the point of surface inundation (where depth to water table equals zero).

Rather than relying on separate empirical fitting, this threshold was derived deterministically by algebraically rearranging the mechanistic state-space equation. Setting the target head to the elevation of the slack floor (h_target), the required change in head becomes $\Delta h = h_{target} - h_{t-1}$.

Substituting this into the mass-balance equation and isolating for precipitation yields the critical threshold parameter:

$$h_{target} - h_{t-1} = \beta_1 P_{flood} - \beta_2 PET_t - \beta_3 h_{t-1}$$

Assuming that atmospheric evaporative demand is negligible during peak winter saturation events ($PET_t \approx 0$)—a standard hydrological assumption for temperate coastal winters—the equation resolves cleanly to:

$$P_{flood} = \frac{(h_{target} - h_{t-1}) + \beta_3 h_{t-1}}{\beta_1}$$

This algebraic inversion provides site managers with a dynamic, site-specific forecasting tool. By simply inputting a measured summer minimum dipwell reading (h_t−1), the equation outputs the estimated millimetres of winter precipitation required to achieve ecological flooding conditions in the subsequent season.

3.6.4 Long-Term Climate Trajectory and Threshold Exceedance Analysis

To assess the long-term trajectory of slack hydrological viability, annual summer minima and winter maxima were extracted separately for C1, C2 and C3 from the block-averaged time series generated during the clustering analysis. C4 was excluded as its water table depths have no direct relevance to slack ecology. The hydrological year was defined as 1 October to 30 September, consistent with Section 3.6.2, and a minimum of three valid monthly observations per season was required for a year to be included.

For summer minima, ordinary least squares linear regression was fitted to the annual series for each cluster over the 2004–2026 observation window and extrapolated to 2040, with 95% confidence intervals derived analytically. Summer trajectories were plotted against the eco-hydrological viability thresholds established by Curreli et al. (2013): −0.61 m for wet slack communities (SD15b) and −0.98 m for dry slack communities (SD16).

For winter maxima, no reliable projection method was identified — direct linear regression on annual winter maxima produced R^2 values below 0.03 across all clusters, and transfer function projection via the seasonal models of Section 3.6.2 produced physically implausible trajectories for the eastern clusters due to extrapolation beyond the fitted range. The winter analysis therefore presents the observed record only. Threshold exceedance frequency was calculated for each cluster as the proportion of hydrological years in which the winter maximum exceeded the wet slack flooding threshold of −0.10 m (SD15b) and the dry slack threshold of −0.25 m (SD16), providing a direct measure of current ecological flooding adequacy for each structural domain.

Critical rainfall threshold equations, peak flood transfer functions, and summer drought transfer functions were implemented in **11_forecasting_thresholds.py**.

3.7. Characterising Water Table Losses: Depth-Dependent Evapotranspiration and Cluster Water Balance

3.7.1 Depth-Dependent Evapotranspiration Model

To test whether the atmospheric draw coefficient varies significantly with water table depth — and whether a depth-dependent formulation is warranted over the standard fixed-β_2 SSM — an extended model was implemented in which β_2 was replaced by the depth-dependent term $\beta_2 \cdot \exp(-\lambda \cdot d)$, where d is the depth of the water table below ground surface (metres) and λ is a decay parameter (m^{-1}) controlling the rate at which evapotranspiration influence declines as the capillary fringe retreats from the root zone. Depth below ground surface was computed at each timestep as $d = -h + \bar{u}$, where h is the water table position (negative convention) and $\bar{u}$ is the cluster mean upstand height. Because the modified predictor $-\exp(-\lambda \cdot d) \cdot PET$ remains linear in β_2 for any fixed λ, the three coefficients β_1, β_2 and β_3 were recovered by no-intercept OLS at each value of λ. A systematic grid search over $\lambda \in [0, 6]$ m^{-1} at intervals of 0.05 m^{-1} was conducted independently for each cluster, with the optimal λ selected by maximising iterative simulation NSE. Setting $\lambda = 0$ recovers the standard SSM exactly, providing a direct baseline for comparison.

The grid search produced negligible NSE improvement in all clusters except C3 (Western Block, ΔNSE = +0.037 at λ = 0.45 m^{-1}). For the Forest cluster (C4) the optimal λ was zero — the standard fixed-β_2 formulation was not improved upon by any degree of depth-dependent decay — confirming that the Forest's anomalously low β_2 reflects canopy interception operating at the surface rather than capillary disconnection at depth. The depth-dependent formulation was therefore not adopted for the primary analysis; the standard fixed-β_2 SSM is retained throughout. The diagnostic significance of the C4 null result is discussed further in Section 5.4.2.

The depth-dependent PET grid search and benchmark comparison were implemented in **15_depth_dependent_pet.py***.

3.7.2 Mean Monthly Water Balance Decomposition

To quantify and compare the hydrological inputs and losses operating on each hydrogeological cluster, a mean monthly water balance decomposition was applied to the four principal clusters (C1–C4) using cluster-level SSM coefficients derived in Section 3.4.1. All terms are expressed in metres of water table change per month (m/month), the natural unit of the state-space model and directly comparable across clusters receiving the same climate forcing.

The decomposition isolates three components from the state-space equation $\Delta h = \beta_1 \cdot P - \beta_2 \cdot PET - \beta_3 \cdot h_prev$. Mean monthly recharge was computed as $\beta_1 \times \bar{P}$, where $\bar{P}$ is the mean monthly precipitation over the monitoring period from the RAF Valley record. Atmospheric draw was computed as $\beta_2 \times \overline{PET}$, where $\overline{PET}$ is the mean monthly Thornthwaite PET. Gravity drainage was computed as $\beta_3 \times |\bar{h}|$, where $|\bar{h}|$ is the mean absolute water table depth for each cluster derived from the cluster-average hydrograph. Canopy interception is not included as a separate loss term — it is implicitly captured in the Forest cluster's lower β_1 and lower β_2, and adding it explicitly would double-count losses already embedded in the

fitted coefficients. Total losses are therefore the sum of atmospheric draw and drainage only. The net monthly balance — recharge minus total losses — quantifies the boundary subsidy required from ridge-derived lateral recharge to maintain long-run water table equilibrium.

An indicative volumetric conversion is presented alongside the primary head-space decomposition (Table 3b; Figure 8a), using assumed specific yield values: C1 = 0.08 (shallow sand over glacial till) and C2–C4 = 0.12 (deeper clean dune sand), consistent with the literature range for unconfined sandy aquifer materials (Fetter, 2001). Empirical upper bounds on Sy derived from the WTF method are presented in Section 3.7.3 and used to bracket the volumetric estimates from above (Table 3d; Figure 8b). For the Forest cluster, canopy interception is shown as an explicit segment on the inputs bar of Figures 8 and 8a, computed as $0.24 \times \bar{P}$ following Freeman (2008), reducing the boundary subsidy rather than being additive.

The mean monthly water balance decomposition and volumetric conversion were produced by 16_water_bal.py (outputs: 16_water_bal_table.csv, 16_water_bal_vol_table.csv).

3.7.3 Specific Yield Estimation: Water Table Fluctuation Method

o provide an empirical upper bound on the specific yield (Sy) values used in the volumetric water balance conversion (Section 3.7.2; Table 3b), an independent estimation was performed using the water table fluctuation (WTF) method of Healy and Cook (2002):

$$Sy = R / \Delta h$$

where R is the net recharge reaching the water table (m) and Δh is the corresponding water table rise (m). The method was applied to the cluster-mean hydrographs derived in Section 3.4.1, using monthly net recharge computed as P − PET for months where both the water table rose (Δh > 5 mm) and net recharge was positive (P − PET > 10 mm). Only months satisfying both criteria were included. Monthly data cannot isolate individual storm events, and the WTF method at monthly resolution is known to overestimate effective Sy because it cannot separate true gravity drainage from capillary fringe release (Healy and Cook, 2002; Scanlon et al., 2002). The resulting estimates should therefore be interpreted as upper bounds on Sy rather than point estimates of the long-term effective value.

For the Forest cluster (C4), a correction was applied for canopy interception prior to computing net recharge: $R_{effective} = (1 - 0.24) \times P - PET$, following the site-specific interception fraction of Freeman (2008). Without this correction, the attenuated water table response beneath the plantation would yield an artificially elevated Sy estimate, because the suppressed Δh in the denominator reflects canopy boundary conditions rather than aquifer storage properties.

The WTF method was subsequently applied to all individual reference and extended network wells using the same event-selection criteria. Cluster assignments for extended wells were taken from the Pearson affinity sitewide audit (Section 3.3.3), with the interception correction applied to all C4-assigned wells. Two extended wells were excluded from spatial interpolation on physical grounds: CEH12, which sits on the bedrock ridge in a forested area and whose WTF response reflects fractured rock rather than dune sand storage; and CEH15, a Forest cluster well in a low-lying slack within the plantation, where water table dynamics

are dominated by slack topography and the interception correction is unreliable given the partially open canopy setting. Both wells are retained in the output table with a Ridge_Flag column.

Results of the cluster-level and well-level WTF analyses are presented in Section 4.2.4. The cluster-level analysis was implemented in 17_wtf_specific_yield.py and the well-level spatial analysis in 18_wtf_spatial.py.

3.8. Computational Methods and Software Implementation

Data Pre-processing and Time-Series Management All data pre-processing, statistical modelling, and spatial mapping were executed within a Python programming environment. Time-series data manipulation, interpolation of missing temporal records, and standardisation were performed utilizing the pandas and NumPy libraries.

Statistical and Predictive Modelling The core parameterization of the aquifer mechanics was executed using the *statsmodels* module. For the continuous state-space modelling (Section 3.4), the Ordinary Least Squares (OLS) algorithm was constrained to fit through the origin (omitting the intercept constant) to enforce physical mass balance. Conversely, seasonal transfer functions predicting peak flood and summer drought minimums used unconstrained OLS regressions. The *statsmodels* architecture was further used to extract 95% confidence intervals and p-values for the drainage coefficients ($-\beta_3$) to statistically verify the impact of the clear-felling intervention. Iterative forecasting metrics, including the Nash-Sutcliffe Efficiency (NSE) and Root Mean Square Error (RMSE), were computed via *NumPy* array operations to benchmark the state-space model against traditional regression approaches.

Geospatial Processing and Cartographic Visualization The integration of the topographical data with the monitoring network was managed through a dedicated spatial stack. *GeoPandas* and *Fiona* were used to parse external KML feature boundaries (e.g., the clear-fell perimeter and hydrological boundaries) and standardize all spatial geometries to the British National Grid coordinate reference system (EPSG:27700).

The high-resolution Digital Elevation Model (DEM) was ingested and masked using the *rasterio* library. To visualize the localized dune topography effectively, the DEM arrays were rendered using a dual-slope normalization technique (`TwoSlopeNorm`) provided by *matplotlib*, anchoring the visual transition at specific elevation thresholds to highlight critical hydrological features. Contextual basemaps were fetched via *contextily* when necessary, and the label placement and occlusion avoidance was handled using the adjustText library.

Spatial Groundwater Analysis. A steady-state 2D finite-difference model was implemented to produce gridded head surfaces, lateral flux fields, and scenario difference maps across the full dune system extent. The computational domain was discretised at 50 m resolution on the British National Grid (EPSG:27700), with active cells defined by the site boundary polygon extracted from the Features KML layer. Sea boundary cells were assigned Dirichlet head conditions (h = 0 m AOD) at the southern coast (Caernarfon Bay), eastern margin (Menai Strait), and western estuary below the ridge termination; Neumann no-flow conditions were applied implicitly at the ridge and northern boundary. The governing

equation is the Helmholtz form of the 2D steady-state groundwater flow equation (Bear, 1972; Freeze and Cherry, 1979):

$$\nabla\cdot(T\cdot\nabla h) - \beta_3\cdot h = -(\beta_1\cdot P_eff - \beta_2\cdot PET)$$

where $T = K\cdot b$ is transmissivity, $K = 6.0$ m/day (Betson et al., 2002), and b is the spatially varying aquifer thickness. The β coefficient fields were IDW-interpolated from the SSM well-level estimates to the full grid. The aquifer thickness surface was constructed by IDW interpolation (power = 1) of four hard borehole constraints (Betson et al., 2002) combined with cluster-level prior thicknesses derived from the β_1 recharge sensitivity proxy, with per-well overrides at CEH14 (ridge crest, 5 m) and the estuarine margin wells (1 m). The sparse linear system of approximately 4,200 equations was assembled using standard 5-point central finite differences with harmonic-mean transmissivity at cell interfaces, and solved using the direct sparse solver scipy.sparse.linalg.spsolve (Virtanen et al., 2020). Darcy lateral flux vectors were computed as $Q = -K\cdot b\cdot\nabla h$ (Bear, 1972) from a DEM-blended head surface. Scenario difference maps were computed as $\Delta h = h(\text{scenario}) - h(\text{baseline})$; climate scenarios used the PDE solver applied to modified forcing fields, while forest management scenarios used an analytical spatial shift derived from the BACI-observed displacement with distance-decay propagation to adjacent clusters (exponential decay, e-folding distance 500 m).

3.9. Data requirements and replicability

A deliberate objective of this study was to demonstrate the hydrological insight achievable from minimal, low-cost monitoring infrastructure. All primary data inputs consist of manual dipwell measurements, recorded with a standard electrical contact meter or tape measure, and publicly available monthly climate summaries from the Met Office historic station dataset. No specialist survey equipment, telemetry, or laboratory analysis was required. Potential evapotranspiration was estimated using the Thornthwaite and Mather (1957) method specifically because it requires only mean monthly air temperature — a variable recorded at every Met Office station and at the majority of automatic weather stations worldwide. Methods requiring wind speed, humidity, or radiation data were deliberately avoided as these would limit replicability to sites with more comprehensive climate instrumentation.

The analytical framework presented here could be deployed at any comparable lowland wetland, dune slack, or shallow unconfined aquifer system with a dipwell network and access to a nearby climate station. The principal constraint on analysis quality is record length rather than measurement sophistication; the state-space modelling and BACI analyses both benefit substantially from the long baseline available here, but meaningful cluster separation and mechanistic coefficient estimation are likely achievable with records of eight to ten years or more.

4. Results

Climate and Well Network Characterisation

4.1.1 Climate Record

The RAF Valley climate station provides a continuous monthly record from December 1930 to February 2026 (Figure 3, first panel) (Section 3.1.2). Over the well-record period (April 2005 to February 2026), mean annual precipitation was 890 mm and mean annual Thornthwaite PET was 652 mm, giving a mean P/PET ratio of 1.36 (Table 1). The wettest year on record within this period was 2008 (1,040 mm), followed by 2012 (1,022 mm); the driest complete year was 2010 (736 mm). Four years recorded a P/PET ratio below 1.2 — indicating near-balance between precipitation and evaporative demand — namely 2007, 2010, 2011 and 2022, with 2022 recording the lowest ratio of the monitoring period at 1.12. There is no meaningful difference in mean annual precipitation between the pre-felling (2006–2017: 887 mm) and post-felling (2018–2025: 893 mm) periods, confirming that the post-2018 hydrological signal detected in the BACI analysis (Section 4.6) is not driven by a concurrent change in rainfall amount.

PET shows strong seasonality but negligible inter-annual trend across the record, with the 12-month rolling mean remaining stable at approximately 50–55 mm/month throughout (Figure 3, second panel). Consequently, inter-annual variability in the water balance is driven almost entirely by precipitation rather than evaporative demand.

The levelled cumulative water balance series (Figure 3, third panel), expressed relative to the study-period mean of 19.46 mm/month, oscillates with a clear seasonal signal — rising sharply each winter as precipitation exceeds PET and falling each summer — while tracking the multi-year rhythm of wetter and drier phases. No sustained multi-year drift away from the baseline is evident, confirming that the site maintains a consistent long-run surplus throughout the monitoring period. The water table clearly follows this movement (Figure 3, fourth panel). Linear regression of the cumulative balance against the unsmoothed network mean water level at zero lag yielded $R^2 = 0.515$, confirming that approximately half the variance in network mean water table depth is explained by antecedent climate forcing alone. The residual variance reflects the spatial heterogeneity of the network, local boundary subsidies, and land cover effects explored in subsequent sections. This close correspondence between cumulative climate forcing and water table behaviour provided the empirical motivation for the state-space modelling framework described in Section 3.4.

The seasonal structure of the climate record has direct implications for aquifer behaviour. Mean monthly rainfall peaks in October–November (around 96–100 mm/month) and falls to its minimum in April–May (around 50–52 mm/month), whilst Thornthwaite PET peaks in July–August and is negligible from November to February. The seasonal opposition between peak PET and minimum rainfall concentrates atmospheric evaporative demand on the aquifer during the period of lowest recharge input, driving summer water table drawdown that must be overcome by the following winter's recharge before any slack flooding can occur. This seasonal asymmetry — not the annual rainfall total — is the primary control on dune slack hydrology and is reflected throughout the subsequent analyses.

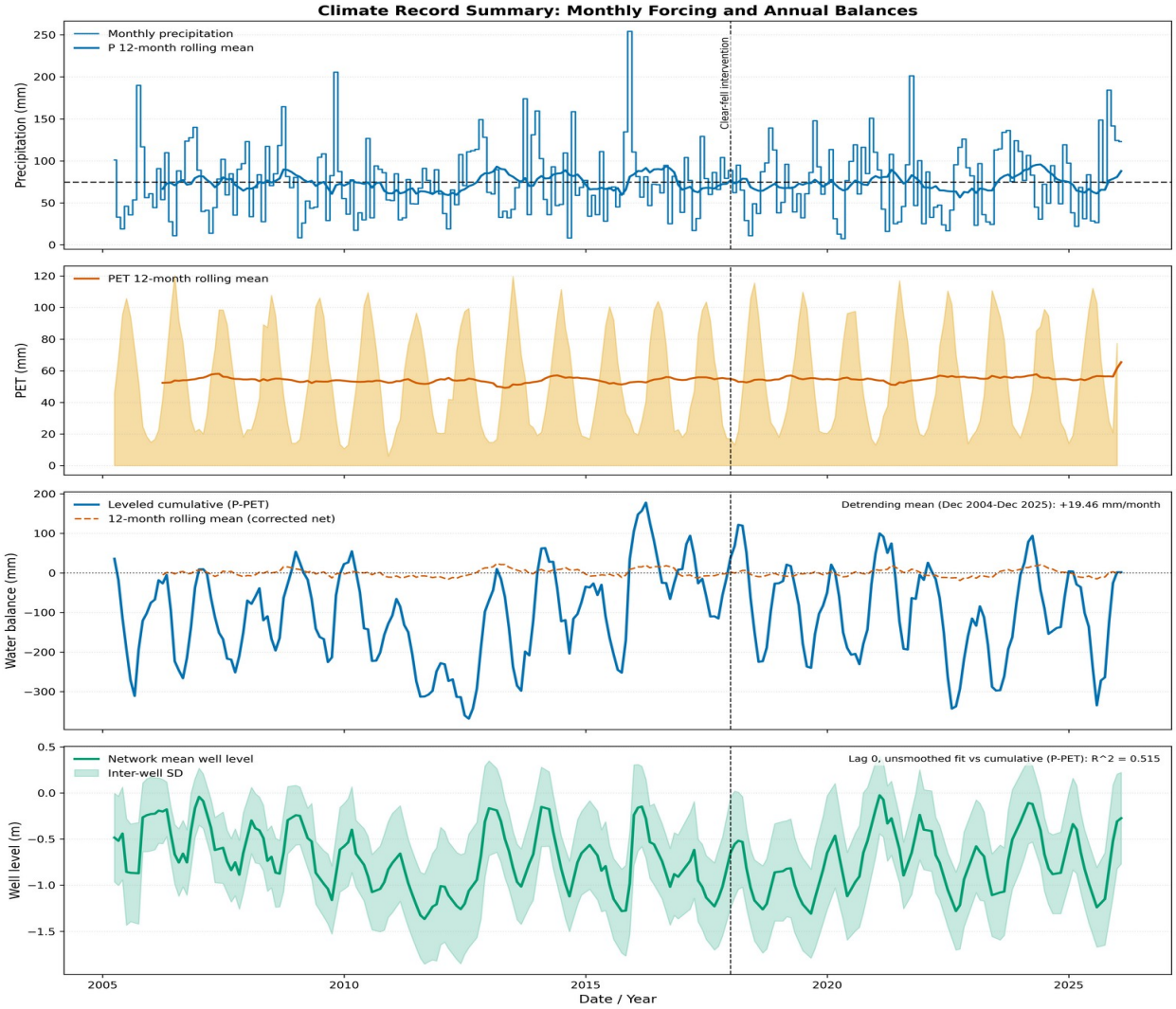


Figure 3. *Climate record summary for the well-monitoring period (April 2005 – February 2026), derived from RAF Valley Met Office station. Upper panel: monthly precipitation (mm) with long-term monthly mean indicated by dashed line and 12-month rolling mean as solid line. Second panel: monthly Thornthwaite PET (mm) as filled area with 12-month rolling mean overlaid. Third panel: levelled cumulative water balance (P − PET, mm), corrected by subtracting the long-term mean monthly net balance to centre the series on zero; the dashed orange line shows the 12-month rolling mean of the corrected net balance. The intercept correction value is shown inset. Lower panel: network mean monthly water level (m below pipe top) across the 69-well reference network, with inter-well standard deviation as shaded envelope; the R² inset gives the coefficient of determination from linear regression of the cumulative balance against the unsmoothed network mean at zero lag. Vertical dashed line marks the December 2017 clearfell intervention in all panels.* *(00_01_climate_timeseries_short.png)*

Year	Annual P mm	Annual PET mm	Months complete	P PET ratio
2005	650.3	577.3	9	—
2006	954	655.4	12	1.46
2007	773.7	653.7	12	1.18
2008	1039.7	648.7	12	1.6
2009	871.3	647	12	1.35
2010	736.2	628.8	12	1.17
2011	760.6	647.5	12	1.17
2012	1021.7	637	12	1.6
2013	844	633.6	12	1.33
2014	943	662.5	12	1.42
2015	974	631.2	12	1.54
2016	858.4	652	12	1.32
2017	871	659.4	12	1.32
2018	873.2	653	12	1.34
2019	832	654.4	12	1.27
2020	944.2	656.6	12	1.44
2021	906.6	649	12	1.4
2022	747.2	668.2	12	1.12
2023	986.2	672.9	12	1.47
2024	911.2	657.8	12	1.39
2025	943	675.5	12	1.4
2026	247.2	77.4	1	—
Long-term mean	890.6	652.2		1.36

Table 1. *Annual climate summary for RAF Valley station over the well-monitoring period. Annual precipitation (P) and Thornthwaite PET are in mm. P/PET ratio below 1.0 indicates evaporative demand exceeds precipitation. Months_complete gives the number of months with valid data in that year. Partial years (2005: April–December only; 2026: January–February only) are included for completeness but no annual P/PET ratio is calculated for these years (shown as —). Long-term mean row is calculated from complete years 2006–2025 only. (00_01_annual_climate_summary_short.csv).*

4.1.2 Well Network

The 69-well reference network (Section 3.1.1) spans a median record length of 150 months (12.5 years), with individual records ranging from 100 to 201 months (Figure 4, upper left). The network mean water level across all wells and months was 0.74 m below pipe top, with individual well means ranging from 0.04 m above pipe top to 2.07 m below, reflecting the considerable topographic and hydrogeological diversity of the site (Figure 4, upper right). The median seasonal amplitude — defined as the difference between mean August and mean February water levels — was 0.59 m, with the water table deepest in late summer and shallowest in late winter, consistent with the expected seasonal recharge cycle for a maritime dune system (Figure 4, lower right). The network mean time series (Figure 4, lower left) shows a clear seasonal cycle throughout the record, with the inter-well standard deviation envelope widening visibly after the December 2017 clearfell intervention, reflecting the divergence of the impact zone wells from the broader network that is explored in detail in Section 4.6.

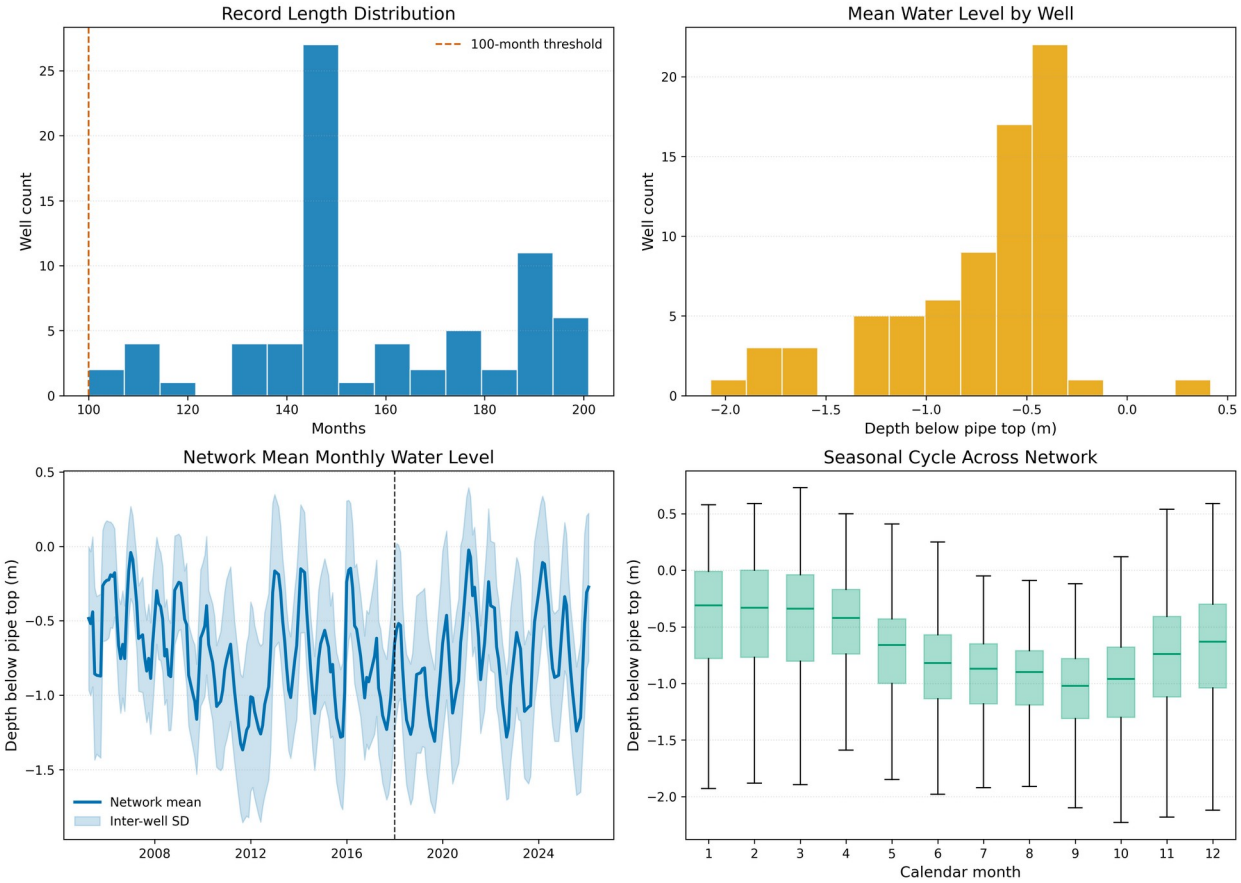


Figure 4. *Well network characterisation for the 69-well reference network. Upper left: frequency distribution of record lengths (months), with the 100-month minimum threshold indicated by dashed line. Upper right: frequency distribution of mean water level by well (m below pipe top; negative values indicate water standing above pipe top). Lower left: network mean monthly water level (m below pipe top) with inter-well standard deviation as shaded envelope; vertical dashed line marks the December 2017 clearfell intervention. Lower right: network-wide seasonal cycle shown as monthly box plots pooling all well-month observations; boxes show interquartile range, horizontal line shows median, whiskers extend to 1.5 × IQR, outliers suppressed. (00_02_well_network_summary_short.png)*

Mapped against the eco-hydrological thresholds of Curreli et al. (2013), the network-wide seasonal extremes reveal a system already under considerable stress. The mean annual summer minimum across the reference network was −1.11 m below pipe top, ranging from −0.33 m to −2.28 m across individual wells. Only one well maintains a mean summer minimum shallower than the wet slack viability threshold of −0.61 m; 35 wells fall within the dry slack zone (−0.61 to −0.98 m); and 37 wells record mean summer minima deeper than the dry slack threshold of −0.98 m, of which 21 lie below the SD16 recovery limit of −1.20 m — placing them beyond the reach of a single scraping event under average conditions. For winter maxima, 22 wells exceed the SD15b wet slack flooding threshold of −0.10 m in an average year and 38 exceed the SD16 dry slack threshold of −0.25 m, though only 11 record mean winter maxima above pipe top. These network-level baselines provide the reference against which cluster-specific dynamics, intervention effects, and climate trajectory are assessed in subsequent sections.

4.2. Network Clustering and Spatial Architecture

The clustering analysis (Section 3.2) identified k = 6 as the preferred cluster solution (Ward linkage merge distance at k = 6 of 1.70; silhouette coefficient = 0.406). The six clusters are spatially coherent and hydrogeologically interpretable (Table 2; Figures 5–7); their mechanistic characterisation including LCSC values is presented in Section 4.2.2.

The dendrogram confirms a primary binary division separating eastern wells (C1, C2) from western, forest and specialist wells (C3, C4, C5, C6) at a Ward linkage distance of approximately 6. This top-level split reflects the division between the till-and-estuary-influenced eastern dune aquifer and the deeper, more uniform western sand block. The forest cluster (C4) clusters separately from adjacent C3 wells despite geographical proximity, confirming that plantation land cover modifies groundwater behaviour independently of the broader dune aquifer geology. The separation also reflects the position of the C4 wells in the northern part of the study area, closest to the rock ridge, where the water table receives lateral recharge descending from the ridge. C5 (CEH3) and C6 (Llyn Rhos-ddu) are structurally isolated outliers whose position adjacent to fixed-head tidal and lacustrine boundaries respectively places them outside the free-draining aquifer assumptions that govern the remaining clusters.

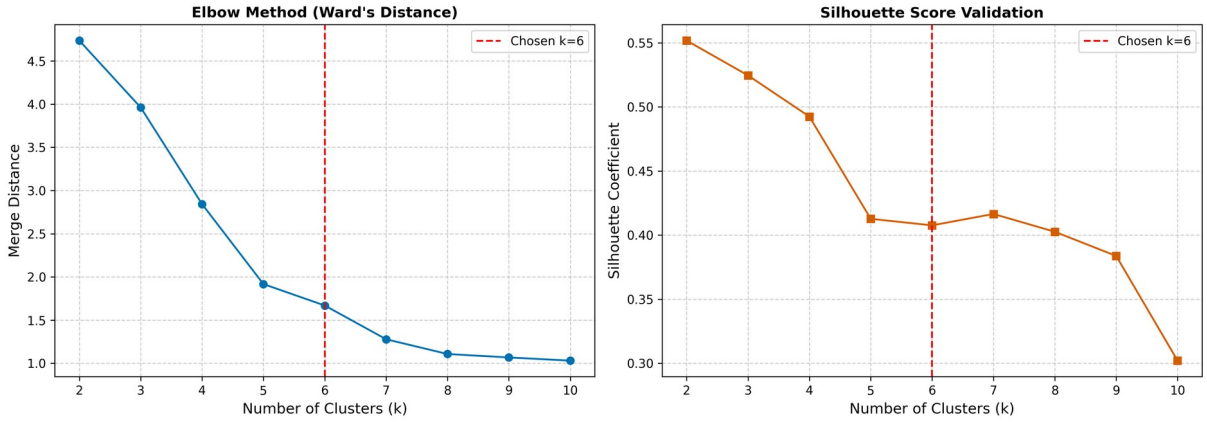


Figure 5. *Cluster validation plots: elbow method (Ward's merge distance) and silhouette coefficient for k = 2–10, with chosen k = 6 indicated. Note that the silhouette coefficient peaks at k = 2 reflecting the fundamental binary contrast between eastern and western aquifer domains; k = 6 represents a domain-informed selection providing sufficient resolution to distinguish the geomorphological units relevant to conservation management. (02_02_validation_plots.png)*

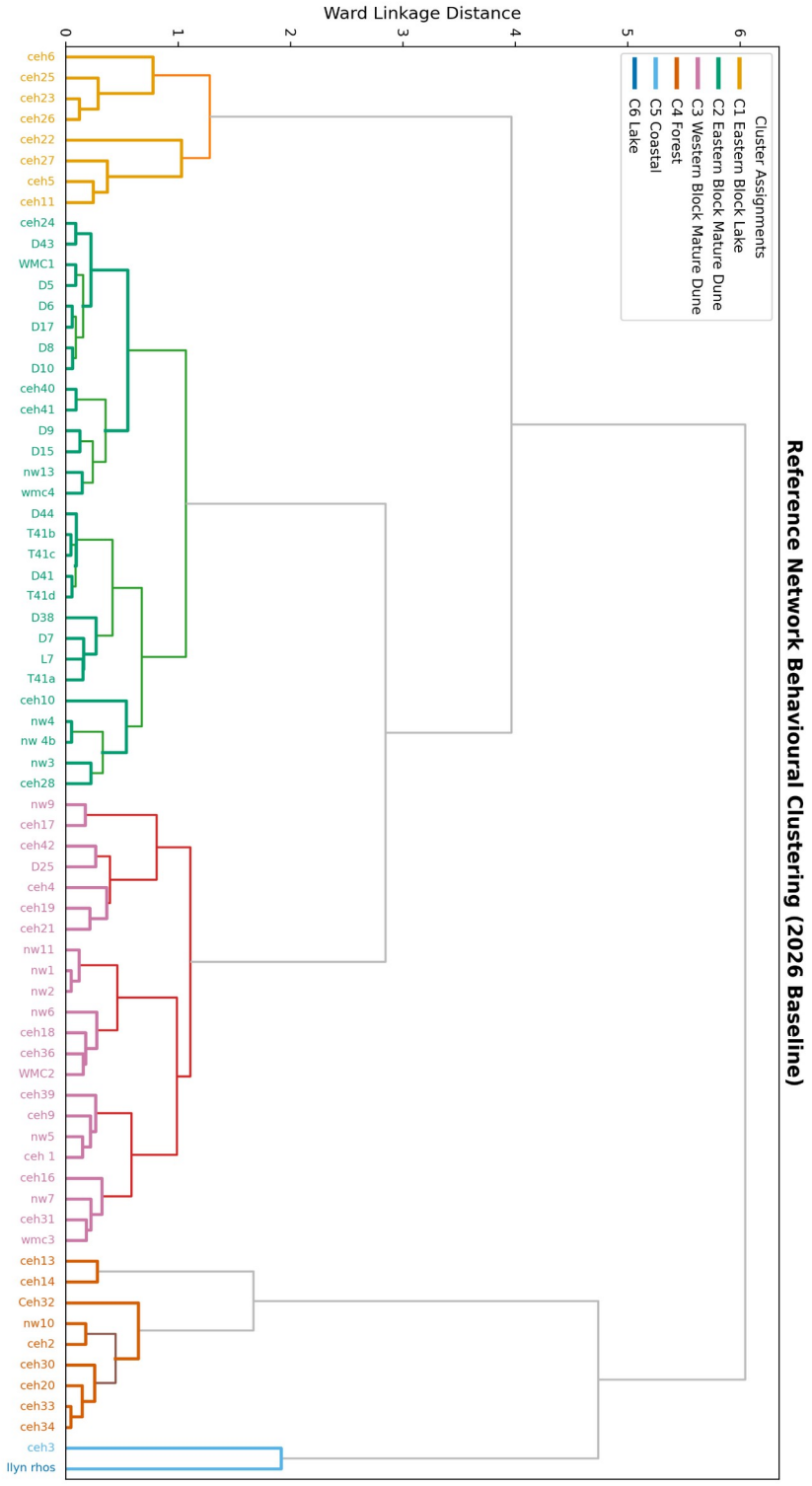


Figure 6. Clustering results: Ward's hierarchical clustering dendrogram for the 69-well reference network. Wells are labelled by identifier and coloured by assigned cluster. The primary binary split at a Ward linkage distance of approximately 6 separates the eastern till-influenced clusters (C1, C2) from the western sand, forest and specialist clusters (C3, C4, C5, C6). The Forest cluster (C4) separates from the adjacent Western Block (C3) at a lower linkage distance, confirming that plantation land cover imposes a distinct hydrogeological signature independent of the shared sandy substrate. (02_01_dendrogram.png)

4.2.1 Cluster Hydrographs

The six cluster average hydrographs (Figure 7, panel b) reveal a clear stratification of groundwater behaviour that reflects the hydrogeological architecture identified by the clustering analysis. C6 (Lake; Llyn Rhos-ddu) sits persistently just above zero throughout its record, confirming near-permanent surface water conditions at the lake margin and justifying its treatment as a fixed-head boundary in subsequent modelling. Among the open dune clusters, C1 (Eastern Block Lake-buffer) shows the shallowest mean depth and greatest seasonal responsiveness, consistent with its low storage capacity and hydraulic connectivity to Llyn Rhos-Ddu. C2 (Eastern Block Mature Dune) tracks C1 closely but sits systematically deeper, reflecting greater distance from the lake boundary. The fundamental geological contrast between the Eastern and Western Blocks is visible as a persistent vertical offset between C1/C2 and C3 (Western Block Mature Dune), with C3 running approximately 0.3–0.4 m deeper on average despite occupying a broadly similar topographic position — a contrast that reflects the deeper sand accumulation and higher storage capacity of the Western Block aquifer.

The C4 (Forest) is the most distinctive cluster, sitting 0.3–0.8 m deeper than adjacent C3 wells throughout the record and exhibiting the greatest seasonal amplitude of any cluster.This persistent depression relative to the surrounding dune aquifer is consistent with the canopy interception interpretation: the Corsican pine plantation intercepts a substantial fraction of gross precipitation before it reaches the water table, suppressing recharge and depressing the water table below that of the adjacent unforested dune. The C4 cluster mean hydrograph should be interpreted cautiously prior to 2010, when only three of the nine cluster wells were operational (CEH2, CEH13 and NW10). These three early wells are disproportionately located at higher elevations on the ridge margin and carry above-average boundary subsidies; the apparent high mean heads in 2006–2008 therefore reflect an unrepresentative spatial sample rather than a genuine network-wide hydrological signal. Full nine-well coverage was not achieved until mid-2010, and all analyses drawing on C4 cluster statistics are more reliable from that point onwards.C5 (Coastal) tracks broadly with C3 but with a more attenuated seasonal signal, reflecting tidal influence at the south-western margin.

The cumulative water balance panel (Figure 7, panel a) confirms that the cluster separation is not an artefact of differential climate forcing — the 12-month rolling mean of the corrected net balance remains close to zero throughout the record, indicating that all clusters are responding to the same climatic regime. The structural differences between clusters therefore reflect hydrogeological and land cover controls rather than spatially variable precipitation or evapotranspiration.

The network mean water table elevation was 7.35 m AOD, ranging from 2.77 m AOD at the coastal margin (CEH22, C1) to 13.31 m AOD at the ridge flank (CEH14, C4). The Forest cluster (C4) occupies the highest mean water table elevation in the network at 9.52 m AOD, compared with 7.85–7.86 m AOD in the Eastern Block clusters and 6.09 m AOD in the Western Block — a contrast that reflects the topographic position of the plantation on the northern ridge flank rather than any difference in recharge behaviour.

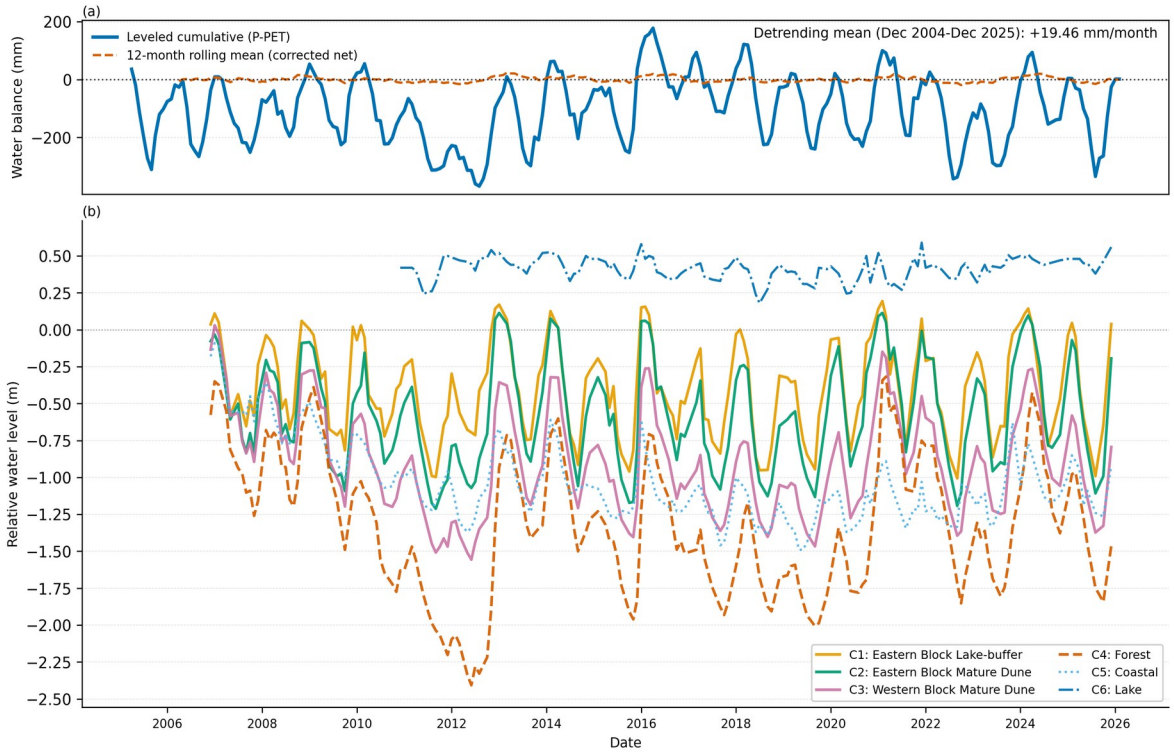


Figure 7. *Cluster-mean relative water level hydrographs for the six hydrogeological clusters, April 2005 to February 2026.* ***(a)*** *Levelled cumulative water balance (P − PET), detrended at the study-period mean of +19.46 mm/month (December 2004 to December 2025); the 12-month rolling mean of the corrected net balance (orange dashed) confirms no long-run directional trend in climate forcing over the monitoring period.* ***(b)*** *Monthly cluster-mean water level relative to pipe-top datum (m), where zero approximates ground surface, computed from all wells assigned to each cluster in the 69-well reference network. C6 (Lake) is available from December 2010 only. Cluster labels and colours follow those used in figure 6. (02_03_cluster_hydrographs_wb.png)*

4.2.2 Cluster Mechanistic Characterisation

The six clusters are characterised mechanistically in Table 2 using a state-space model fitted to the average hydrograph for each cluster (Section 3.4.1). β_1 is the effect of 1 mm of rainfall on the monthly change in water table depth, β_2 is the effect of 1 mm of PET on the monthly change in water table depth, and $-\beta_3$ is the proportional drainage decay rate — the fraction of antecedent water table head dissipated by lateral drainage each month. Table 2 reports mechanistic coefficients for all six clusters; C5 and C6 are excluded from substantive interpretation for the reasons noted below.

The recharge sensitivity coefficient β_1 is highest in the Eastern Block (C1: 1.56, C2: 1.67), indicating that each millimetre of rainfall drives a proportionally larger water table rise in the shallow-till aquifer than in the deep-sand Western Block (C3: 1.20) or Forest cluster (C4: 0.96). The atmospheric draw coefficient β_2, representing the evaporative demand transmitted to the water table per mm of PET, follows the same east-to-west gradient (C1: 4.98, C2: 4.38, C3: 3.59) but is lowest in the Forest cluster (C4: 3.41), which is counter-intuitive given the large canopy leaf area and is discussed further in Section 5.4.2. The drainage decay coefficient $-\beta_3$ decreases from east to west (C1: 0.366, C2: 0.200, C3: 0.118, C4: 0.078),

quantifying the rate at which head-dependent drainage dissipates stored water each month; the low Forest value reflects canopy interception smoothing the recharge signal into a near-linear climate response. The LCSC (= $100/\beta_1$) expresses these contrasts as a directly interpretable management metric: 60–64 mm of rainfall is required to raise Eastern Block water tables by 10 cm, compared with 83 mm in the Western Block and 105 mm beneath the plantation. C5 (coastal) and C6 (lake) are excluded from mechanistic interpretation — C5 because tidal exchange rather than rainfall-PET forcing governs water table dynamics, and C6 because near-fixed head conditions at Llyn Rhos-Ddu violate the drainage-decay assumptions of the model.

An extended form of the state-space model incorporating a depth-dependent evapotranspiration coefficient (Section 3.7.1) was evaluated at cluster level; as the results of this analysis bear primarily on the mechanistic interpretation of the Forest cluster β_2 coefficient, they are discussed in Section 5.4.2 rather than reported here.

Cluster	Label	n	β_1	β_2	$-\beta_3$	LCSC (%)	R^2
C1	Eastern Block Lake-buffer	8	1.565	4.978	0.366	63.9	0.7
C2	Eastern Block Mature Dune	28	1.674	4.379	0.200	59.7	0.6
C3	Western Block Mature Dune	22	1.201	3.592	0.118	83.3	0.5
C4	Forest	9	0.957	3.414	0.078	104.5	0.5
C5	Coastal	1	0.977	1.470	0.017†	102.7	0.3
C6	Lake	1	0.853	0.149	0.134‡	117.3*	0.3

Table 2. *Cluster mechanistic characterisation. The coefficients represent the effect of a 1 mm increase in monthly rainfall (β_1), a 1 mm increase in monthly PET (β_2), or a 1 m increase in antecedent water table depth ($-\beta_3$) on the monthly change in water table depth, respectively. LCSC = $100/\beta_1$; values express the millimetres of rainfall required to raise the cluster average water table by 10 cm. R^2 is one-step. *C6 LCSC > 100% reflects near-fixed head dynamics. †C5 $-\beta_3$ non-significant ($p = 0.40$); tidal exchange rather than gravity drainage governs recession. ‡C6 β_2 non-significant ($p = 0.21$); near-fixed head boundary condition precludes mechanistic inference. C5 and C6 are excluded from mechanistic interpretation. (03_03_cluster_mechanistic_coefficients.csv).*

4.2.3 Recharge, Loss and the Boundary Subsidy: A Cluster-Level Water Balance

Water balance decomposition applied to the four principal clusters (Section 3.7.2) reveals markedly different recharge efficiencies across clusters receiving identical climate forcing, while total losses remain broadly similar. The head-space decomposition is presented in Table 3a and Figure 8; an indicative volumetric conversion using assumed specific yield is presented in Table 3b and Figure 8a. The derivation and defence of the specific yield values is addressed in Section 4.2.4.

Recharge into the water table ($\beta_1 \cdot P$) declines systematically from the Eastern Block (C1: 0.112 m/month; C2: 0.120 m/month) through the Western Block (C3: 0.086 m/month) to the Forest (C4: 0.069 m/month). Total losses range from 0.304 m/month in C4 to 0.449 m/month in C1 and do not decline proportionally with recharge. The consequence is a

persistent monthly deficit in every cluster, ranging from 0.227 m/month in C3 to 0.337 m/month in C1. This deficit is independently confirmed by the boundary subsidy audit (Section 4.4.2; Figure 15a), in which positive model intercepts concentrated along the northern forest margin identify an unmeasured lateral flux, with the largest subsidy recorded at CEH14 (α = +0.222 m/month).

Cluster	LCSC (%)	Recharge (m/month)	Atm. Draw (m/month)	Drainage (m/month)	Total Loss (m/month)	Boundary Subsidy (m/month)
C1 Eastern Lake-buffer	63.9	0.11	0.28	0.17	0.45	0.34
C2 Eastern Mature Dune	59.7	0.12	0.24	0.12	0.36	0.24
C3 Western Mature Dune	83.3	0.09	0.2	0.12	0.31	0.23
C4 Forest	104.5	0.07	0.19	0.11	0.3	0.24

Table 3a.*Mean monthly head-space water balance decomposition by hydrogeological cluster, Newborough Warren 2005–2026. Recharge ($\beta_1 \cdot \bar{P}$), atmospheric draw ($\beta_2 \cdot PE\bar{T}$) and gravity drainage ($\beta_3 \cdot |\bar{h}|$) are expressed as metres of water table change per month (m/month) — the natural unit of the state-space model. All coefficients are cluster-median fits from the authoritative SSM (Table 2). The boundary subsidy represents the monthly deficit between local recharge and total losses that must be sustained by ridge-derived lateral recharge to maintain long-run water table equilibrium; it is confirmed independently by the positive model intercepts from the boundary subsidy audit (Section 4.4.2). For the Forest cluster (C4), canopy interception ($0.24 \cdot \bar{P}$; Freeman, 2008) is shown as a separate segment on the inputs bar of Figure 8, reducing the boundary subsidy rather than being additive: subsidy = total losses − recharge − interception. (Source: 16_water_bal_table.csv)*

An indicative volumetric expression of the head-space water balance (Table 3a) is presented in Table 3b and Figure 8a. Head-space coefficients were converted to annual fluxes (mm/year) using assumed specific yield: C1 = 0.08 (shallow sand over glacial till, silty matrix) and C2–C4 = 0.12 (deeper clean dune sand), consistent with the literature range for unconfined sandy aquifer materials (Fetter, 2001). The boundary subsidy expressed volumetrically ranges from 314 mm/year (C4) to 351 mm/year (C2), representing 35–39% of mean annual rainfall (890 mm). These values should be treated as conservative lower-bound estimates; empirical upper bounds derived from the water table fluctuation method are presented in Section 4.2.4.

Cluster	Sy	Recharge mm/yr	Atm. Draw mm/yr	Drainage mm/yr	Total Loss mm/yr	Subsidy mm/yr	Subsidy % rainfall
C1 Eastern Lake-buffer	8%	108	266	166	432	324	36%
C2 Eastern	12%	173	348	176	524	351	39%

Mature Dune							
C3 Western Mature Dune	12%	124	285	166	451	327	37%
C4 Forest	12%	99	274	164	438	314	35%

Table 3b. *Indicative annual volumetric water balance by cluster, 2005–2026. Head-space SSM coefficients (Table 3a) converted to mm/year using assumed specific yield: C1 = 8% (shallow sand over glacial till, silty); C2–C4 = 12% (deeper clean dune sand). Formula: flux (mm/yr) = monthly head-space term (m/month) × 12 × Sy × 1000. Values generated by pipeline script 16_water_bal.py and tabulated in 16_water_bal_vol_table.csv. Subsidy = ridge-derived lateral recharge required to close the annual deficit; independently corroborated by CEH14 boundary intercept α = +0.222 m/month ≈ 320 mm/yr at Sy = 0.12. Subsidy % rainfall = subsidy as a percentage of mean annual rainfall (890 mm). Sy values are assumed — direct field measurement by slug test or water table fluctuation method at 2–3 representative wells per cluster is a recommended priority for future work.*

4.2.4 Specific Yield Estimation

WTF event-based median Sy estimates for the four principal clusters, derived from 38–49 qualifying monthly rising-limb events per cluster, are presented in Table 3c and Figure 8b. Uncorrected median values range from 0.215 (C4 Forest) to 0.259 (C3 Western Mature Dune), with broad interquartile ranges reflecting the inherent noise in monthly cluster-mean data. The Forest cluster returns the lowest uncorrected median, consistent with the expectation that canopy interception suppresses the net recharge signal reaching the water table, thereby reducing the observed Δh relative to open dune clusters receiving the same gross rainfall. After applying the interception correction of Freeman (2008), the Forest cluster median rises to 0.227, within the range of the open dune clusters (0.223–0.259). This convergence confirms that the aquifer has broadly uniform storage properties across the site irrespective of surface land cover, and that the attenuated water table response beneath the plantation reflects the surface boundary condition rather than any difference in aquifer storage capacity.

All four WTF medians exceed the assumed Sy values used in Table 3b (C1: 0.08; C2–C4: 0.12), with assumed values falling below the lower quartile of the WTF distributions in every cluster. The implied volumetric subsidies under WTF Sy (702–785 mm/year, equivalent to 79–88% of mean annual rainfall; Table 3d) are implausibly large, consistent with the known tendency of monthly WTF to overestimate effective Sy through conflation of true gravity drainage with capillary fringe release (Healy and Cook, 2002; Scanlon et al., 2002). The assumed values are therefore retained as conservative lower bounds for the primary volumetric water balance (Table 3b), with the WTF medians and their volumetric equivalents (Table 3d; Figure 8b) providing an independently derived upper constraint. The CEH14 boundary intercept (α = +0.222 m/month ≈ 320 mm/year at Sy = 0.12) corroborates the assumed-value subsidy estimates and is inconsistent with the WTF upper bounds, providing a further basis for preferring the conservative assumption.

Cluster	n events	Sy assumed	Sy WTF median	Q25	Q75	Interception corrected?
C1 Eastern Lake-buffer	49	0.08	0.223	0.141	0.313	No
C2 Eastern Mature Dune	48	0.12	0.234	0.151	0.348	No

C3 Western Mature Dune	40	0.12	0.259	0.136	0.344	No
C4 Forest (uncorrected)	38	0.12	0.215	0.155	0.329	No
C4 Forest (corrected)	44	0.12	0.227	0.132	0.344	Yes — Freeman (2008)

Table 3c. *Specific yield (Sy) estimates by hydrogeological cluster derived from the water table fluctuation (WTF) method (Healy and Cook, 2002). Event-based median Sy computed from monthly rising-limb events satisfying Δh > 5 mm and net recharge P − PET > 10 mm over the monitoring period April 2005 to February 2026. For the Forest cluster (C4), two estimates are reported: uncorrected (using gross rainfall) and corrected (net recharge adjusted for 24% canopy interception following Freeman, 2008). The corrected C4 median (0.227) falls within the range of the open dune clusters (0.223–0.259), confirming broadly uniform aquifer storage properties across the site. IQR = interquartile range; wide IQR reflects noise inherent in monthly cluster-mean data rather than genuine spatial heterogeneity in aquifer storage. Assumed Sy values (used in Table 3b) are shown for comparison and fall below the lower quartile of the WTF distributions in all clusters, confirming they represent conservative lower bounds. Monthly WTF estimates are known to overestimate effective Sy through conflation of gravity drainage with capillary fringe release and should be interpreted as upper bounds rather than point estimates of long-term effective storage (Healy and Cook, 2002; Scanlon et al., 2002). Direct field measurement by slug test or pumping test at two to three representative wells per cluster is recommended as a priority for future work. Generated by 17_wtf_specific_yield.py (output: 17_wtf_01_sy_estimates.csv).*

Cluster	Sy WTF	Recharge	Atm. Draw	Drainage	Total Loss	Subsidy	Sub % rainfall
C1 Eastern Lake-buffer	0.223	396	747	435	1182	785	88%
C2 Eastern Mature Dune	0.234	380	715	366	1082	702	79%
C3 Western Mature Dune	0.259	289	625	410	1035	746	84%
C4 Forest	0.227	173	539	383	922	702	79%

Table 3d. *Annual volumetric water balance by hydrogeological cluster using WTF-derived specific yield, Newborough Warren 2005–2026. Head-space SSM coefficients (Table 3a) converted to mm/year using WTF event-based median Sy values (Table 3c): C1 = 0.223; C2 = 0.234; C3 = 0.259; C4 = 0.227. The C4 value is interception-corrected following Freeman (2008). These values represent empirical upper bounds on effective Sy; boundary subsidies of 702–785 mm/year (79–88% of mean annual rainfall) are implausibly large for a physically closed water balance, consistent with the known tendency of monthly WTF to overestimate effective Sy through conflation of gravity drainage with capillary fringe release (Healy and Cook, 2002; Scanlon et al., 2002). Table 3b (assumed Sy: C1 = 8%; C2–C4 = 12%) provides the preferred conservative lower-bound estimates and should be treated as the primary volumetric reference. Table 3d is presented for completeness to bracket the plausible range of volumetric fluxes. Generated by 16_water_bal.py (output: 16_water_bal_vol_wtf_table3d.csv).*

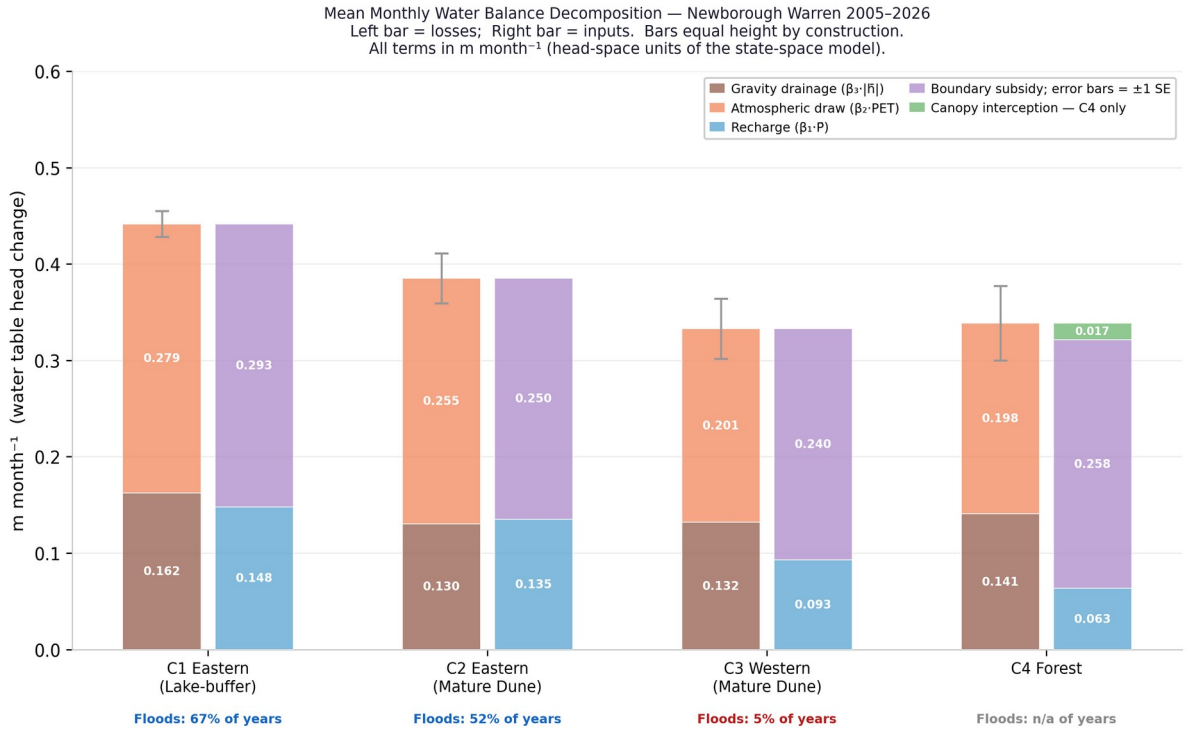


Figure 8*. Mean monthly head-space water balance decomposition by hydrogeological cluster, Newborough Warren 2005\u20132026. For each cluster, the left bar shows losses decomposed into gravity drainage and atmospheric draw; the right bar shows inputs decomposed into local recharge, ridge-derived lateral recharge subsidy, and canopy interception (C4 only; Freeman, 2008). The interception term reduces the subsidy rather than being additive: subsidy = total losses minus recharge minus interception. Left and right bars are equal height by construction. All terms are head-space quantities in m/month. Error bars show plus or minus 1 SE of propagated subsidy uncertainty. Flooding frequency shown below each cluster label. Generated by 16_water_bal.py.*

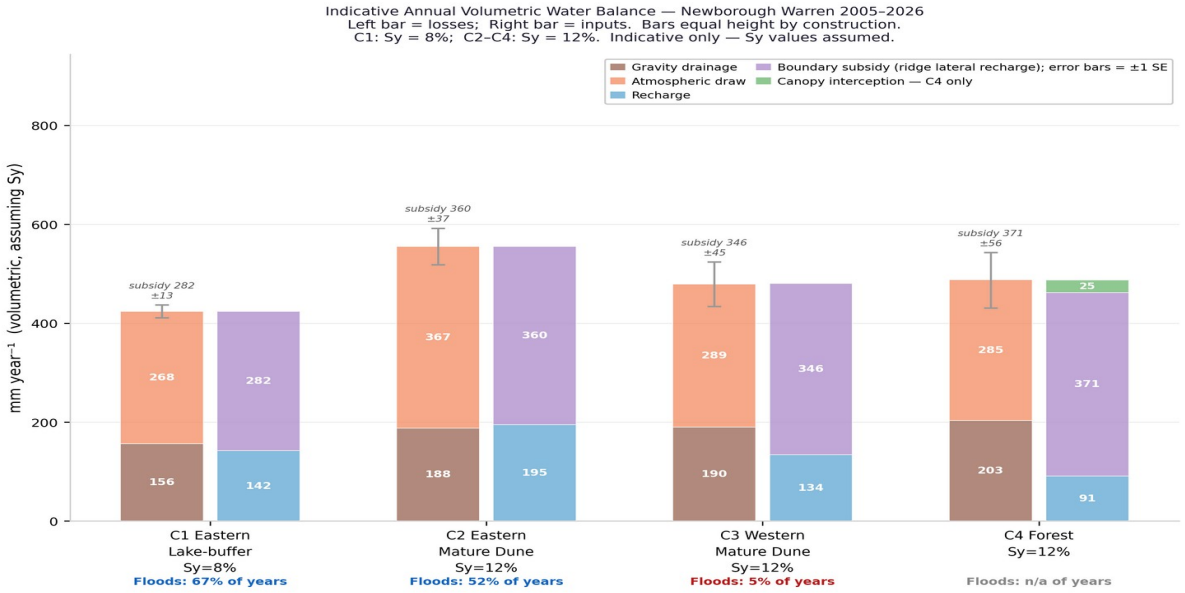


***Figure 8a**. Indicative annual volumetric water balance by hydrogeological cluster, Newborough Warren 2005–2026. Head-space SSM coefficients (Table 3a) converted to mm/year using assumed specific yield: C1 = 8%; C2–C4 = 12%. Left bar = losses (gravity drainage + atmospheric draw); right bar = inputs (recharge + ridge lateral recharge subsidy + canopy interception for C4 only). Bars equal height by construction. Error bars on losses bar = ±1 SE of propagated subsidy uncertainty scaled by Sy. Subsidy independently corroborated by CEH14 boundary intercept ≈ 320 mm/yr at Sy = 0.12. Indicative only - Sy values assumed; see Table 3c and Figure 8b for WTF upper bounds. (Generated by 16_water_bal.py)*

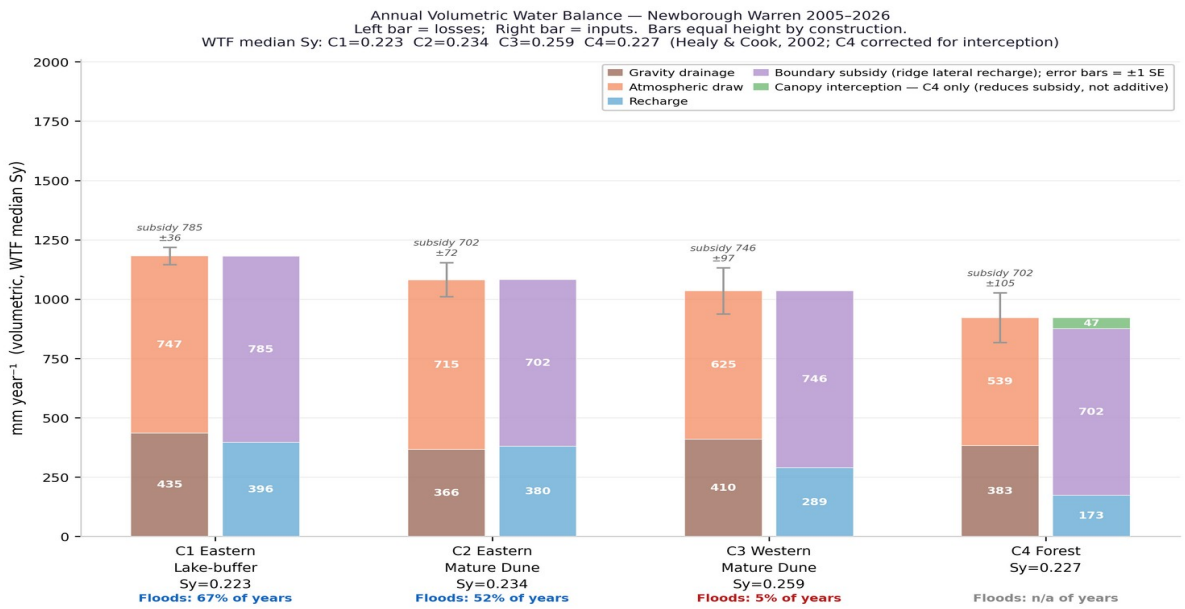


Figure 8b. Annual volumetric water balance by hydrogeological cluster, Newborough Warren 2005–2026 (WTF empirical upper bound). Head-space SSM coefficients converted to mm/year using WTF median Sy values (Table 3c): C1 = 0.223; C2 = 0.234; C3 = 0.259; C4 = 0.227. Same layout as Figure 8a. Boundary subsidies of 702–785 mm/year (79–88% of mean annual rainfall) are implausibly large, reflecting the known tendency of monthly WTF to overestimate effective Sy through capillary fringe release. This figure and Table 3d are presented as an upper bound to bracket the plausible volumetric flux range alongside the conservative estimates of Figure 8a and Table 3b. Generated by 16_water_bal.py (output: 16_water_bal_volumetric_wtf_ms.png).

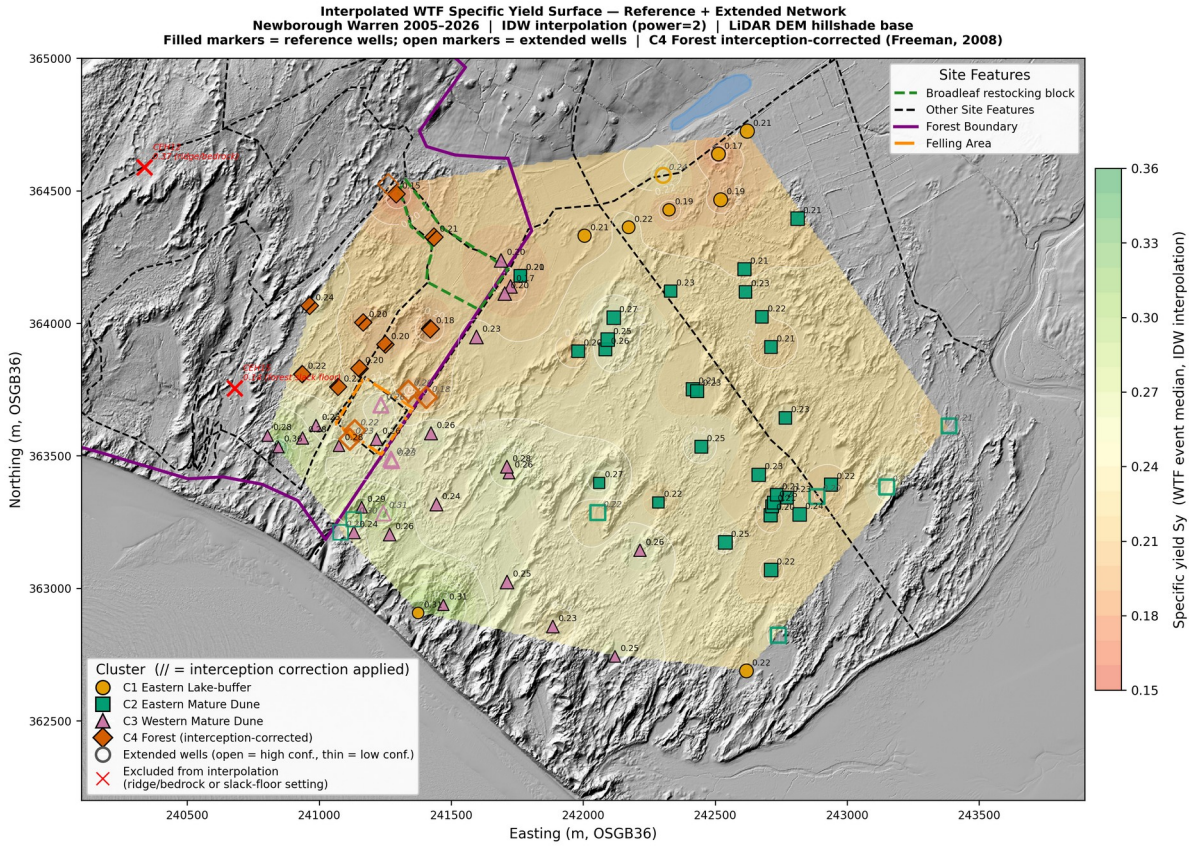


Figure 9. *Interpolated WTF specific yield surface, Newborough Warren 2005–2026, reference and extended monitoring networks combined. IDW interpolation (power=2) of event-based median Sy per well, masked to the convex hull of the combined network. OpenStreetMap basemap with KML site feature overlays (forest boundary, clearfell area, drainage network). Filled markers = reference wells (n=67); open markers = extended network wells (n=18, excluding two flagged wells). Two wells are excluded from interpolation and shown as red crosses: CEH12, located on the bedrock ridge in a forested area, where the WTF response reflects fractured rock rather than dune sand storage; and CEH15, a Forest cluster well in a low-lying slack within the plantation, where water table dynamics are dominated by slack topography and the interception correction is unreliable given the partially open canopy setting. Extended well cluster assignments derived from Pearson affinity sitewide audit (Section 4.3; Best_Match_Cluster). Forest cluster (C4) wells interception-corrected following Freeman (2008). The broadly uniform surfaceconfirms that inclusion of the extended network does not alter the principal finding of homogeneous aquifer storage properties across the site. Generated by 18_wtf_spatial.py (output: 18_wtf_04_sy_contour_extended.png), .LiDAR DEM hillshade base with KML site feature overlays (forest boundary, clearfell area).Individual well Sy estimates for all 86 wells are provided in Table S4.*

Individual well WTF Sy estimates (Table S4) were derived from 7–55 qualifying rising-limb events per well (mean 34 across both networks; reference wells: 22–55, mean 39; extended wells: 7–28, mean 18). Cluster-mean values from the combined network analysis — C1: 0.220, C2: 0.229, C3: 0.259, C4: 0.204 — are consistent with the reference-only estimates (C1: 0.217, C2: 0.227, C3: 0.257, C4: 0.202), confirming that inclusion of the extended network does not alter the principal finding of homogeneous aquifer storage properties across the site.. The spatial distribution of Sy, shown in Figure 9 across the combined reference and extended network, is broadly uniform across the site, with the highest values concentrated in the south-western Western Block (C3), consistent with the deepest clean sand accumulation at that location. The Forest cluster (C4) returns the lowest individual well

values after interception correction (cluster mean 0.202), modestly below the open dune clusters; residual spatial variability in canopy closure means these estimates carry greater uncertainty than those for open dune wells. The spatial coherence of the Sy surface — with no abrupt discontinuities coinciding with the cluster margins — provides further confirmation that the cluster behavioural differences reflect surface hydrology and land cover rather than heterogeneity in aquifer storage architecture. Extended well cluster assignments are derived from the Pearson affinity sitewide audit described in Section 4.3; the spatial pattern of Sy is unchanged by their inclusion.

4.3. Checking the Clusters Pearson Affinity Analysis

All 69 wells within the continuous reference network confirmed their primary cluster membership through the Pearson Affinity Analysis (Section 3.3), with zero instances of structural divergence. This stability is characterised by three distinct membership tiers based on the correlation delta (Δr) between assigned and neighbouring cluster centroids:

- Core Members (23 wells): Wells with an unambiguous dominant affinity to their assigned cluster ($\Delta r > 0.05$), representing stable hydrogeological anchors within their designated geomorphological zone.
- Fuzzy Members (46 wells): Wells exhibiting closely competing affinities between their primary and secondary clusters ($\Delta r < 0.05$), identifying physical ecotones where the groundwater regime transitions smoothly between adjacent structural blocks.
- No Spy classifications were identified within the reference network, confirming that the Ward's linkage assignments are spatially coherent and that no well's temporal dynamics are better explained by a non-adjacent cluster.

Classification confidence was further characterised by the Multi-Cluster Affinity flag, raised where three or more cluster centroids achieved a correlation of $r > 0.90$. This criterion was met at 27 wells. Of these, 21 are concentrated within the gradational C1/C2/C3 transitional zone of the mid-eastern dune sector, and 6 lie along the C2/C3/C4 forest–dune boundary. The prevalence of MCA flags confirms the continuous, highly transmissive nature of the sand matrix, where boundary wells respond to system-wide climatic forcing rather than localised topographic drivers, and identifies the precise locations of greatest structural ambiguity within the network (Figure 11a).

The extended network analysis (Section 3.3.3) classified all 20 extended wells against the reference centroids, yielding 3 Ext_Core and 17 Ext_Fuzzy assignments (Figure 11b). The high proportion of Fuzzy classifications reflects the shorter and earlier records of these wells rather than genuine hydrogeological ambiguity — with less temporal overlap against the centroid time series, affinity margins are naturally compressed.

Within the extended network, FE1, FE2, FE4 and LIS1 all returned strongest affinity with the C4 Forest centroid ($r = 0.990$, 0.972, 0.974 and 0.977 respectively), confirming that the characteristic plantation groundwater signature persists across these wells despite the December 2017 clearfell. FE3 showed strongest affinity with C3 ($r = 0.985$) rather than C4 ($r = 0.936$), consistent with its position at the plantation boundary where the western dune

signal competes with the forest signal. WMC3, which is in the reference network, similarly showed strongest affinity with C3 (r = 0.973) with C4 a close second (r = 0.949) (Figure 10).

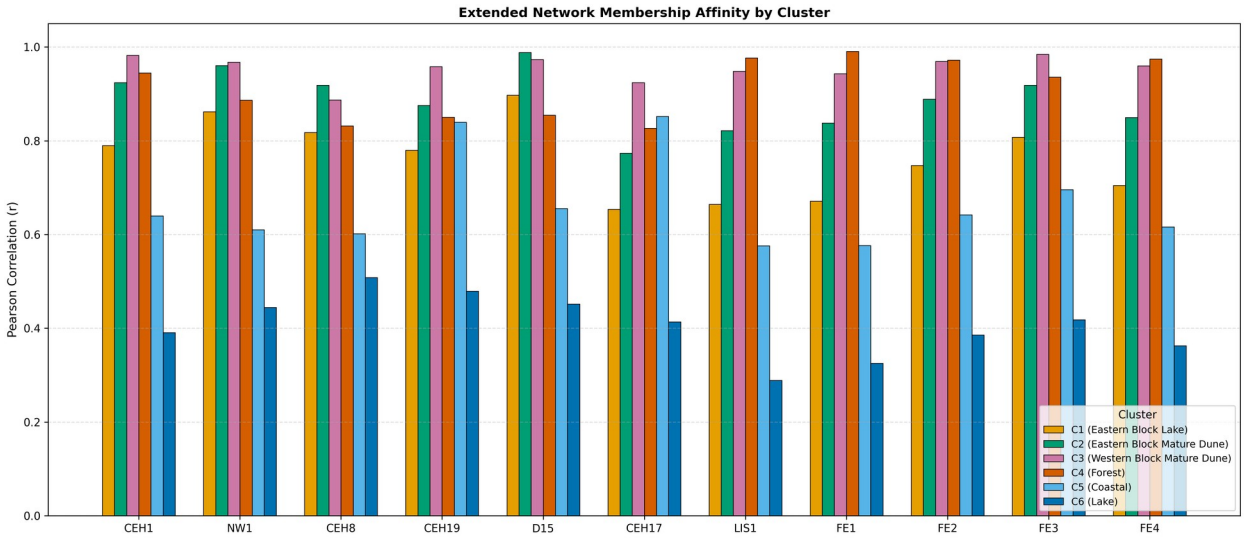


Figure 10. *Pearson affinity analysis of selected reference network wells, showing the correlation coefficient (r) between each well's z-scored time series and each of the six cluster centroids. Wells are selected to illustrate the range of affinity confidence from strong single-cluster dominance (Core members) to near-equal competing affinities (Fuzzy members), along with wells in or close to the clearfell experiment. (06_pear_01_affinity_chart_extended.png)*

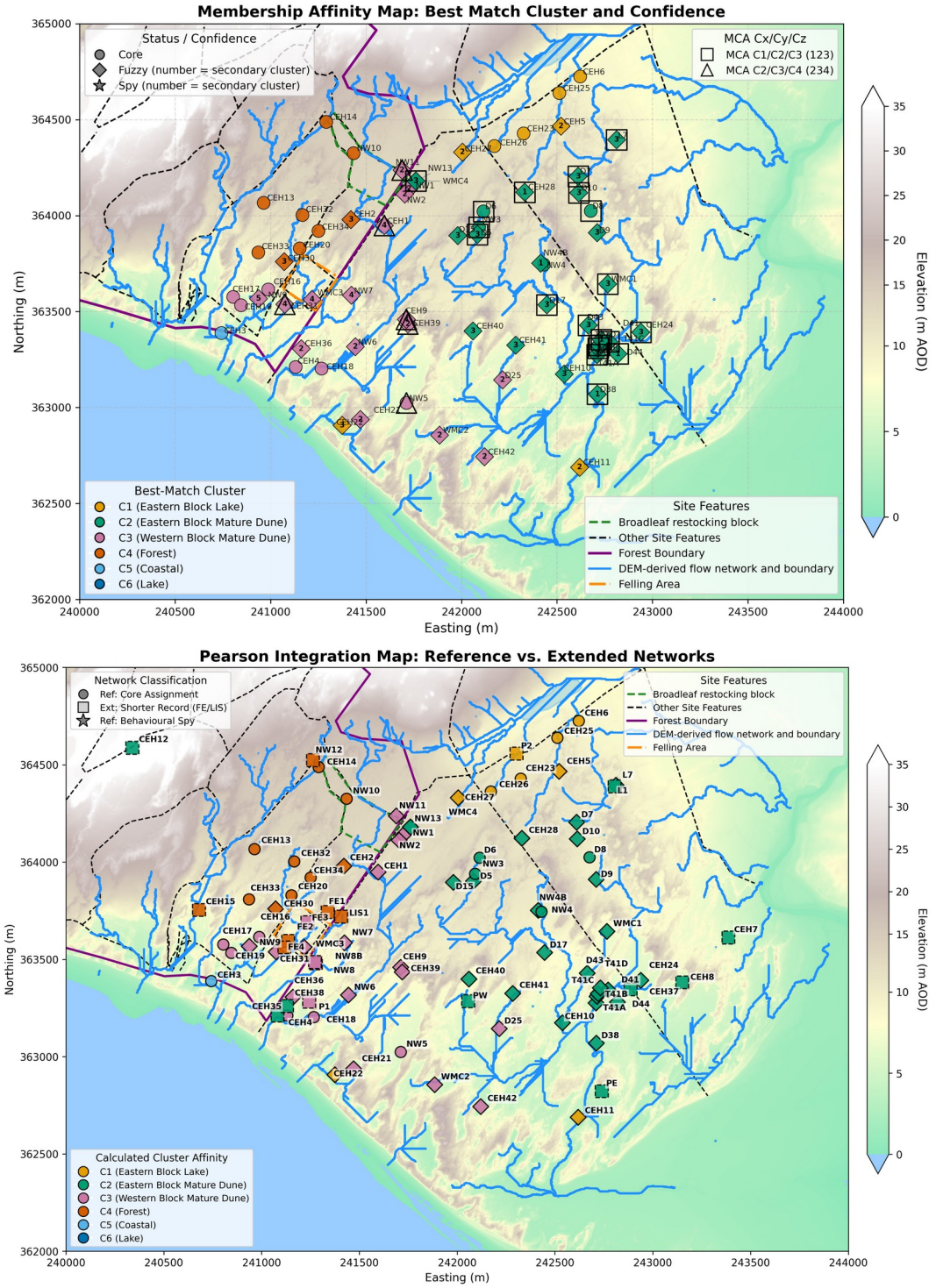


Figure 11. *Spatial distribution of the six primary hydrogeological clusters validated via Pearson affinity analysis. Points are coloured by best-match cluster following z-score standardization. Symbology denotes classification confidence: circles represent Core members (Δr > 0.05) and diamonds represent Fuzzy members (Δr < 0.05). Panel (a): Reference network wells with records of at least 100 months extending to February 2026. Large hollow overlays identify wells with Multi-Cluster Affinity (MCA), exhibiting high convergence (r > 0.90) with three or more cluster average hydrographs. (05_pear_01_spatial_confidence_map.png Panel (b): Pearson affinity classifications for all 89 wells across the combined reference and extended networks. (06_pear_02_integration_map.png) (LiDAR data: © NRW & OS; see Section 10 for full attribution).*

4.4. Model Benchmarking

4.4.1 State-Space Model Comparison with Traditional Linear Models

The SSM was compared to a TLM as outlined in Section 3.4.2. The physical distinction between the two models is most clearly illustrated at forest well CEH19. In one-step diagnostic mode — where each prediction is corrected with the observed water level from the previous month — both models track the observed record closely, achieving comparable R^2 values of 0.828 (TLM) and 0.831 (SSM). The difference emerges when the models are run autonomously over 100 months without observational correction. The TLM, lacking any mechanism to represent head-dependent drainage, drifts progressively from the observed record and achieves an iterative NSE of −0.113 — slightly worse than simply predicting the long-term mean at every step. The SSM, by contrast, maintains physical consistency through its storage decay term and achieves an iterative NSE of 0.445 (Figure 12).

Across the full 69-well reference network, the TLM achieves a median iterative NSE of −0.01, effectively equivalent to a mean-value forecast, while the SSM achieves +0.62 — a median gain of +0.59 NSE units. The SSM returned a positive iterative NSE at 68 of 69 wells compared with 33 of 69 for the TLM (Table 4). In one-step diagnostic mode both models perform almost identically (median R^2 0.81 and 0.82 respectively; Table 4), confirming that the TLM can replicate historical patterns adequately when updated with real observations but lacks the physical memory to sustain accurate autonomous forecasts.

The largest single improvement was recorded at NW3 (TLM NSE = −1.66, SSM NSE = +0.62, improvement = +2.28; Table 4). NW3 occupies an elevated position within the dune slack system, producing a large seasonal head variation and strongly head-dependent drainage dynamics; D6, on a lower slack surface within the same system, shows a substantially smaller improvement (+0.936), consistent with its more subdued drainage response. This intra-slack contrast illustrates that the magnitude of SSM gain reflects local topographic position relative to the water table, not simply cluster membership.

The SSM improvement as measured by NSE varies markedly across the six hydrogeological clusters (Figure 13a, 13b). The largest gains occur in C1 (Eastern Block Lake-buffer, median improvement +0.749) and C2 (Eastern Block Mature Dune, +0.730), where strong head-dependent drainage makes the storage decay term essential for autonomous simulation. C3 (Western Block, +0.565) shows a substantial but smaller improvement reflecting its deeper sand buffer. The Forest cluster (C4) is the clear exception: the TLM already achieves a median iterative NSE of +0.281 with all 9 wells returning positive iterative NSE, and the SSM improvement is only +0.166. This reflects C4's characteristically small $-\beta_3$ (0.078 — the lowest in the network), a consequence of canopy interception smoothing the recharge signal into a near-linear climate-to-water-table relationship that the TLM handles adequately. The SSM's storage decay term becomes indispensable precisely where drainage is strongly head-dependent — and the Forest cluster's attenuated dynamics do not meet that threshold. The sole well where the SSM fails entirely is CEH3 (C5 coastal, iterative NSE = −8.48), confirming that continuous tidal exchange violates the drainage-decay assumptions of both models at this boundary.

Spatial visualisation of the improvements (Figure 13a, 13b) confirms this pattern geographically, with the largest R^2 and NSE gains concentrated along the north-eastern forest–Warren boundary and the C1 lake-buffer sector, smallest in the Forest cluster, and

absent at the coastal margin — physically delineating the subsurface structural boundaries of the aquifer without invasive coring.

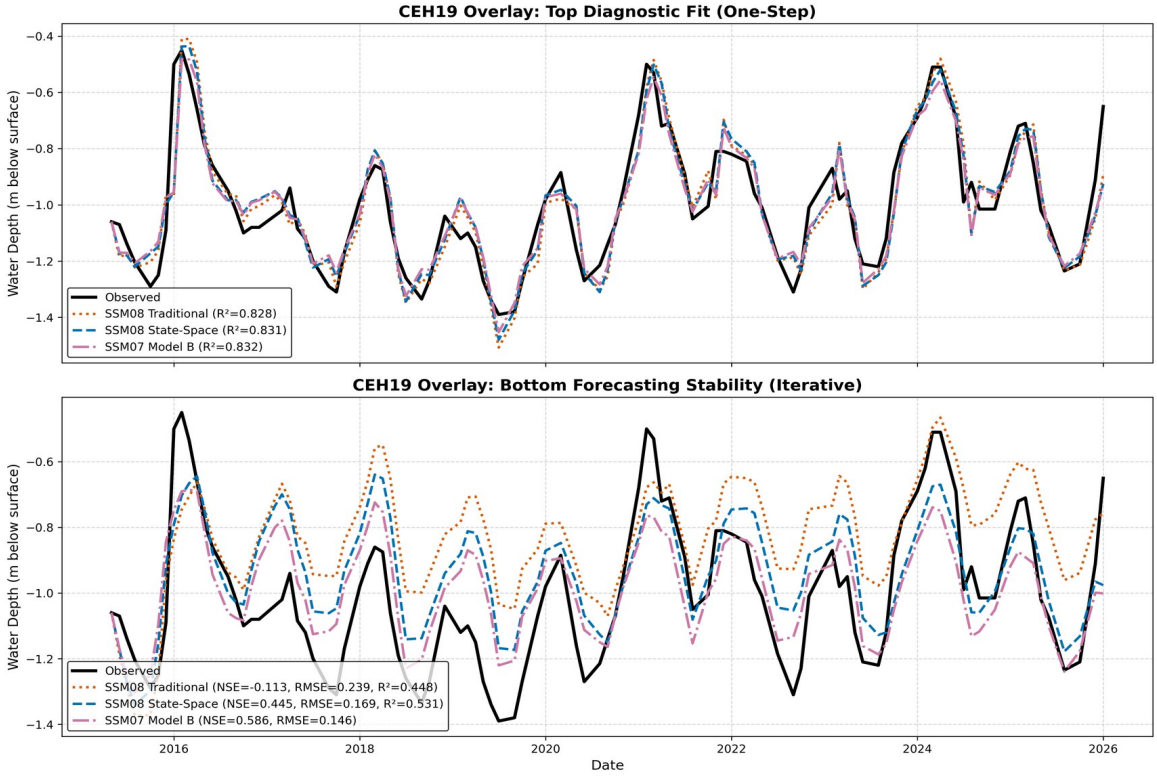


***Figure 12**. Dual-model performance comparison at forest well CEH19. Upper panel: one-step diagnostic fit — both TLM (R^2 = 0.828) and SSM (R^2 = 0.831) track observed water levels closely when corrected with real observations at each step. Lower panel: iterative 100-month autonomous simulation without observational correction — the TLM (NSE = −0.113) drifts progressively from the observed record as its constant drainage assumption accumulates error, while the SSM (NSE = 0.445) maintains physical consistency through its storage decay term. When an intercept term is additionally fitted as in Model B (Section 4.4.2), iterative NSE further improves to 0.565. (08_lcsc_01_ceh19_with_model_b.png)*

Metric	TLM	SSM	Improvement Δ (SSM − TLM)
Median one-step R^2	0.81	0.82	0.01
Median iterative R^2	0.44	0.62	0.16
Median iterative NSE	-0.01	0.62	0.59
Wells with iterative NSE > 0	33 / 69	68 / 69	—
Max NSE improvement (NW3)	-1.66	0.62	2.28
CEH19 iterative NSE	-0.11	0.45	0.56

***Table 4.** Model benchmarking summary comparing the Traditional Linear Model (TLM) and State-Space Model (SSM) across the 69-well reference network. Performance evaluated over the most recent 100 months at each well. One-step R^2 uses observed h(t−1) at each step; iterative NSE uses simulated h(t−1) over the full 100-month evaluation window without observational correction. NW3 is the well recording the largest single NSE improvement; CEH19 is the illustrative forest well shown in Figure 10. (08_lcsc_04_table3_benchmark_summary.csv)*

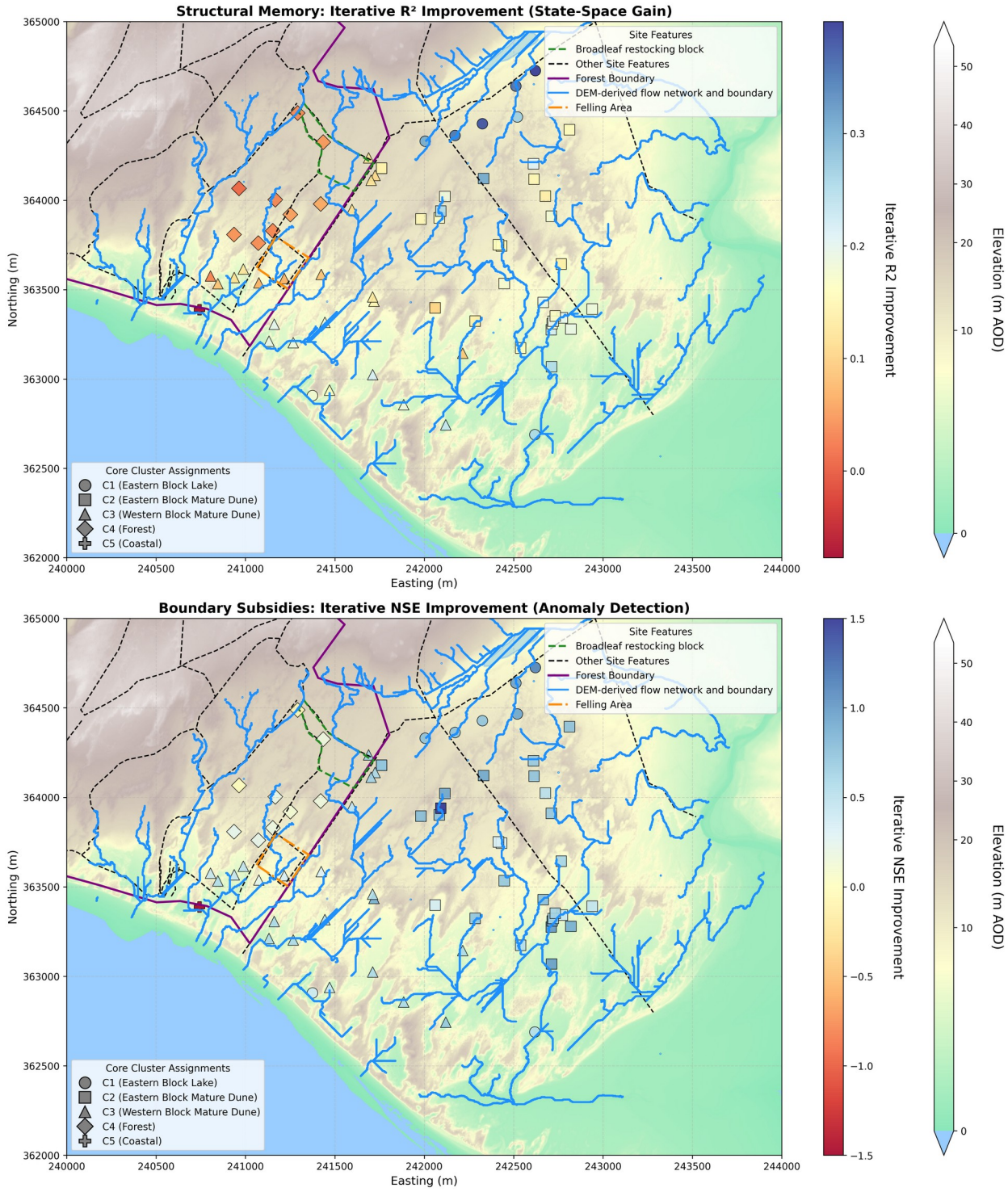


Figure 13. *SSM gain over TLM across all 69 reference wells. Panel (a): iterative R² improvement — largest gains in the Eastern Block and lake-buffer sector, smallest in the Forest cluster where the TLM already performs adequately. (08_lcsc_02_r2_improvement_map.png) Panel (b): NSE improvement — corresponding spatial pattern with gains frequently exceeding 1.0 NSE unit in the north-eastern and central eastern dune zones. The spatial distribution of improvement physically delineates subsurface structural boundaries without invasive coring. (08_lcsc_03_nse_improvement_map.png) LiDAR data: © NRW & OS; see Section 9 for full attribution.*

4.4.2 Boundary Subsidies and Deficits Mapping , using State Space Models with and without an Intercept

The SSM with and without intercept models were built as outlined in Section 3.4.3. State-Space Model B intercepts (α) were predominantly positive, identifying localised boundary fluxes concentrated along the northern forest–dune boundary and the eastern mid-dune sector (Figure 15a). The largest subsidy in the network (α = +0.222 m/month) was recorded at CEH14, a structural transition well situated at the northern forest margin. Figure 14 illustrates the behaviour at CEH14 directly: in one-step diagnostic mode both Model A and Model B track the observed record closely (R^2 = 0.867 and 0.873 respectively), but in iterative simulation Model B (NSE = 0.311) maintains a systematically higher water table trajectory than Model A (NSE = 0.237), reflecting the persistent positive boundary flux that Model A cannot represent without an intercept term. The divergence between the two models in the lower panel quantifies the boundary subsidy effect — the gap between the two trajectories represents the water table elevation that ridge-derived recharge sustains above what local rainfall and drainage alone would produce.

Positive intercepts also characterised the C4 Forest cluster wells CEH34 (α = +0.142), CEH13 (α = +0.112) and CEH2 (α = +0.112), consistent with their position in the principal lateral recharge pathway descending from the northern rock ridge. The largest negative intercept was recorded at CEH3 (α = −0.377), consistent with its position at the tidal boundary where continuous seaward drainage dominates the water balance.

The NSE intercept effect map (Figure 15b) shows that the boundary term meaningfully improves iterative forecasting performance at a small but physically interpretable subset of wells. The most extreme improvement was recorded at CEH3 (+8.93 NSE units), where Model A performs extremely poorly because continuous tidal exchange completely violates the mass-balance assumption; the intercept partially compensates but neither model is adequate at this boundary well and CEH3 should be regarded as outside the operational domain of the rainfall-PET-drainage framework. Among the remaining wells, the largest improvements were at CEH17 (+0.266), CEH19 (+0.141), CEH22 (+0.133) and CEH14 (+0.074) — tidal, coastal-margin and ridge-boundary wells where a persistent boundary flux is physically real. In total, 23 wells show a beneficial effect (NSE improvement > 0.01), 36 are effectively neutral (|effect| ≤ 0.01), and 9 show a marginally detrimental effect, of which only 2 exceed an effect size of 0.02 — negligible in practical terms. The mass-balanced Model A is nonetheless retained for all site-wide simulation and forecasting on the grounds of physical parsimony and to avoid introducing unjustified constant flux assumptions where the boundary signal is absent or marginal. However, the spatially structured pattern of positive intercept effects along the northern forest margin identifies precisely the locations where a more complex model incorporating explicit boundary flux terms would be most warranted in future groundwater modelling work.

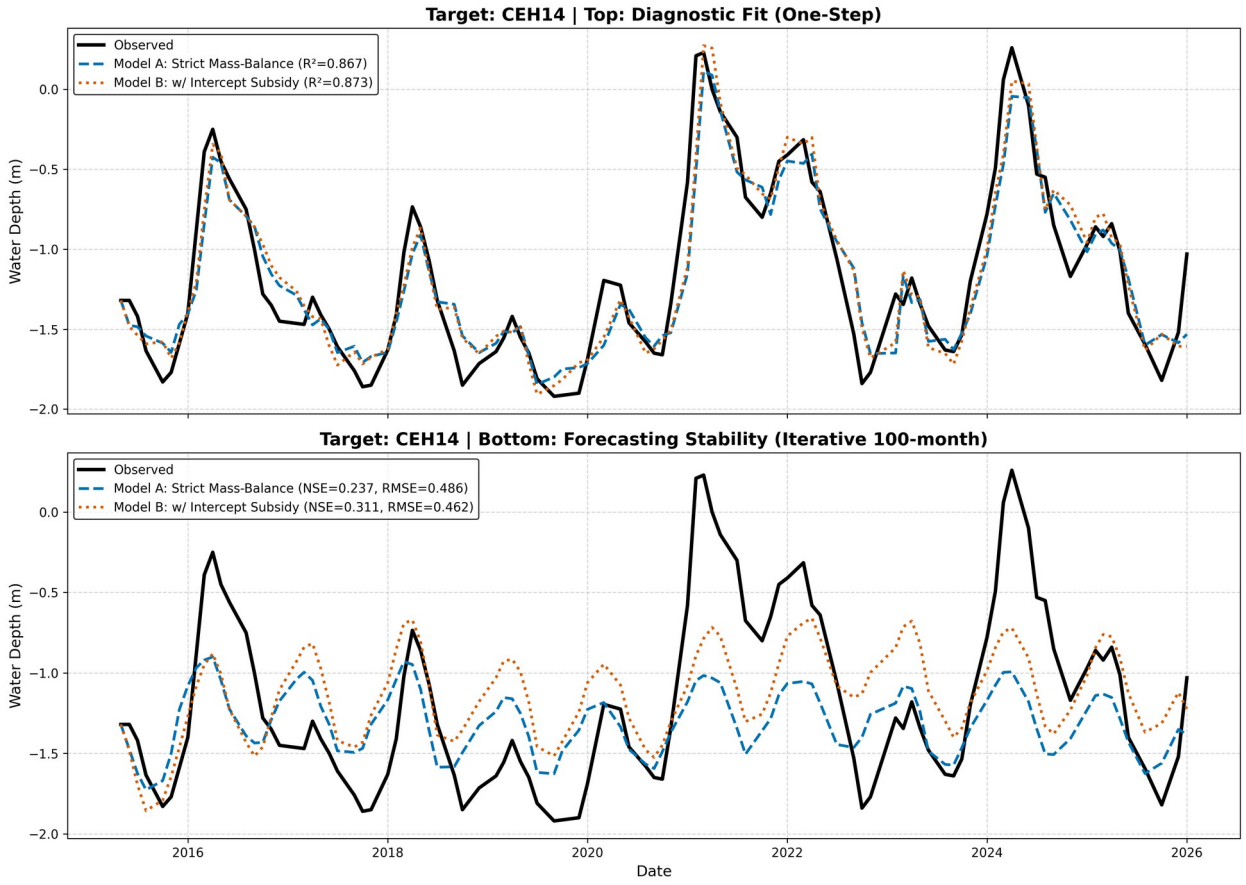


***Figure 14**. Dual-model performance comparison at boundary well CEH14. Upper panel: one-step diagnostic fit showing Model A (strict mass balance, R² = 0.867) and Model B (fitted intercept, R² = 0.873). Lower panel: iterative 100-month forecasting stability showing Model A (NSE = 0.237, RMSE = 0.486) and Model B (NSE = 0.311, RMSE = 0.462). CEH14 carries the largest boundary subsidy in the network (α = +0.222 m/month); Model B provides a marginal improvement in both diagnostic and iterative performance at this well, consistent with a genuine steady-state recharge input from the adjacent rock ridge.*
(07_intercept_01_ceh14_showdown.png)

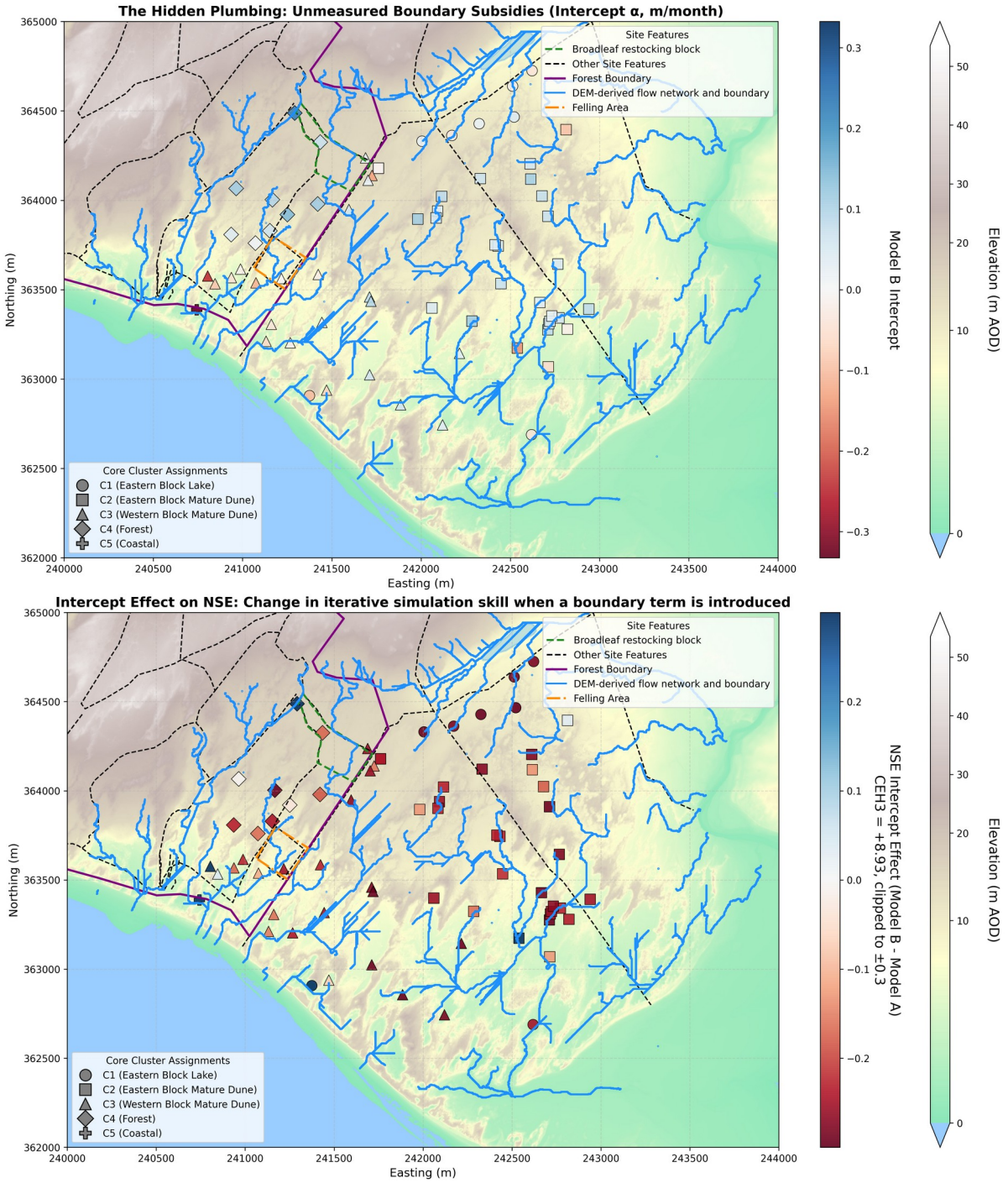


Figure 15. *Spatial evaluation of boundary intercepts (α) and iterative simulation stability. Panel (a): Model B intercepts identifying zones of positive (boundary subsidy) and negative (localised sink) boundary flux. (07_intercept_02_plumbing_map.png) Panel (b): Intercept effect on iterative NSE (NSE_B − NSE_A), showing the change in iterative simulation skill when a constant boundary term is introduced (07_intercept_03_nse_penalty_map.png). Positive values indicate that Model B improves iterative performance relative to the mass-balanced Model A; negative values indicate marginal degradation. The large positive effect at CEH3 reflects fixed-head tidal control and places this well outside the operational domain of the rainfall-PET-drainage framework; the broadly near-zero pattern across the main dune body confirms that Model A is appropriate for the majority of the network. (LiDAR data: © NRW & OS; see Section 9 for full attribution.*

4.5. Dune Scraping Intervention Analysis

Tier 1 CUSUM analysis at control wells CEH4 and CEH22 identified an accelerating background deterioration trend across the monitoring period, independent of management actions (Section 3.5.4; Figure 17). CEH4 diverged progressively from the regional mean following the December 2017 clearfell, with the Tier 1 CUSUM reaching −9.88 m by early 2026, confirming that the control well itself has been drawn down by the felling pulse. CEH4 is also subject to progressive coastal erosion of the ground surface, which independently contributes to its deteriorating trajectory and limits its suitability as a stable long-term control." CEH22 showed a more severe and earlier onset of deterioration, its CUSUM reaching −14.55 m by early 2026 and dominated by a coastal drawdown signal that accelerated markedly from 2021 onwards (Figure 17). These Tier 1 trajectories establish that both local controls are experiencing substantial background decline that must be accounted for before any scraping benefit can be assessed.

Against this background, Tier 2 paired analysis yielded a net benefit of +0.143 m at CEH36 during the Pure Scraping era (April 2015 to December 2018), the only well and era combination to show a positive net benefit after Tier 1 correction (Figure 18). The robustness of the +0.143 m net benefit finding was assessed using two additional independent methods. A synthetic control was constructed as an optimally-weighted combination of 11 donor wells calibrated on the pre-scraping period; the Pure Scraping era step change estimated from the synthetic control gap (+0.163 m) is closely consistent with the raw BACI estimate (+0.136 m), confirming that the scraping signal at CEH36 is detectable independently of CEH4 and is not an artefact of CEH4's own coastal erosion or felling pulse trajectory. An SSM forward residual was computed by calibrating the model on the pre-scraping period and running it forward iteratively; the mean residual during the Pure Scraping era was near-zero (−0.034 m), consistent with the interpretation that the scraping benefit is structural — reflecting permanent ground surface lowering — rather than a deviation from the climate-driven water table trajectory. The +0.143 m net benefit therefore represents a robust finding, with the two BACI-based methods converging closely and the SSM residual confirming the structural basis of the signal (Figure 16).

During the subsequent Felling Pulse era the net benefit at CEH36 reversed to −0.047 m, consistent with the clearfell-driven drawdown in CEH4 progressively eroding the scraping gain. At CEH18, net benefits were −0.017 m during the Felling Pulse era and −0.047 m after the October 2023 scraping, indicating that background deterioration cancelled the localised scraping benefit entirely. CEH18 and CEH21 showed no scraping-specific signal under any method during the 2015 era, consistent with the fact that neither well was scraped until October 2023; their 2015–17 era records represent unmanipulated monitoring against which the subsequent 2023 intervention responses are assessed. A net benefit is not reported for CEH21 as its paired control CEH22 is not a stable baseline; raw Tier 2 BACI shifts were +0.045 m during the Coastal Drawdown era and +0.061 m after scraping. The annual summer minimum records for all three treatment wells, partitioned by management era and plotted against the Curreli et al. (2013) ecological thresholds, are presented in Figure 20. The progressive deepening of the CEH4 control well across all eras — from 0.732 m pre-scraping to 1.037 m post-rescraping — is consistent with the system-wide climate drying trend documented in Section 4.8.1 and confirms that management benefit must be assessed against this background trajectory rather than in absolute terms.

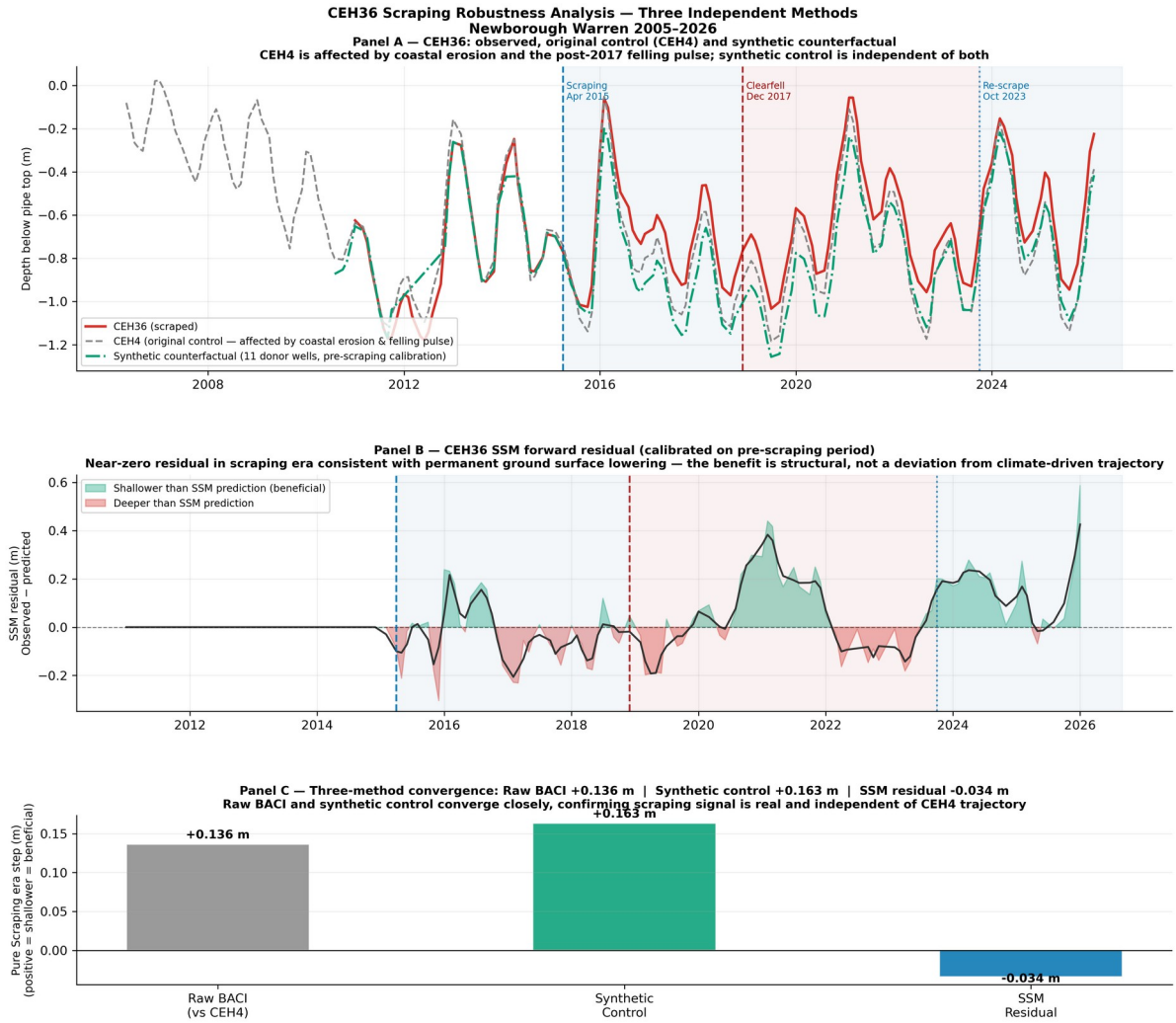


Figure 16. *Three-method robustness assessment of the scraping signal at CEH36. (a) Observed water level at CEH36 (red), original control CEH4 (grey dashed) and synthetic counterfactual constructed from 11 donor wells calibrated on the pre-scraping period (green dashed), with intervention dates marked. CEH4's progressive deepening relative to CEH36 following the April 2015 scraping and December 2017 clearfell is visible as divergence between the observed and control series; the synthetic counterfactual, which is independent of CEH4's coastal erosion and felling pulse trajectories, tracks CEH36 more closely throughout. (b) SSM forward residual at CEH36, computed by calibrating the model on the pre-scraping period and running forward iteratively; shading indicates periods where CEH36 is shallower (green, beneficial) or deeper (pink) than the SSM prediction. The near-zero residual during the Pure Scraping era is consistent with permanent ground surface lowering — the scraping benefit is structural rather than a deviation from the climate-driven trajectory — while the return to positive residual following the October 2023 re-scraping confirms a second topographic benefit at this well. (c) Pure Scraping era step changes estimated by three independent methods: Raw BACI vs CEH4 (+0.136 m), synthetic control (+0.163 m), and SSM residual (−0.034 m). The close agreement between the raw BACI and synthetic control estimates confirms that the scraping signal at CEH36 is real and independent of CEH4's own trajectory; the near-zero SSM residual reflects the structural rather than hydrodynamic nature of the benefit. (09_scrape_08_ceh36_robustness.png)*

Era-specific $-\beta_3$ estimates are presented in Table 5, where values are followed by significance indicators in parentheses: (*) $p < 0.05$, (**) $p < 0.01$, (***) $p < 0.001$, (ns) not significant. A not-significant result with a confidence interval spanning zero indicates the coefficient is indistinguishable from zero in that era. Across all three treatment wells, $-\beta_3$ showed a broadly increasing trajectory from baseline through the Felling Pulse era, though the pattern differed by site. At CEH36, the coefficient was 0.160 (*) at baseline, dipped slightly to 0.145 (ns) during Pure Scraping — the confidence interval spanning zero reflecting the short era length — then rose to 0.190 (***) during the Felling Pulse. At CEH18 the increase was more pronounced, rising from 0.133 (***) at baseline to 0.226 (**) during the Felling Pulse, before falling back to 0.203 (ns) after the October 2023 scraping. At CEH21, $-\beta_3$ increased from 0.156 (**) at baseline to 0.198 (**) during Coastal Drawdown and remained essentially unchanged at 0.197 (ns) after scraping.

The critical finding, however, is that $-\beta_3$ increased at both control wells over the same period: CEH4 from 0.148 (***) at baseline to 0.212 (***) during the Felling Pulse era — with no meaningful change during the Pure Scraping era (0.152, ns) — and CEH22 from 0.235 (***) at baseline to a peak of 0.351 (***) during the Coastal Drawdown era before settling at 0.295 (***) after scraping (Table 5; Figure 19). The fact that drainage acceleration is detectable at unmanipulated controls confirms it is a network-wide structural signal rather than a localised response to scraping or felling within the treatment footprints.

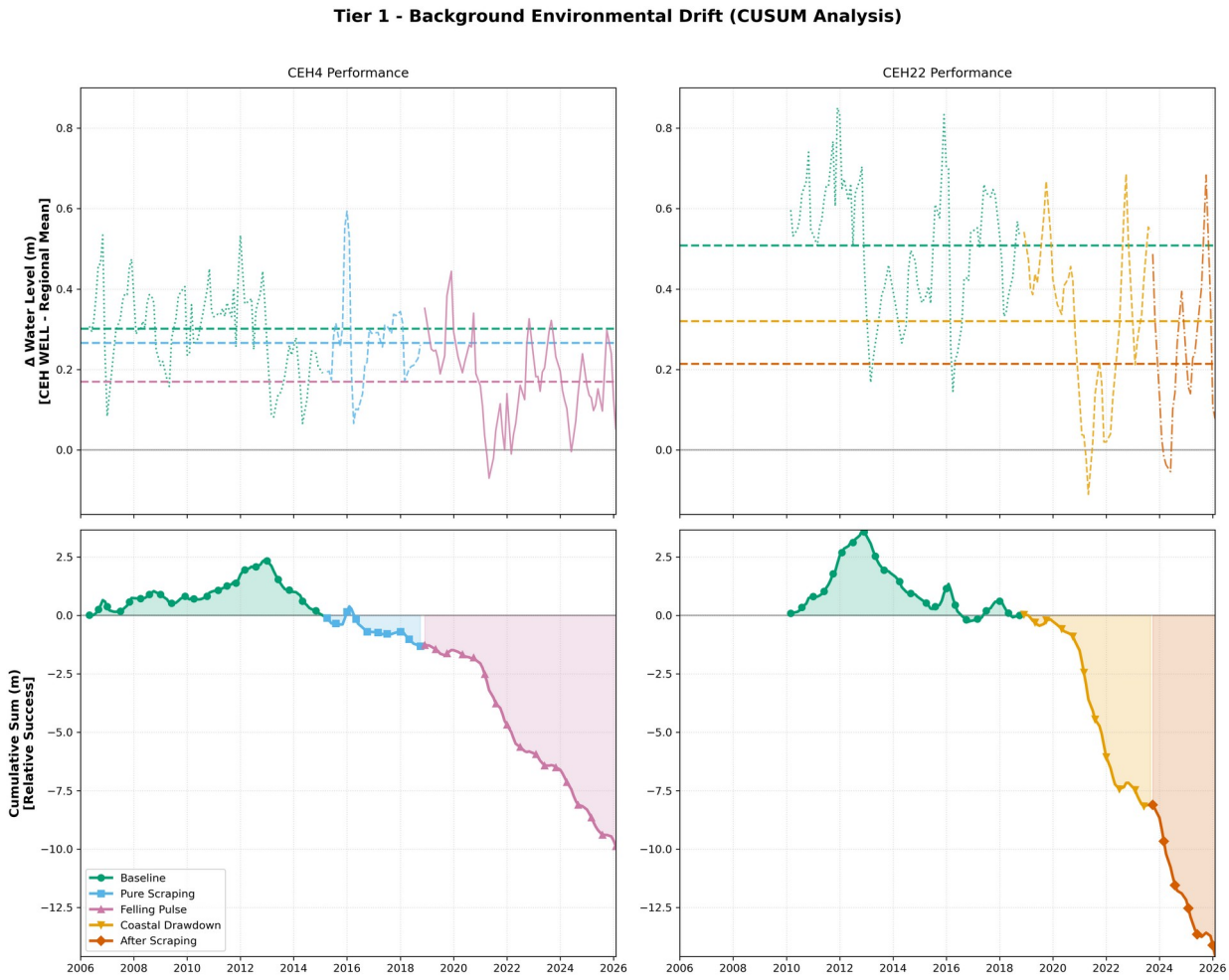


Figure 17. *Tier 1 background environmental drift analysis. CUSUM time series for control wells CEH4 (left) and CEH22 (right) decomposed by management era relative to the regional mean. Upper panels show monthly Δ water level (treatment well minus regional mean) with era mean levels indicated by dashed horizontal lines. Lower panels show cumulative CUSUM trajectories. A rising CUSUM indicates the well is performing above the regional mean (shallower water table); a falling CUSUM indicates below-average performance (deeper water table). The accelerating negative CUSUM at both control wells confirms a system-wide deterioration trend independent of local management interventions. (09_scrape_05_tier1_background_drift.png)*

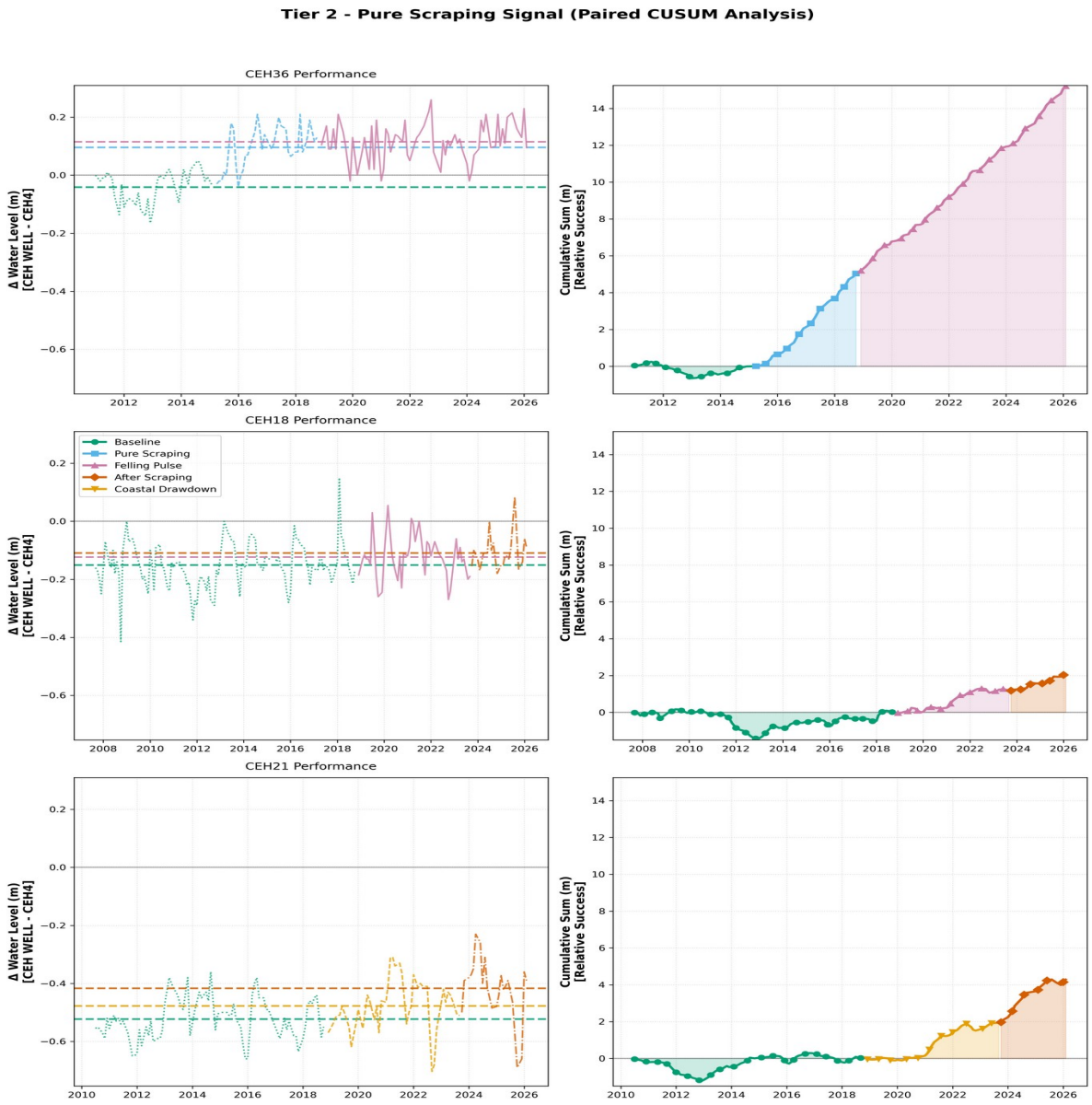


Figure 18. *Tier 2 pure scraping signal — paired CUSUM analysis at treatment wells CEH36 (left), CEH18 (centre) and CEH21 (right), each differenced against its matched control well (CEH36 and CEH18 against CEH4; CEH21 against CEH22). Upper panels show monthly Δ water level relative to the paired control with era mean levels indicated by dashed horizontal lines. Lower panels show cumulative BACI CUSUM trajectories. A rising CUSUM indicates the treatment well is becoming shallower relative to its control (positive scraping benefit); a falling CUSUM indicates deepening relative to control. CEH36 shows a clear positive signal during the Pure Scraping era (2015–2018); CEH18 and CEH21 show more complex responses reflecting the confounding influence of the post-2018 felling pulse and coastal drawdown respectively. For CEH36, additional contextual markers denote the period of active coastal foredune accumulation (shaded region, pre-2018) and the timing of the adjacent CEH18 scraping intervention (vertical dashed line, October 2023), illustrating the spatially constrained hydraulic influence of topographical interventions at this site. (09_scrape_06_tier2_scraping_signal.png)*

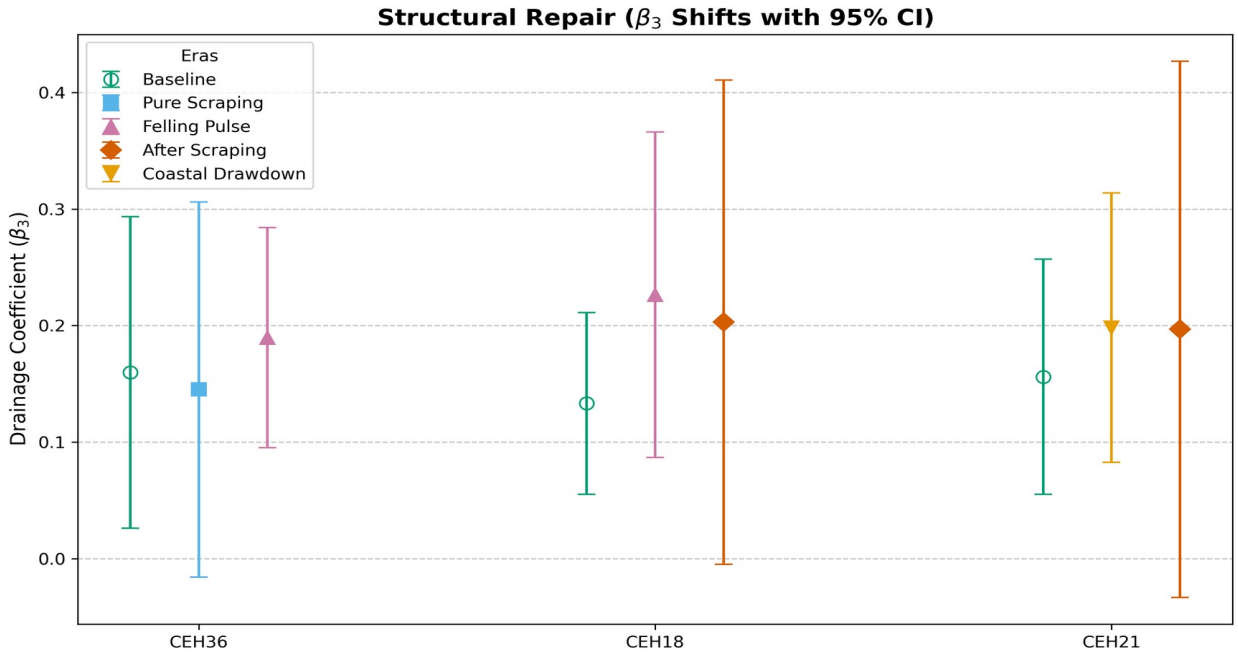


Figure 19. *Era-specific β_3 storage decay coefficients with 95% confidence intervals at scraping treatment wells CEH36, CEH18 and CEH21. Each symbol represents a management era: Baseline (hollow circle), Pure Scraping (filled square), Felling Pulse (triangle), After Scraping (diamond) and Coastal Drawdown (inverted triangle). Where confidence intervals overlap between eras, β_3 changes are not statistically distinguishable. The progressive increase at CEH18 from Baseline through Felling Pulse reflects the compounding influence of the December 2017 clearfell on drainage dynamics at this well. (09_scrape_07_beta3_confidence.png)*

Well	Role	Era	β_3	95% CI	p-value	Sig.
CEH36	Treatment	Baseline	0.160	[0.026, 0.293]	0.02	*
CEH36	Treatment	Pure Scraping	0.145	[−0.016, 0.306]	0.076	ns
CEH36	Treatment	Felling Pulse	0.190	[0.095, 0.284]	< 0.001	***
CEH18	Treatment	Baseline	0.133	[0.055, 0.211]	< 0.001	***
CEH18	Treatment	Felling Pulse	0.226	[0.087, 0.366]	0.002	**
CEH18	Treatment	After Scraping	0.203	[−0.005, 0.411]	0.055	ns
CEH21	Treatment	Baseline	0.156	[0.055, 0.257]	< 0.001	**
CEH21	Treatment	Coastal Drawdown	0.198	[0.083, 0.314]	< 0.001	**
CEH21	Treatment	After Scraping	0.197	[−0.033, 0.427]	0.089	ns
CEH4	Control	Baseline	0.148	[0.064, 0.232]	< 0.001	***
CEH4	Control	Pure Scraping	0.152	[−0.004, 0.309]	0.057	ns
CEH4	Control	Felling Pulse	0.212	[0.119, 0.305]	< 0.001	***
CEH22	Control	Baseline	0.235	[0.137, 0.333]	< 0.001	***
CEH22	Control	Coastal Drawdown	0.351	[0.216, 0.486]	< 0.001	***
CEH22	Control	After Scraping	0.295	[0.140, 0.451]	< 0.001	***

Table 5. *Era-specific β_3 storage decay coefficients at scraping treatment and control wells with 95% confidence intervals. Treatment wells (CEH36, CEH18, CEH21) shaded blue; control wells (CEH4, CEH22) shaded grey. β_3 increases at control wells reflect the system-wide hydraulic adjustment associated with the 2018 plantation clearance rather than scraping. Significance: *** p < 0.001, ** p < 0.01, * p < 0.05. (09_scrape_04b_table4_beta3_era_summary.csv)*

The annual summer minimum depth records for the three scraping treatment wells and the CEH4 control well, partitioned by management era, are shown in Figure 20. CEH36 shows a clear reduction in summer minimum depth during the April 2015 scraping era, with the era mean rising from 1.079 m (pre-scraping) to 0.858 m — a net improvement of 0.221 m consistent with the CUSUM analysis above. This benefit partially erodes post-felling (era mean 0.949 m) before a further partial recovery following the October 2023 rescraping (0.865 m). CEH18 and CEH21 were not scraped in 2015 — their 2015–17 era represents unscraped monitoring — but CEH18 shows a measurable improvement following the October 2023 rescraping (era mean 0.950 m vs post-felling 1.223 m). CEH21 shows limited response to the 2023 intervention. The progressive deepening of the CEH4 control well across all eras (0.732 m pre-scraping to 1.037 m post-rescraping) confirms that the system-wide background climate drying signal must be subtracted from any apparent scraping benefit.

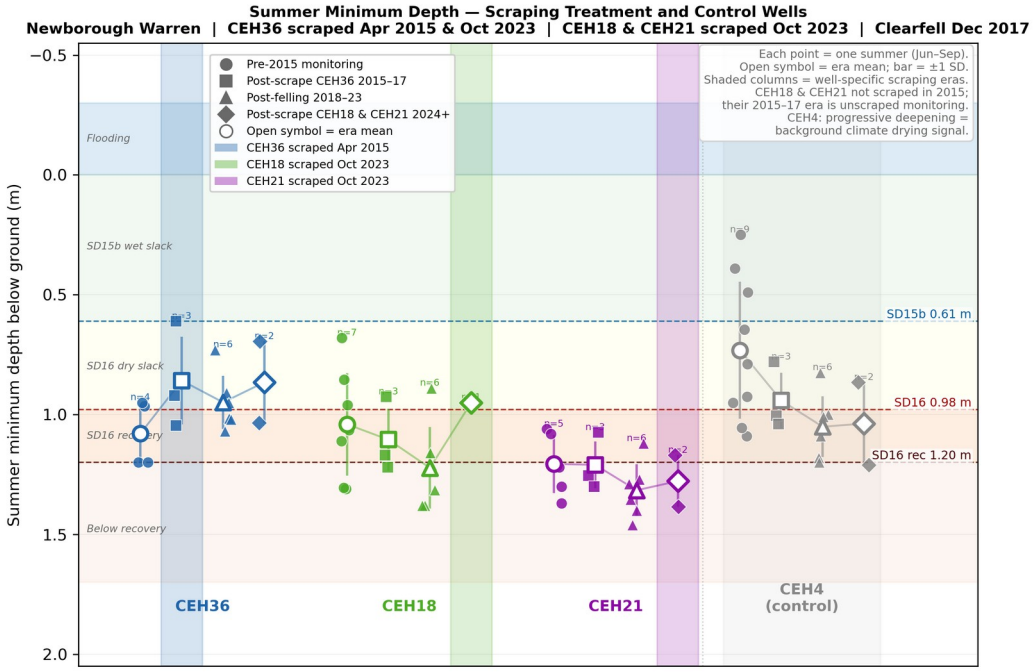


Figure 20. *Annual summer minimum depth below ground surface at scraping treatment wells (CEH36, CEH18, CEH21) and control well (CEH4), partitioned by management era. Each point represents one summer (June–September). Open symbols show era means; vertical bars = ±1 SD. Coloured vertical bands indicate each well's specific scraping period: CEH36 scraped April 2015 (blue, 2015–17 era) and October 2023; CEH18 and CEH21 scraped October 2023 only (green and purple bands, post-rescraping era). CEH18 and CEH21 were not scraped in 2015 — their 2015–17 era column represents unscraped monitoring. CEH4 (grey) provides a climate-corrected control; its progressive deepening reflects the background climate drying signal (Section 4.8.1). Ecological zone colouring follows Curreli et al. (2013) thresholds: SD15b wet slack (0–0.61 m), SD16 dry slack (0.61–0.98 m), SD16 recovery (0.98–1.20 m). (21_forestry_03_scraping_eras.png)*

4.6. The December 2017 Experimental Clearfell: Monitoring and Retrospective Analysis

4.6.1 Experimental Context and Monitoring Design

The December 2017 clearfell of approximately 8 ha at the plantation–open dune boundary was an operational management experiment designed to test whether canopy removal would raise water tables in the adjacent dune slack system. The monitoring network reported in this study was established independently of the intervention, with wells positioned in anticipation of planned forestry operations prior to their commencement. This independent design provides pre-intervention baselines uncontaminated by the experimental process and a monitoring record of 21 years — substantially exceeding the operational post-intervention monitoring period.

Wells within the felling footprint (FE series) were installed in August 2015 and form part of the extended network. Wells FE3 and FE4 have incomplete records in the dataset available to the author; their post-felling SSM coefficients are reported where data permit but they are excluded from paired before-after comparisons.

Two features of the monitoring record are noted at the outset as they bear directly on interpretation. First, a dune scraping event at CEH36 in April 2015 — 32 months before the clearfell — occurred approximately 400 m seaward of the felling footprint in the adjacent open dune slack system. CEH36 forms part of the edge zone in the BACI monitoring network and its water table behaviour in the pre-clearfell baseline period therefore reflects post-scraping conditions rather than undisturbed pre-intervention hydrology. This affects the raw BACI displacement figures (Section 4.6.2) and the edge zone ANCOVA result (Section 4.6.3), but not the primary impact zone ANCOVA, which is based solely on FE2, FE4 and WMC3 relative to the eight control wells. Second, the 30-month window between the scraping and the clearfell encompasses only three summer minima, which is insufficient to establish a stable post-scraping baseline from which to measure the subsequent clearfell effect. These features are examined in detail in Section 4.6.4.

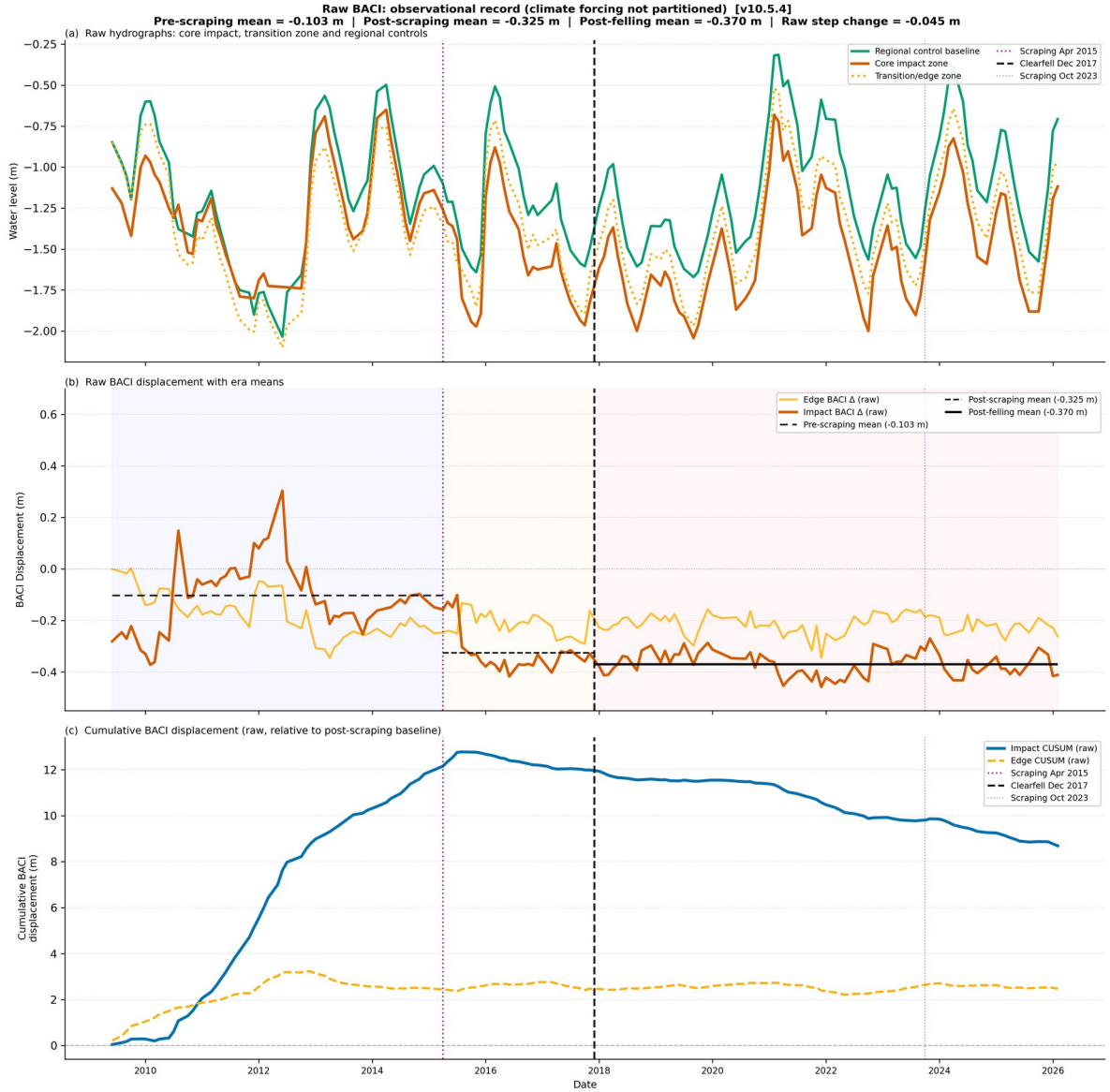


Figure 21. *Raw BACI: observational record (climate forcing not partitioned). Upper panel: raw hydrographs for core impact (FE2, FE4, WMC3), edge zone and regional control zones (8-well pool). Middle panel: raw BACI displacement with pre-scraping, scrape-era and post-felling era means. Lower panel: cumulative raw BACI CUSUM relative to the post-scraping pre-felling baseline. (10_cfell_01b_raw_baci.png)*

4.6.2 BACI Displacement Analysis

Prior to the April 2015 dune scraping event, core impact wells (FE2, FE4, WMC3) sat +0.049 m above the regional control baseline on average, indicating a slight pre-existing hydrological advantage relative to the broader dune system (Figure 21, upper panel). Following the scraping event the raw BACI displacement fell markedly, with the post-scraping pre-felling mean reaching −0.158 m by the time of the December 2017 clearfell — a raw shift of −0.207 m attributable to the scraping intervention within the clearfell footprint. The edge zone showed no comparable response to the scraping event, with a pre-scraping mean of −0.020 m and a post-scraping pre-felling mean of −0.051 m.

The post-felling raw mean displacement of the core impact zone was −0.201 m, giving a raw step change of −0.043 m from the post-scraping pre-felling baseline (Figure 21, middle panel). The combined raw hydrological cost of both interventions, measured from the pre-scraping baseline to the post-felling mean, was −0.251 m. The edge zone showed negligible raw felling-specific displacement (step change +0.002 m; combined cost −0.030 m).

The raw cumulative BACI CUSUM is shown in Figure 21, lower panel. The CUSUM tracks the cumulative sum of monthly deviations from the post-scraping pre-felling baseline mean, so a rising curve indicates the impact zone is consistently performing above the baseline and a falling curve indicates it is consistently below. The curve rises strongly during the pre-scraping era when the impact zone sat above controls, peaks at +11.95 m in February 2016 during the anomalously wet 2015–16 period, and declines gradually through the post-scraping pre-felling period as the scraping-induced displacement takes hold. From December 2017 the curve declines more steeply as the post-felling displacement compounds month by month, reaching +8.027 m by February 2026 and still declining with no sign of stabilisation. The positive value at the end of the record does not indicate a net hydrological benefit — it reflects the large pre-intervention surplus that accumulated before the scraping and felling events, which the post-felling deficit is progressively eroding.

These raw displacement statistics do not partition the intervention signals from background climate variability or from each other. The pre-felling period encompassed substantially different climate conditions from the post-felling period, and the impact and control zones respond differentially to wet and dry conditions. The climate-corrected analysis is presented in Section 4.6.3.

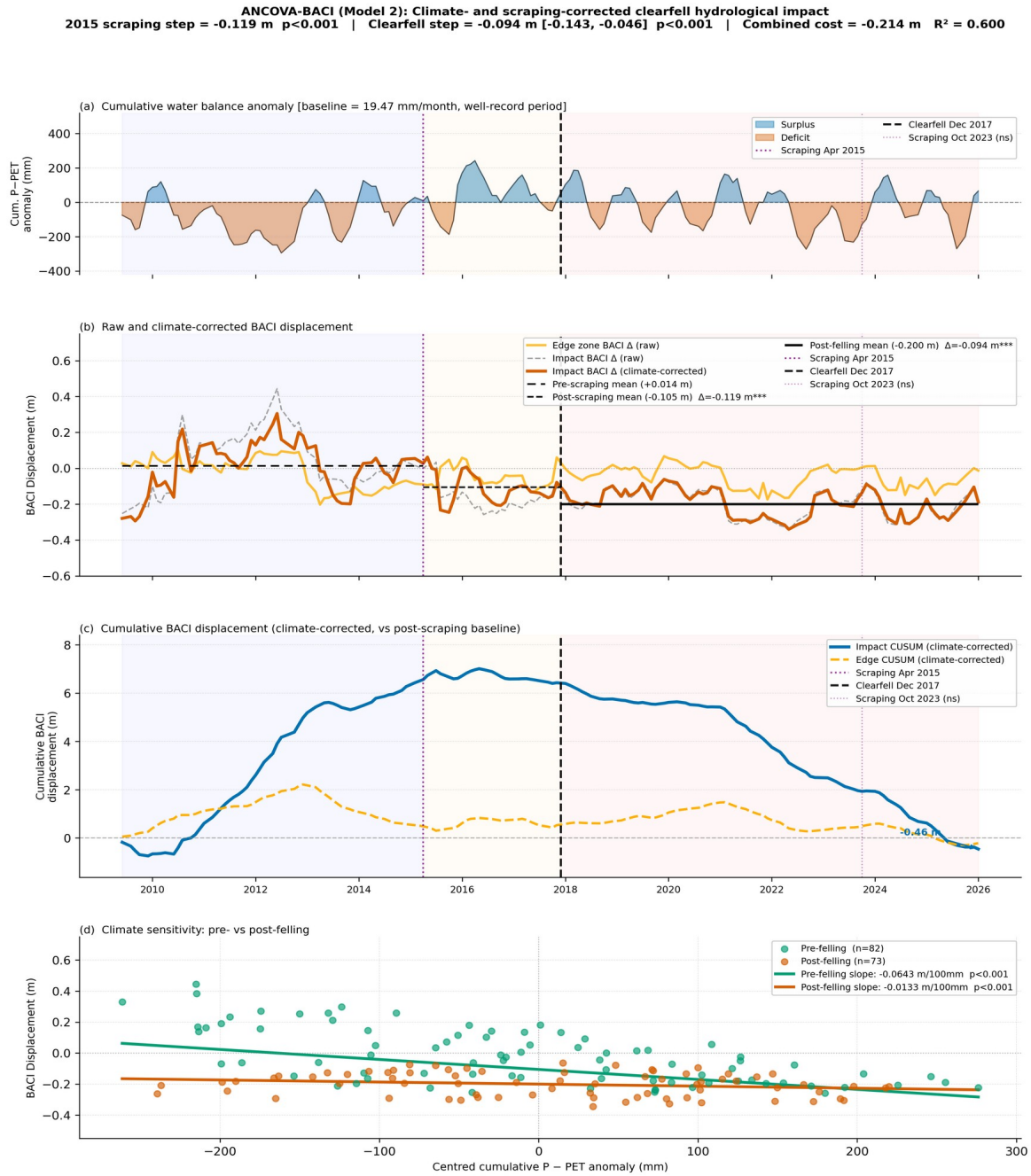


Figure 22. *ANCOVA-BACI analysis of the December 2017 clearfell [v10.5.4 R²=0.729], n = 155. (a) Centred cumulative water balance anomaly. (b) Raw and climate-corrected BACI displacement with fitted era means; scraping step = −0.137 m ***, clearfell step = −0.093 m ***. (c) Climate-corrected CUSUM starting at zero at clearfell date, reaching −6.82 m by February 2026; no zero crossing. (d) Climate sensitivity pre- vs post-felling; 79.3% reduction. † October 2023 scraping term tested but not retained (p = 0.258, ΔAIC = +0.66). (10_cfell_01_dual_control_baci.png)*

4.6.3 ANCOVA-BACI Climate-Corrected Analysis

To separate the intervention signals from background climate variability, an ANCOVA-BACI model was fitted with the centred cumulative water balance anomaly (P − PET − 19.47 mm/month, cumulated from the start of the BACI record) as a continuous climate covariate, alongside step terms for the 2015 scraping event and the December 2017 clearfell, and an interaction term to test whether climate sensitivity changed after felling (Section 3.5.5). The control pool comprises eight wells — CEH32, CEH34, CEH33, NW10, CEH19, CEH9, NW7 and NW6 — of which CEH9, NW7 and NW6 are open warren wells unaffected by either intervention; their inclusion raises ANCOVA R^2 from 0.600 (original 5-well pool) to 0.729. The October 2023 scraping event was tested as an additional step term but was not retained ($p = 0.258$, $\Delta AIC = +0.66$).

For context, cumulative climate forcing alone explains 51.5% of network mean water table variance (Section 4.1), establishing climate as the dominant background signal. The ANCOVA-BACI model explains 72.9% of monthly BACI displacement variance ($R^2 = 0.729$, $n = 155$), with the additional 21.4 percentage points attributable to the scraping and clearfell step terms.

At mean long-run climate conditions, the 2015 scraping event lowered the core impact zone displacement by −0.137 m ($p < 0.001$). The December 2017 clearfell imposed a further step change of −0.093 m [−0.131, −0.055] ($p < 0.001$), giving a combined climate-corrected hydrological cost of −0.230 m relative to the pre-scraping baseline (Table 6; Figure 22). These figures should be treated as **upper bounds** on the respective intervention effects for the reasons developed in Section 4.6.4.

The climate covariate was highly significant in the pre-felling period (slope = −0.0618 m/100mm, $p < 0.001$), confirming that the intact plantation canopy was amplifying the climate signal in the impact zone relative to the open dune controls. The interaction term indicated that climate sensitivity was reduced by 79.3% following the clearfell (pre-felling slope: −0.0618 m/100mm; post-felling slope: −0.0127 m/100mm, ns). The apparent step deepenings in the raw BACI displacement during the wet winters of 2020/21 and 2023/24 are fully explained by this differential climate response and do not represent additional structural changes (Figure 22b).

The climate-corrected CUSUM starts at zero at the clearfell date (December 2017) and accumulates post-felling deviations from the post-scraping pre-felling fitted mean (Figure 22c). The CUSUM declines monotonically following the clearfell, reaching −6.82 m by February 2026 with no zero crossing — the post-felling cumulative deficit has not been reversed within the monitoring period.

The edge zone showed no significant climate-corrected response to the clearfell (step = −0.028 m, $p = 0.246$, ns) but a significant response to the scraping event (step = −0.061 m, $p = 0.018$), confirming that CEH36's altered baseline dominates the edge zone signal (Table 6). The mechanistic underpinning of these displacement patterns is examined through the SSM coefficient analysis in Section 4.6.5.

Zone	Wells	Scraping step	Clearfell step	95% CI	p	R^2
Impact	FE2, FE4, WMC3 (65–102 m)	−0.137 m ***	−0.093 m ***	[−0.131, −0.055]	< 0.001	0.729
Edge	FE1, FE3, CEH31, LIS1, CEH20, CEH30, CEH16, NW8b (65–241 m)	−0.061 m *	−0.028 m ns	[−0.076, +0.019]	0.246	0.191

Table 6. *ANCOVA-BACI results by zone (Script 10 v10.5.4). Control pool: CEH32, CEH34, CEH33, NW10, CEH19, CEH9, NW7, NW6 (8 wells). Negative steps = water table deepening. R^2 = 0.729, n = 155.*

4.6.4 Robustness Analysis and Confound Assessment

The ANCOVA-BACI result presented in Section 4.6.3, whilst statistically significant within its own framework, is subject to three confounds that limit causal attribution to the clearfell specifically.

Confound 1 — Scraping baseline contamination. CEH36 was scraped in April 2015, 32 months before the clearfell, approximately 400 m seaward of the felling footprint in the open dune slack system. CEH36 is part of the edge zone in the BACI monitoring network. Its pre-clearfell water table record therefore reflects post-scraping hydrology rather than undisturbed pre-intervention conditions, affecting the raw BACI displacement figures and the edge zone ANCOVA result. The scraping and clearfell signals cannot be cleanly separated given the 30-month window between them.

Confound 2 — Wet 2015–16 baseline. The scrape era (April 2015 – December 2017) was anomalously wet (920 mm/yr) relative to the long-run mean (864 mm/yr) and the post-clearfell period (863 mm/yr). The ANCOVA covariate accounts for the cumulative water balance trend but cannot fully remove the effect of an elevated pre-clearfell baseline created by a specific wet sequence. The wet 2015–16 period elevated water tables across the whole network before the clearfell, inflating the reference level against which post-clearfell conditions are measured.

Confound 3 — Common post-2018 climate signal. SSM residuals normalised against regional controls show all zones — including controls — exhibiting a common declining residual from 2022 onwards, consistent with reduced lagged ridge recharge following the drier 2018–2021 period (Section 5.4.3). This network-wide signal is indistinguishable from a clearfell-specific effect within the ANCOVA framework.

Three independent robustness analyses

Three independent robustness analyses were conducted to test whether a clearfell-specific signal could be detected once these confounds were accounted for (Section 3.5.5):

- **SSM residual analysis (Option 4).** The SSM was calibrated on the pre-scraping period for each well and run forward iteratively. Residuals normalised against the regional control mean showed no statistically significant step change at the clearfell date in any

zone. Only WMC3 showed a significant post-clearfell departure (−0.165 m, p = 0.026), but this pre-dated the clearfell (scrape era: −0.096 m) and the step between eras was not significant (p = 0.355).

- **Synthetic control analysis (Option 2).** A synthetic counterfactual for each impact well constructed from 11 donor wells showed non-significant gaps at all zones in the post-clearfell era (Core: +0.164 m ns; Edge: +0.141 m ns; Scraping zone: −0.022 m ns). Core and edge zones were consistently at or above their synthetic counterfactual levels.
- **Cluster transition analysis.** Rolling 48-month β_1 and β_3 estimates for the felled zone showed no consistent post-clearfell convergence toward C3 Western Block values, with the felled zone occupying an intermediate position between C3 and C4 before and after the intervention.

Spatial transect analysis

A spatial transect from the uncleared plantation interior (CEH2, 414 m from clearfell centroid) through the regional control (CEH34, 285 m) to the core impact zone (WMC3, 92 m) and edge wells (184–191 m) provides the most direct visual evidence for the null result (Figure 23). Post-felling step changes relative to the scrape era baseline are uniformly negative across the full transect — CEH2 −0.111 m, CEH34 −0.093 m, WMC3 −0.140 m, NW8b −0.125 m — with no spatial gradient consistent with a clearfell-specific effect. A genuine clearfell response would be strongest at WMC3 and absent at CEH2 and CEH34; the observed pattern is the opposite. Note that CEH2 is approximately 100 m downslope of the 1996 broadleaf restocking block and its progressive deepening relative to the network mean may reflect the influence of the maturing broadleaf canopy on lateral inflow to the adjacent pine zone (Section 4.6.7), adding a further caveat to its use as a stable reference well.

The convergence of three independent robustness analyses and the spatially uniform transect response on the same result is the principal analytical finding. The ANCOVA-BACI displacement of −0.093 m is most parsimoniously explained by the contrast between the anomalously wet pre-clearfell baseline and subsequent drier conditions, amplified by the scraping confound at CEH36. **It should be treated as an upper bound on any clearfell-specific effect rather than a precise causal attribution.**

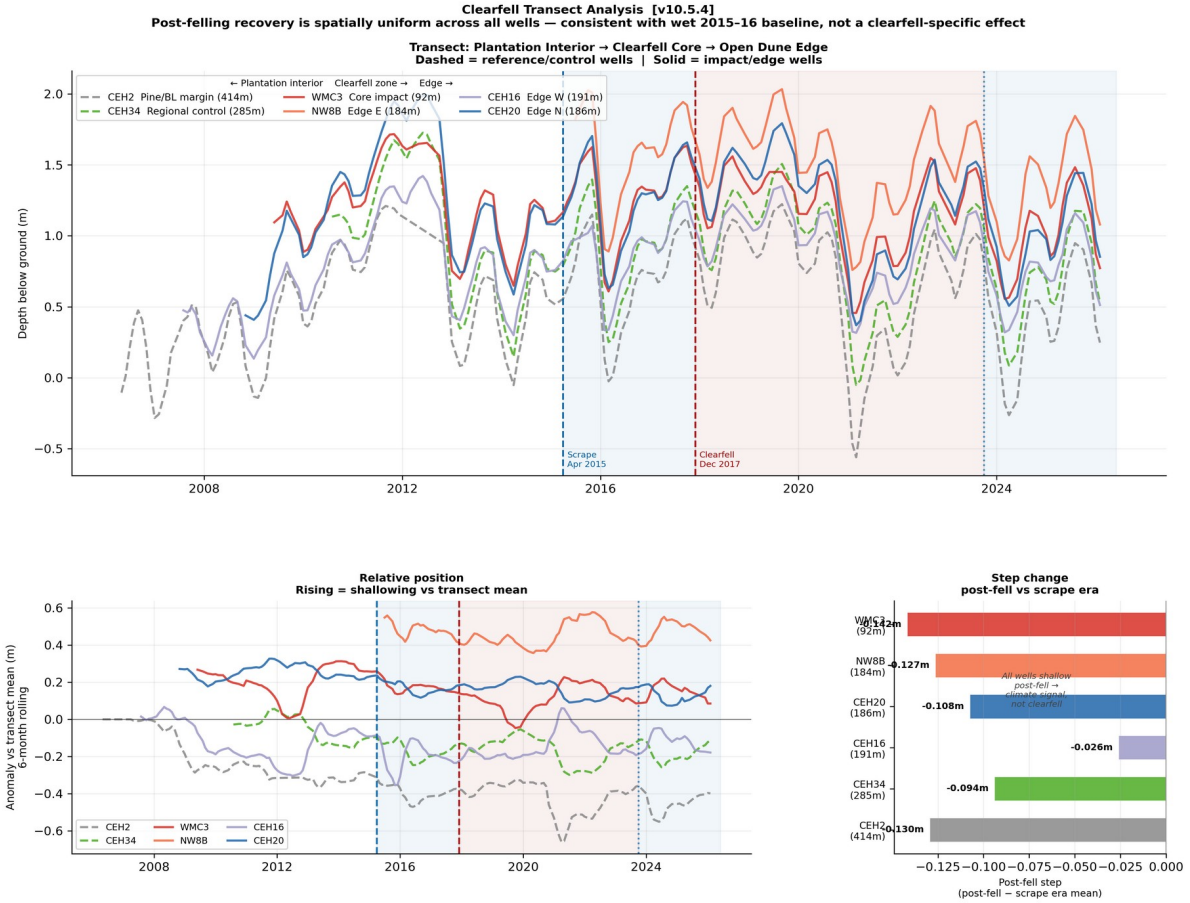


Figure 23. *Clearfell transect analysis [v10.5.4]: plantation interior → clearfell core → open dune edge. Upper panel: depth hydrographs. Middle panel: anomaly relative to transect mean (6-month rolling). Lower right: post-felling step changes — all wells shallower post-fell regardless of distance from clearfell, consistent with a common climate baseline effect. (10b_01_clearfell_transect.png)*

4.6.5 Drainage Slope Diagnostic and Storage Decay Analysis

Era-specific β_3 estimates for all 16 clearfell network wells are presented in Table 7 (Section 3.5.3). Of the 14 wells with paired pre- and post-felling records, 13 showed an increase in β_3 after felling. FE3 is the single exception, with β_3 decreasing from 0.413 to 0.147; however its pre-felling value is the outlier rather than the post-felling one — a very short pre-felling record (n = 6) produced a wide confidence interval and an unstable coefficient estimate, and the post-felling value of 0.147 is consistent with the broader edge zone network. FE4 and LIS1 lack sufficient pre-felling data for paired comparison; their post-felling values (0.114 and 0.082 respectively) are within the expected range for their respective zones.

Although individual well β_3 changes are frequently non-significant owing to wide confidence intervals from short pre-felling records, the consistency of direction across 13 of 14 paired wells is itself highly significant (sign test $p < 0.001$). The probability of 13 of 14 wells all increasing in the same direction by chance alone is less than 0.02%, making the directional pattern robust regardless of individual well significance.

The β_3 increases were consistent across all three spatial tiers of the experimental design (Figure 25c), with zone mean gains of +0.038 (core impact), +0.055 (edge) and +0.029

(regional control). The extension of increases to the five regional control wells, located several hundred metres from the felling boundary, is the key diagnostic finding. Under the original interpretation — that β_3 increases reflect genuine drainage acceleration — it would be physically implausible for control wells outside the felling footprint to show the same structural change as wells within it. The extension to controls instead supports the interpretation that these increases are SSM fitting artefacts arising from the calibration window straddling the transition between two equilibrium states (Section 5.4). As the system adjusts from the forest to the post-clearfell equilibrium, the model interprets the systematic head trajectory as an elevated drainage signal, producing spurious β_3 increases that are not confined to the intervention footprint precisely because they are not driven by it.

This reinterpretation does not undermine the ANCOVA-BACI result. The 79.3% reduction in climate sensitivity of the impact zone (Section 4.6.3; Figure 22) provides convergent, independent evidence that the hydrological behaviour of the impact zone changed after felling — a result derived from the ANCOVA model rather than from β_3 estimation. The BACI framework expresses impact and edge zone changes relative to the control baseline, so the step changes quantified in Section 4.6.3 represent any felling-specific component above and beyond system-wide shifts captured by the controls. The β_3 increases, because they extend uniformly to those controls, cannot be interpreted as felling-specific.

Well	Zone	$-\beta_3$ Before	$-\beta_3$ After	$\Delta-\beta_3$	Before	After
FE2	Core Impact	0.006	0.128	0.123	ns	*
FE4	Core Impact	—	0.114	—	—	*
WMC3	Core Impact	0.079	0.167	0.087	ns	**
FE1	Edge Zone	−0.008	0.092	0.1	ns	ns
FE3	Edge Zone	0.413	0.147	−0.266	ns	*
LIS1	Edge Zone	—	0.082	—	—	ns
CEH20	Edge Zone	0.046	0.081	0.035	ns	ns
CEH30	Edge Zone	0.044	0.109	0.065	ns	*
CEH16	Edge Zone	0.09	0.143	0.053	*	**
NW8B	Edge Zone	0.061	0.168	0.107	ns	**
CEH31	Edge Zone	0.107	0.167	0.06	*	**
CEH32	Regional Control	0.009	0.06	0.051	ns	ns
CEH34	Regional Control	0.015	0.054	0.04	ns	ns
CEH33	Regional Control	0.03	0.077	0.048	ns	ns
NW10	Regional Control	0.06	0.103	0.043	ns	*
CEH19	Regional Control	0.147	0.202	0.055	**	***

Table 7. *Before and after clearfell estimates of the storage decay coefficient ($-\beta_3$) for all 16 experimental wells, derived from OLS regression of the residual drainage component against lagged water level. Wells are grouped by zone: core impact (FE2, FE4, WMC3), transition/edge (FE1, FE3, LIS1, CEH20, CEH30, CEH16, NW8B, CEH31) and regional control (CEH32, CEH34, CEH33, NW10, CEH19). Before = pre-December 2017; After = full post-felling period (December 2017 to February 2026). $\Delta-\beta_3$ = After − Before. Significance codes: (***) $p < 0.001$, (**) $p < 0.01$, (*) $p < 0.05$, (ns) not significant. Dashes indicate insufficient pre-felling record for paired comparison (FE4, LIS1: no pre-felling data; FE3: n = 6, see footnote). † FE1 pre-felling $-\beta_3 = -0.008$ (ns): this marginally negative value reflects an insufficiently constrained pre-felling record (n = 23 months) and is treated as a near-zero estimate; the post-felling value of 0.092 (ns) is consistent with adjacent edge zone wells.*

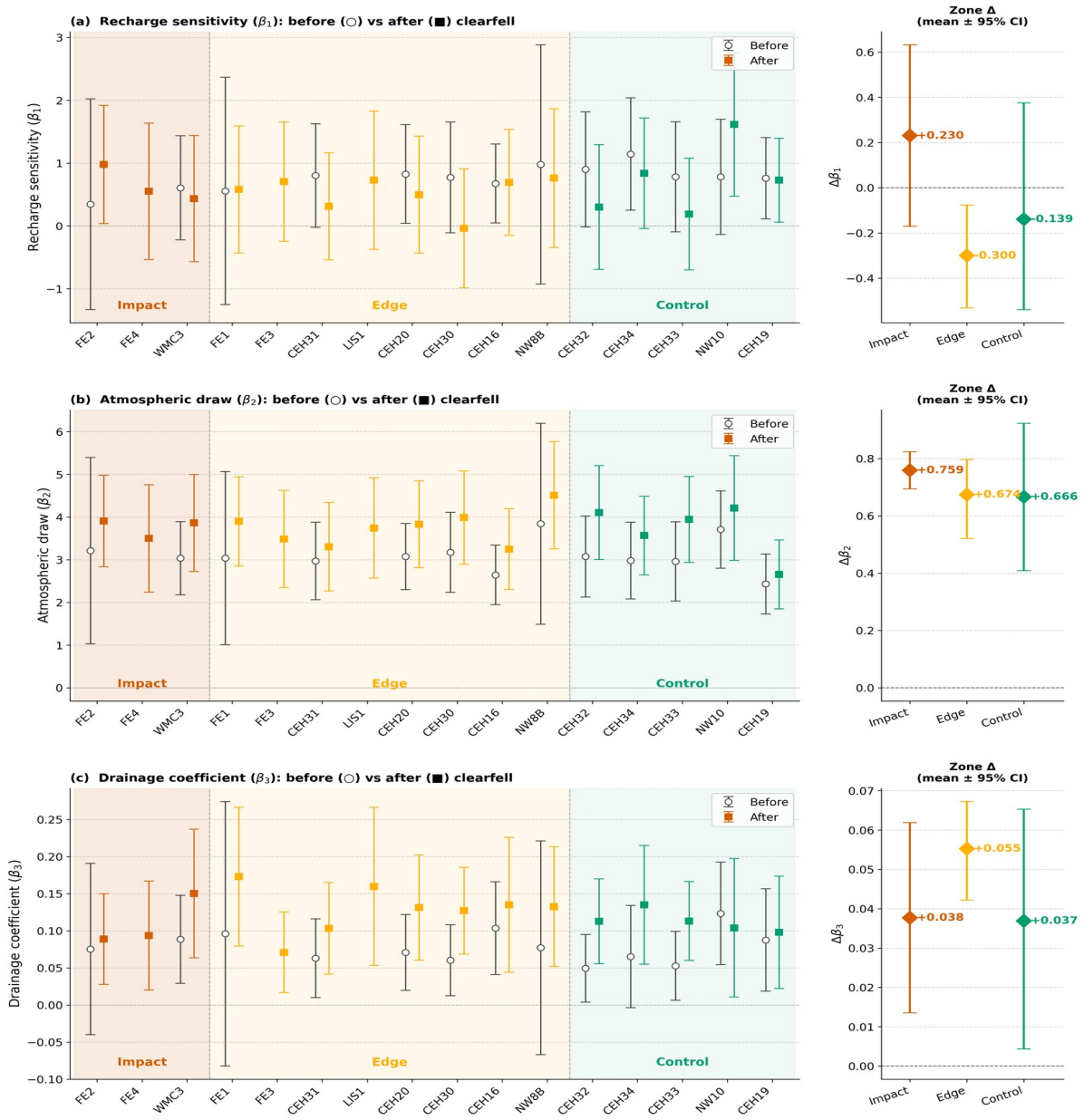


Figure 24. *Before and after clearfell SSM coefficient estimates with 95% CI for all 16 experimental wells [v10.5.4]. (a) β_1 recharge sensitivity: increased at 12/13 paired wells. (b) β_2 atmospheric draw: increased at all 13 paired wells. (c) β_3 storage decay: increased at 13/14 paired wells — interpreted as SSM fitting artefacts of the transition between equilibria, not genuine drainage acceleration (Section 5.4). (10_cfell_03_beta3_ols_slopes.png)*

4.6.6 Changes in Recharge Sensitivity and Atmospheric Draw

Recharge sensitivity (β_1) increased at 12 of 13 paired wells following the December 2017 clearfell (Figure 25a). The largest absolute gains were recorded at NW10 (+1.306), FE2 (+0.970) and FE1 (+0.756). FE4 exhibited a significant post-felling β_1 of 1.135 [0.257,

2.013], although the absence of pre-felling data precluded a paired comparison. This is the most robust finding from the intervention analysis — directionally consistent, spatially coherent, and physically expected from the removal of 142 mm/yr of canopy interception loss.

Atmospheric draw (β_2) increased at all 13 paired wells (Figure 25b), with gains ranging from +0.227 (CEH19) to +0.776 (WMC3) and a median gain of +0.578. Zone mean gains were +0.759 (core impact), +0.674 (edge) and +0.577 (regional control). Under the physical interpretation developed in Section 5.4, these increases reflect the removal of the canopy's role in partitioning atmospheric evaporative demand between the canopy surface and the water table — without the canopy intercepting rainfall and satisfying part of that demand above ground, more of the Thornthwaite atmospheric demand acts directly on the water table. The β_1 and β_2 increases are genuine physical responses to canopy removal, directionally consistent with the known interception physics.

4.6.7 Summer Minimum Depth Distributions by Cluster and Management Phase

The distributional consequences of the December 2017 clearfell across all BACI experimental zones are shown in Figure 26. The pre-felling baseline is limited to 2015–17 for the core impact wells (FE2, FE4, WMC3), which were installed in August 2015; for the edge zone, forest interior and regional control, the pre-2015 baseline extends back to 2007–2014. All four zones sit deep in the below-recovery zone (>1.20 m) throughout the monitoring period — the clearfell has not restored any zone to within the ecologically functional range defined by Curreli et al. (2013). Notably, the largest shift in summer minimum depths occurred during the 2015–17 era — after the April 2015 scraping event and during the anomalously wet 2015–16 period — rather than after the December 2017 clearfell. The post-clearfell shift is modest relative to the 2015–17 movement, consistent with the scraping and climate signals dominating the monitoring record.

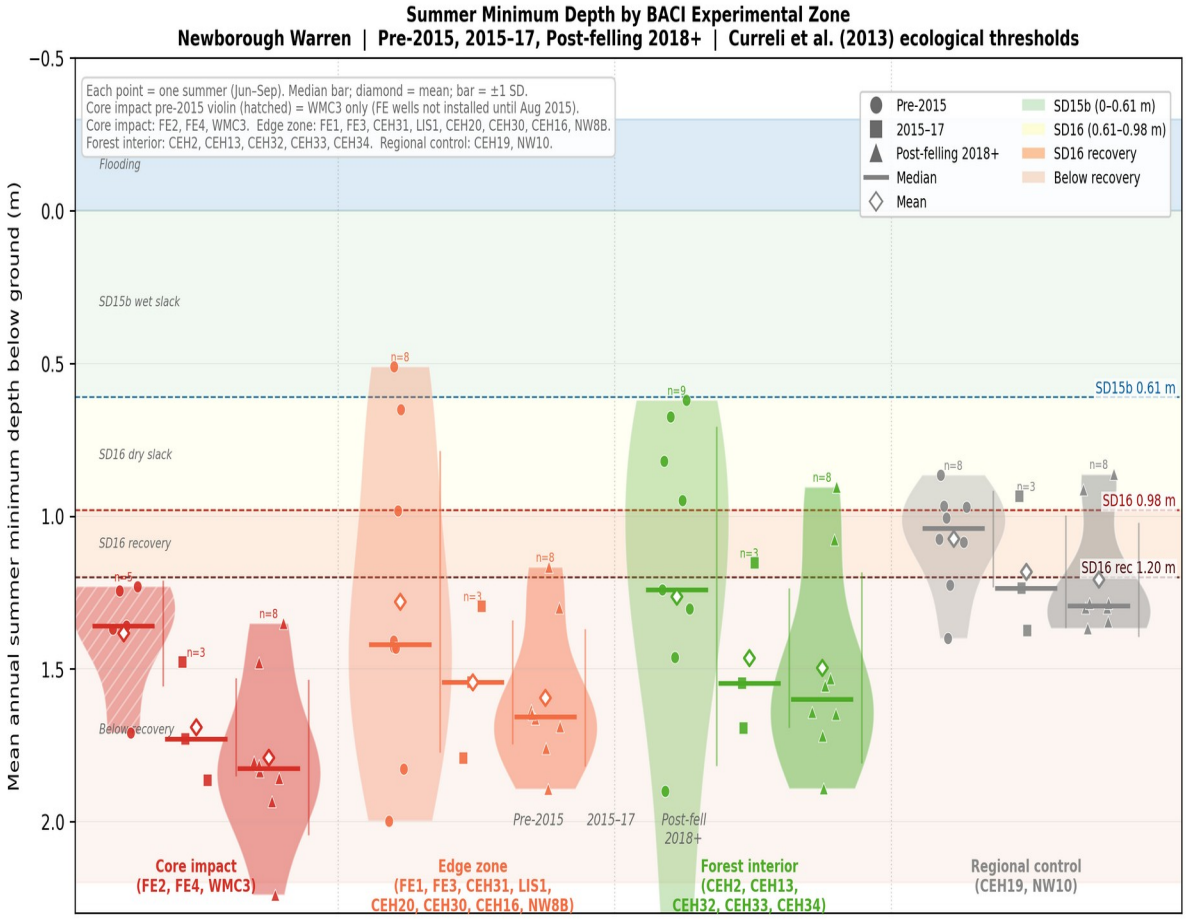


Figure 25. *Summer minimum depth below ground surface by BACI experimental zone, Newborough Warren 2005–2026. Three phases shown: pre-2015 baseline, 2015–17 era, and post-felling 2018+. Core impact: FE2, FE4, WMC3 (within clearfell footprint); edge zone: FE1, FE3, CEH31, LIS1, CEH20, CEH30, CEH16, NW8b (forest margin); forest interior: CEH2, CEH13, CEH32, CEH33, CEH34 (uncleared forest); regional control: CEH19, NW10. Each point = one summer (June–September). Median bar; diamond = mean; bracket = ±1 SD. Violin width indicates frequency of years at each depth. Core impact pre-2015 violin (hatched) = WMC3 only — FE wells not installed until August 2015. Ecological zone colouring follows Curreli et al. (2013). All zones sit below the SD16 recovery threshold (1.20 m) throughout the monitoring period. (21_forestry_04_baci_zone_violin.png)*

4.6.8 Broadleaf Conversion: Observational Evidence from the 1993 Block

The 1993 clearfell and 1996 broadleaf restocking on the north-western plantation margin provides the only within-network evidence on the hydrological effects of species conversion. Dipwell NW10, situated within the broadleaf block and monitored from 2007, shows a recharge sensitivity of $\beta_1 = 1.532$ — substantially above the C4 cluster median of 0.882 and the adjacent Corsican pine well CEH2 ($\beta_1 = 0.939$) — consistent with lower canopy interception under a deciduous canopy allowing more rainfall to reach the water table per millimetre of precipitation. The atmospheric draw coefficient at NW10 ($\beta_2 = 4.119$) is essentially identical to CEH2 ($\beta_2 = 4.222$), a difference of less than 3%, indicating that year-round integrated ET demand at the water table is no greater under broadleaf than under adjacent pine in the current stand configuration. The storage decay coefficient $\beta_3 = 0.099$ is indistinguishable from CEH34 ($\beta_3 = 0.100$), confirming that both wells occupy the same

geological substrate. When NW10 is normalised to the shared climate signal, its summer minimum tracks adjacent pine and warren wells almost identically throughout the monitoring period, with a mean anomaly relative to the pine interior composite of +0.002 m during the bramble-dominated phase (2010–2021) — indistinguishable from zero. An emerging trend of −34.1 mm/yr over 2019–2025 ($p = 0.062$) is consistent with the expected direction of canopy closure deepening summer minima but does not meet conventional significance thresholds and is sensitive to a single influential value. These observations are presented as exploratory findings from a natural experiment.

NW10 is a single dipwell, and the positional confounders — elevation, distance from the ridge, and boundary subsidy (α = +0.072 m/month at NW10 vs +0.112 m/month at CEH2) — are of comparable magnitude to the observed depth difference itself. The spread between CEH2 (mean summer minimum 0.865 m) and CEH34 (1.197 m), both Corsican pine wells within the same cluster, illustrates that inter-well variability under identical canopy can exceed the NW10 signal entirely. No reliable attribution to canopy type is therefore possible from a single dipwell. A minimum of three to four instrumented wells distributed across the broadleaf compartment, selected to span the same elevation and boundary subsidy range as the pine comparators, would be required before any hydrological conclusions about broadleaf conversion could be drawn. This represents a significant missed opportunity in the current monitoring design, and should be a priority for any future network expansion. The limitation is discussed further in Section 6.3.

CEH2, situated approximately 100 m downslope of the broadleaf block within the adjacent Corsican pine plantation, shows a progressive deepening trend relative to the network mean that is not present in other pine interior wells. This is consistent with the maturing broadleaf canopy progressively reducing lateral inflow to the adjacent downslope pine zone as leaf area index increases — extending the hydrological footprint of the restocking block beyond its physical boundary. This observation is noted as a caveat on the use of CEH2 as a stable plantation reference well in the transect analysis (Section 4.6.4), and warrants investigation with additional instrumentation downslope of the broadleaf block.

4.7. Critical Rainfall Threshold Forecasting - Predicting Winter flooding and Summer droughts

4.7.1 Mechanistic State-Space Equations

The cluster-level state-space equations (Section 3.6.1) fitted without intercept produced statistically significant coefficients at all four clusters (all $p < 0.001$ except C4 β_1 where $p = 0.001$), confirming that the physical mass-balance formulation is well-supported by the data. Note that β coefficients here are expressed in m/mm rainfall, consistent with the forecasting tool formulation; values in Table 2 are expressed in m/m (i.e. multiplied by 1000). Model fit declined systematically from the more responsive eastern clusters to the more buffered western and forest clusters: R^2 ranged from 0.660 (C1) to 0.461 (C4), reflecting the progressively greater storage buffering captured by the increasing LCSC values across clusters (Table 2).

Recharge sensitivity (β_1) was similar for C1 (0.0016 m/mm) and C2 (0.0017 m/mm), declining through C3 (0.0012) to C4 (0.0010), consistent with the interception losses documented for the forest zone. Atmospheric draw (β_2) followed the same gradient (C1: 0.0050, C2: 0.0044, C3: 0.0036, C4: 0.0034). The storage decay coefficient ($-\beta_3$) showed the largest range, declining from 0.366 (C1, eastern lake-influenced wells with rapid drainage) to 0.078 (C4, forest wells with high buffering capacity), consistent with the cluster-level LCSC pattern established in Section 4.2.2 (Table 2).

4.7.2 Seasonal Transfer Functions

Winter rainfall totals (P_winter) was non-significant in all three blocks (Western p = 0.818, Eastern p = 0.195, Forest p = 0.557). The antecedent summer minimum (h_min) was the dominant predictor in the Western Block ($p < 0.001$, $R^2 = 0.684$) and Forest models ($p < 0.001$, $R^2 = 0.894$), confirming that the depth of the preceding summer drought is a stronger predictor of winter flooding potential than the actual winter rainfall amount within the observed range. This reflects the aquifer's multi-month memory — a deep summer minimum requires progressively more winter rainfall simply to restore the water table to its pre-drought position before any flooding can occur. Neither predictor reached individual significance in the Eastern Block model ($R^2 = 0.240$), reflecting the greater spatial heterogeneity of that block and its weaker seasonal memory. The negative P_winter coefficient in the Forest block (−0.00020) reflects multicollinearity with h_min and should not be interpreted mechanistically.

Block	Equation	R^2	n	Key finding
Western Block (C3)	h_peak = 0.00010·P_winter + 1.169·h_min + 0.830	0.68	20 yrs	h_min dominant ($p < 0.001$); P_winter not significant ($p = 0.818$)
Eastern Block (C1+C2)	h_peak = 0.00057·P_winter + 0.305·h_min − 0.089	0.24	20 yrs	Low R^2; neither predictor individually significant
Forest (C4)	h_peak = −0.00020·P_winter + 1.314·h_min + 1.251	0.89	19 yrs	h_min dominant ($p < 0.001$); negative P_winter reflects multicollinearity

Table 8. *Peak flood transfer functions fitted at hydrological-year resolution (Section 3.6.2). P_winter = total October–March rainfall (mm); h_min = preceding summer water table minimum (m below surface). (11_forecast_02_table6_winter_transfer.csv)*

4.7.3 Summer Drought Transfer Functions

The summer drought transfer functions performed substantially better than the winter peak functions (Section 3.6.2). Both predictors — antecedent winter peak (h_max_winter) and cumulative summer precipitation (P_summer) — were significant in all three blocks (all $p < 0.05$). Model fit was high for the Western Block ($R^2 = 0.774$) and Forest ($R^2 = 0.921$), and moderate for the Eastern Block ($R^2 = 0.411$). These functions provide a reliable basis for

forecasting summer drought severity from the preceding winter state and are the more operationally useful of the two transfer function pairs (Table 9).

Block	Equation	R^2	n
Western Block	h_min = 0.00096·P_summer + 0.577·h_max_winter − 1.214	0.774	20 yrs
Eastern Block	h_min = 0.00093·P_summer + 0.333·h_max_winter − 1.183	0.411	20 yrs
Forest	h_min = 0.00079·P_summer + 0.717·h_max_winter − 1.180	0.921	19 yrs

Table 9. Summer drought transfer functions (Section 3.6.2). P_summer = total April–September rainfall (mm); h_max_winter = preceding winter water table peak (m below surface). All coefficients significant at $p < 0.05$. (11_forecast_03_table7_summer_transfer.csv)

4.7.4 Spatial Distribution of P_flood and Flooding Status Across the Well Network

Applying the cluster-level P_flood equations to each well using its observed mean summer minimum as the antecedent condition produces a network-wide picture of flooding potential (Figures 22, 23, 24, 25).

The summer minimum depth map (Figure 26) confirms that the Forest cluster (C4) occupies the most ecologically stressed positions across the network, with mean summer minima consistently below 0.98 m despite sitting at the highest topographic elevation. This apparent contradiction — the deepest water table on the highest ground — is the spatial expression of the canopy interception signal identified in Section 5.4: canopy evapotranspiration suppresses net recharge, driving deeper seasonal drawdown than topographic position alone would predict. All C4 wells fall within the SD16 or SD16-recovery zone. By contrast, the majority of C1 Eastern Block wells maintain mean summer minima within or close to the SD15b wet slack zone, reflecting the high-frequency recharge dynamics of the shallow till substrate. Twenty-nine of the 72 monitored wells have mean summer minima deeper than 0.98 m — the SD16 dry slack threshold — of which 16 are below the SD16-recovery threshold of 1.20 m. These are the wells for which slack flooding under average climatic conditions requires rainfall substantially exceeding the mean winter total of 398 mm.

The P_flood map (Figure 28) quantifies the rainfall demand implied by these summer minima. Wells where computed P_flood exceeds 398 mm — the mean annual winter rainfall — are structurally unable to reach the slack floor under average climate conditions without an exceptionally wet winter. These wells are concentrated in the Western Block (C3) and Forest (C4), where the combination of deep summer drawdown and low recharge sensitivity (β_1 = 0.0012 and 0.0010 respectively, the lowest in the network) creates the greatest disconnect between available winter rainfall and the water level recovery required for flooding.

The winter maximum depth map (Figure 27) shows that Eastern Block wells achieve the shallowest winter maxima, consistent with the flashy, fill-and-spill regime of the shallow till aquifer. Western Block and Forest wells rarely approach the 0 m threshold required by SD15b communities. The flooding frequency map (Figure 29) gives the most ecologically direct summary: C1 wells show the highest flooding frequency across the monitoring period; C3 wells show very low frequency; C4 Forest wells show essentially zero, with the water table failing to reach the ground surface in any monitored year.

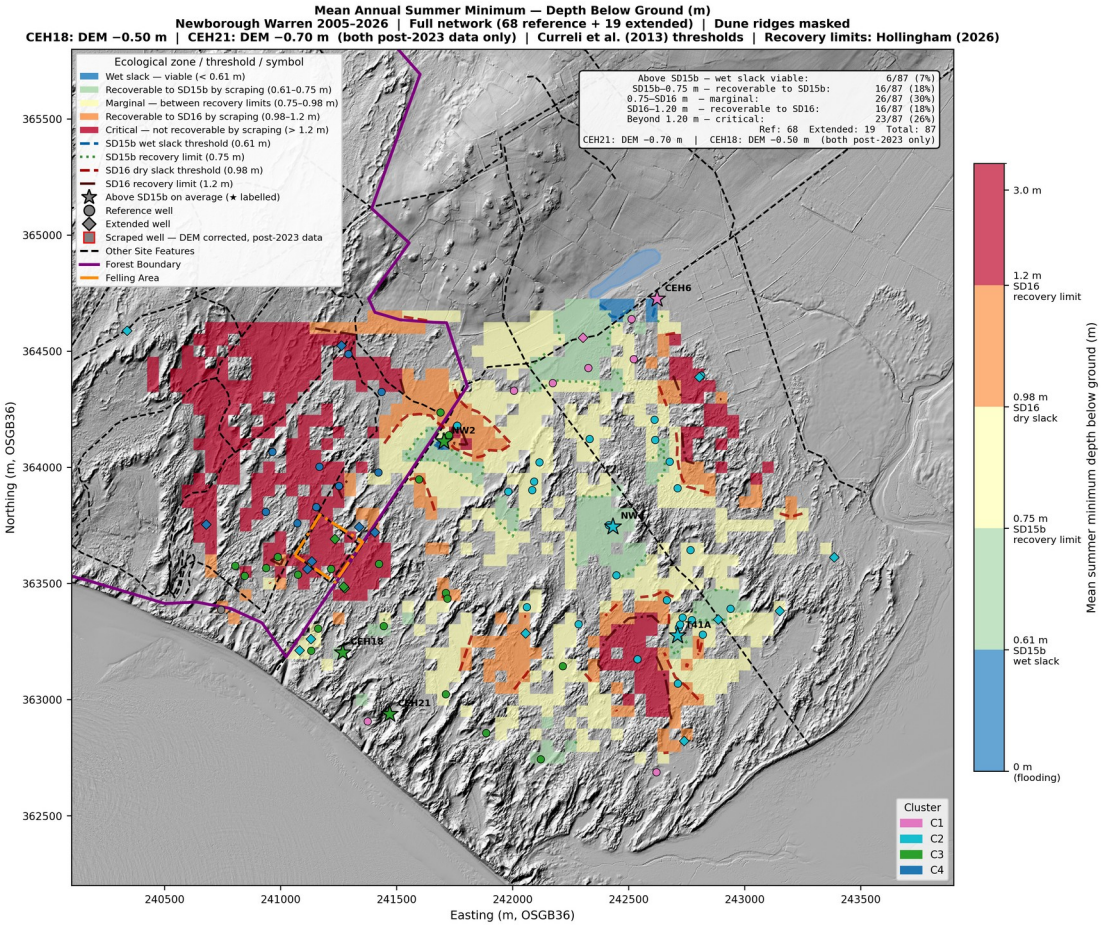


Figure 26. *Spatial distribution of mean annual summer minimum water table depth below ground across the full dipwell network (68 reference wells and 19 extended wells), Newborough Warren 2005–2026. Depth is expressed as metres below the LiDAR ground surface (DEM_Ground_Elev − mean maOD), with positive values indicating the water table lies below the surface. The continuous IDW-interpolated surface is masked across dune ridges (cells where the DEM raster exceeds the interpolated well surface by more than 1 m). Ecological zone boundaries follow the eco-hydrological thresholds of Curreli et al. (2013): SD15b wet slack viability (0.61 m); SD15b recovery limit following one scraping event (0.75 m); SD16 dry slack threshold (0.98 m); SD16 recovery limit (1.20 m). Recovery limits are derived from the BACI scraping benefit of +0.143 m reported in Section 4.5. Wells CEH18 and CEH21 use post-October 2023 data only, with DEM corrections of −0.50 m and −0.70 m respectively to account for material removed during scraping. Well symbols are coloured by cluster; circles indicate reference wells, diamonds indicate extended network wells. (LiDAR data: © NRW & OS; see Section 9 for full attribution.) A ridge-masked variant of this surface is available in the Newborough Warren Hydrogeological Modelling Suite scenario viewer (Hollingham, 2026b).* ***Caution:*** *the relatively shallow summer minima in the forest-adjacent zone (notably around NW2 and CEH2) reflect the structural position of these wells at the ridge recharge boundary rather than any beneficial effect of reduced canopy cover. Plantation clearance in this zone would be expected to deepen rather than raise summer minima, by eliminating the canopy's sacrificial evaporative shielding and increasing atmospheric draw (β_2) at the water table.*

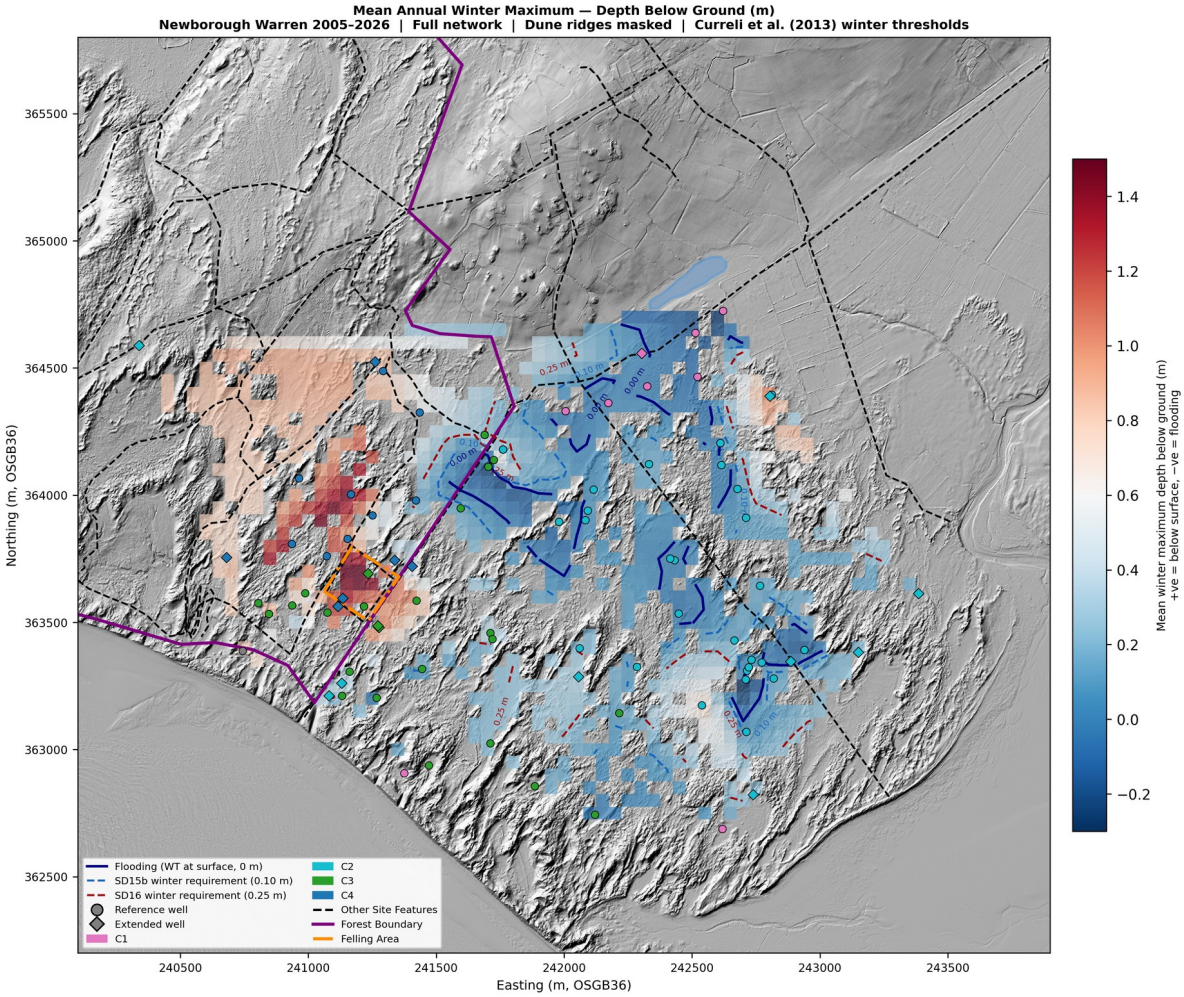


***Figure 27**. Spatial distribution of mean annual winter maximum water table depth below ground across the full dipwell network, Newborough Warren 2005–2026. Winter maximum is defined as the highest monthly maOD value in October–March of each hydrological year (October start); the mean is taken across all years with at least one October–March reading. Depth is expressed using the same DEM-referenced convention as Figure 26. The IDW-interpolated surface is masked across dune ridges. Contours mark the eco-hydrological thresholds of Curreli et al. (2013) for winter water table requirements: 0 m (ground surface — required for SD15b wet slack communities); 0.10 m below ground (minimum winter maximum for SD15b viability); 0.25 m below ground (SD16 dry slack winter threshold). The spatial contrast between the C1 Eastern Block — where winter maxima frequently reach or approach the surface — and the C3 Western Block and C4 Forest, where winter maxima rarely approach 0 m, reflects the contrasting aquifer storage architectures identified in Section 4.2. (LiDAR data: © NRW & OS; see Section 9 for full attribution.)*

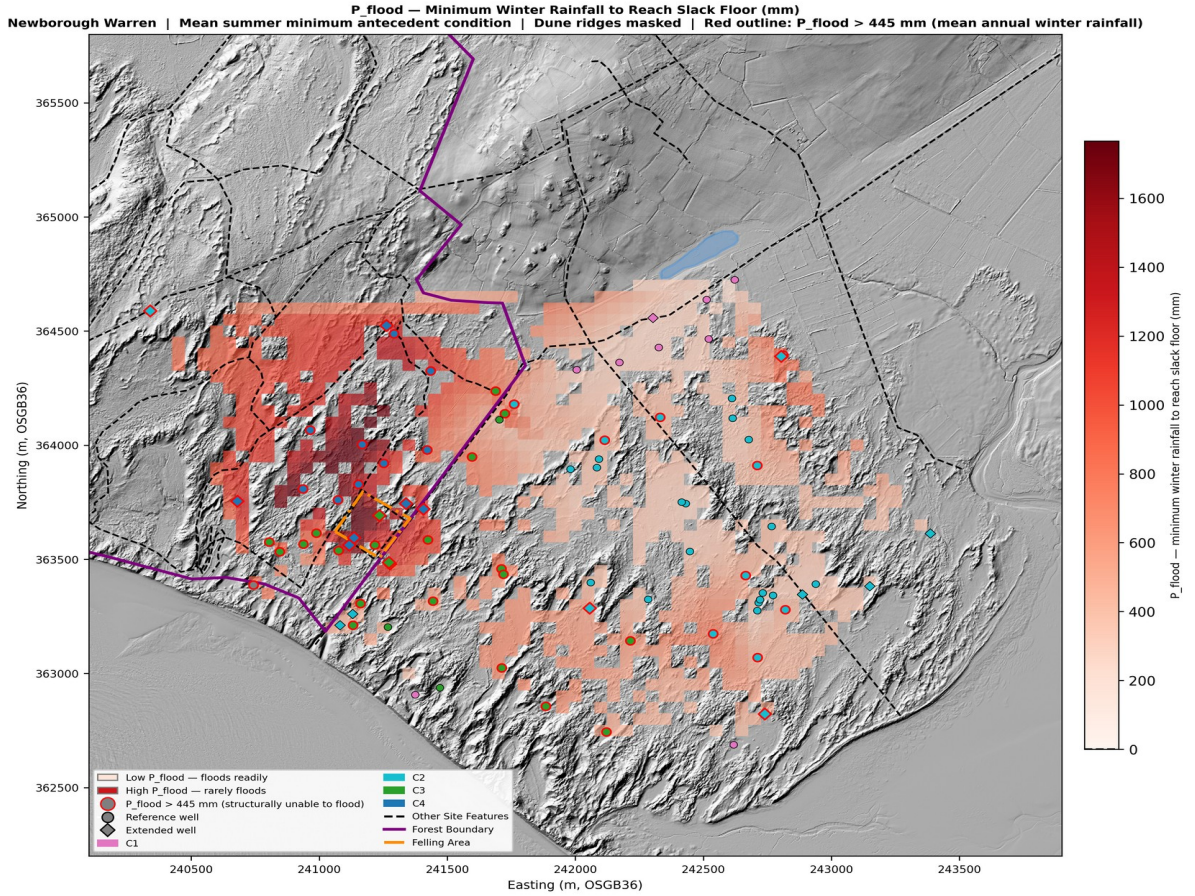


Figure 28. *Spatial distribution of P_flood — the minimum cumulative winter rainfall (mm) required to raise the water table from its observed mean summer minimum to the local slack floor — across the full dipwell network, Newborough Warren. P_flood is computed at each well by applying the cluster-level threshold equations (Table 10) using the observed mean summer minimum as the antecedent condition: P_flood = (h_gap + β_3 · (−h_prev)) / β_1 = h_gap · (1 − β_3) / β_1, where h_gap is the depth from mean summer minimum to the DEM ground surface and h_prev is the antecedent depth expressed in the negative sign convention of the fitted SSM (Section 3.6.3). The IDW-interpolated surface is masked across dune ridges. Wells outlined in red have computed P_flood values exceeding the mean annual winter rainfall of approximately 445 mm and are therefore structurally unable to reach the slack floor under average climate conditions regardless of seasonal rainfall distribution. These wells are concentrated in the Western Block (C3) and Forest (C4). At a typical Western Block well with a mean summer minimum of 1.2 m below the surface, P_flood ≈ 882 mm — exceeding the site mean annual rainfall of 890 mm. The dashed reference line on the colourbar indicates the 445 mm mean annual winter rainfall threshold. (LiDAR data: © NRW & OS; see Section 9 for full attribution.)*

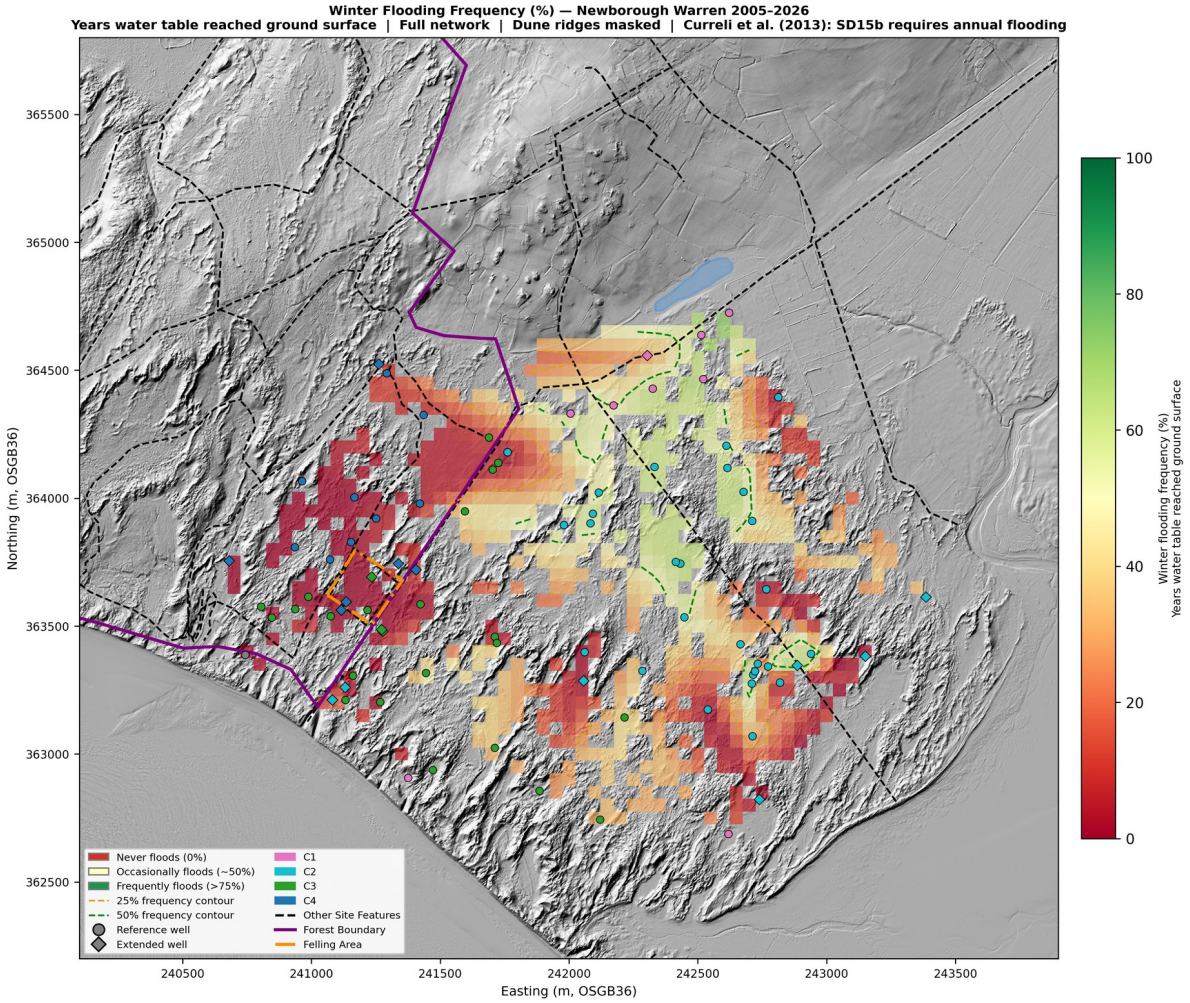


Figure 29. *Winter flooding frequency across the full dipwell network, Newborough Warren 2005–2026, expressed as the percentage of monitored hydrological years in which the winter maximum water table reached or exceeded the local ground surface (pipe top elevation used as ground surface proxy at each dipwell location). Only wells with at least five years of winter data are included. The IDW-interpolated surface is masked across dune ridges. Contours mark the 25% and 50% frequency thresholds. The C1 Eastern Block shows the highest flooding frequency, consistent with the flashy, fill-and-spill dynamics of the shallow till substrate. C3 Western Block wells show very low flooding frequency, and C4 Forest wells record essentially zero — the water table fails to reach the ground surface in any monitored year. This map provides the most ecologically direct assessment of habitat viability: the SD15b wet slack community identified by Curreli et al. (2013) requires the water table to reach the ground surface annually, a condition met only by the C1 Eastern Block under current conditions. (LiDAR data: © NRW & OS; see Section 9 for full attribution.)*

4.7.5 Critical Rainfall Threshold Equations (P_flood)

The structural difference between clusters is immediately apparent from the threshold equations (Table 10; Section 3.6.3). The storage decay coefficient $-\beta_3$ — which governs how strongly antecedent depth drives the required rainfall — declines from 0.366 (C1) through 0.200 (C2) and 0.118 (C3) to 0.078 (C4), reflecting the storage gradient identified in the LCSC analysis (Table 2; Section 4.2.2). Recharge sensitivity β_1 is similar for C1 (0.0016 m/mm) and C2 (0.0017 m/mm), and modestly lower for C3 (0.0012) and C4 (0.0010). As a

worked example, to raise the water table 0.5 m from a current depth of 1.0 m below surface at a C3 well requires P_flood = (0.5 + 0.118 × 1.0) / 0.0012 = 515 mm of winter rainfall, compared with (0.5 + 0.200 × 1.0) / 0.0017 = 412 mm at a C2 well — a difference of 103 mm for identical antecedent conditions. This difference reflects both the lower recharge sensitivity and the weaker antecedent depth dependence of the Western Block relative to the Eastern Block.

Cluster	P_flood equation	Spreadsheet formula
C1: Eastern Block Lake	P_flood = (h_gap + 0.3660·h_prev) / 0.0016	=(h_gap+(0.3660 *h_prev))/0.0016
C2: Eastern Block Mature Dune	P_flood = (h_gap + 0.2000·h_prev) / 0.0017	=(h_gap+(0.2000*h_prev))/0.0017
C3: Western Block Mature Dune	P_flood = (h_gap + 0.1179·h_prev) / 0.0012	=(h_gap+(0.1179*h_prev))/0.0012
C4: Forest	P_flood = (h_gap + 0.0776·h_prev) / 0.0010	=(h_gap+(0.0776*h_prev))/0.0010

Table 10. *Critical rainfall threshold equations derived by algebraic inversion of the cluster-level state-space equations (Section 3.6.3). h_gap = target water table rise required (m); h_prev = current water table depth below surface (m, positive value). Assumes winter PET = 0. C5 and C6 excluded as boundary-condition clusters. (11_forecast_04_table8_critical_thresholds.csv)*

4.8. Climate Trajectory and Threshold Exceedance

The current distribution of mean annual summer minima and winter maxima across the reference well network, plotted against the Curreli et al. (2013) eco-hydrological thresholds and coloured by hydrogeological cluster, is available in the Newborough Warren Hydrogeological Modelling Suite scenario viewer (Hollingham, 2026b).

4.8.1 Summer Minimum Trajectory

The three open dune clusters (C1, C2 and C3) showed statistically significant or near-significant declining summer minimum trends over the 2005–2025 period (Section 3.6.4; Figure 30, panel a). C1 declined at −0.0100 m yr⁻¹ (R^2 = 0.247, p = 0.026), C2 at −0.0115 m yr⁻¹ (R^2 = 0.172, p = 0.061) and C3 at −0.0149 m yr⁻¹ (R^2 = 0.166, p = 0.067). The Forest cluster (C4) showed a comparable rate of decline (−0.0172 m yr⁻¹) but is excluded from Figure 30 as its water table depths have no direct relevance to slack ecology. The R^2 values reflect substantial interannual variability rather than an absence of signal — the trends are physically consistent across all four clusters and span the full range of structural domains.

C1 has been below the wet slack summer viability threshold (−0.61 m) in all 21 monitored years, with the shallowest summer minimum recorded in 2016 (−0.637 m). C2 has similarly remained below the threshold throughout the record, with its shallowest minimum in 2005 (−0.640 m). C3 has never approached the wet slack threshold, ranging between −0.820 m and −1.665 m throughout the record, and falling below the dry slack threshold (−0.98 m) in 17 of 21 years. At the observed rate of decline, the C1 trend crosses the dry slack summer threshold around 2030–2032; C2 is already at or below it for much of the recent record.

The consistency of the declining summer minimum trend across all structural domains provides important evidence on causation. The Western Block (C3) and Forest (C4) clusters

decline at near-identical rates (−0.0149 m yr⁻¹ and −0.0172 m yr⁻¹ respectively) despite their fundamentally different canopy cover and LCSC values (Table 2), and the Eastern Block clusters (C1: −0.0100 m yr⁻¹, C2: −0.0115 m yr⁻¹) decline at a similar although slightly lower rate. If progressive forest evapotranspiration were the primary driver of the long-term decline, Forest cluster wells would be expected to show the steepest trajectory — but they do not. The spatial coherence of the decline across clusters that hierarchical clustering has confirmed as hydrogeologically distinct (Section 4.2) confirms that the summer minimum trend is a system-wide climate signal rather than a progressive forest-induced depression of the regional water table.

Winter maxima tell a complementary story. Linear trends fitted to annual winter maxima across all four clusters were uniformly non-significant (C1: +0.002 m yr⁻¹, p = 0.751; C2: +0.003 m yr⁻¹, p = 0.803; C3: −0.012 m yr⁻¹, p = 0.377; C4: −0.009 m yr⁻¹, p = 0.668), confirming that winter flooding capacity has not declined over the monitoring period. The Eastern Block clusters show marginally positive winter trends, consistent with the stable to slightly increasing annual rainfall totals in the RAF Valley record. The hydrological deterioration at Newborough Warren is therefore entirely concentrated in the summer season — driven by increasing evaporative demand and drier summers rather than declining winter recharge — and is consistent across all structural domains regardless of land cover.

4.8.2 Observed Winter Maximum Flooding Record

The observed winter flooding record reveals a stark spatial contrast between the eastern and western structural domains (Figure 30, panel b). The Eastern Block lake-buffer cluster (C1) exceeded the wet slack winter flooding threshold (−0.10 m) in 14 of 21 monitored years (67%), confirming that the eastern lake-influenced slacks retain substantial winter flooding capacity. The Eastern Block mature dune cluster (C2) exceeded the threshold in 11 of 21 years (52%), indicating episodic but reliable flooding in the eastern mature dune zone. The Western Block (C3) exceeded the wet slack threshold in only 1 of 21 years (5%) — occurring in 2007 — and the dry slack threshold (−0.25 m) in only 2 of 21 years, confirming that the western block fails to achieve the minimum winter flooding conditions required to maintain even dry slack communities in all but exceptional years.

4.8.3 Recent Climate Forcing and Observed Hydrological Response

The RAF Valley climate record shows a statistically significant warming trend in mean summer maximum temperature across the full 95-year record (slope +0.013°C yr⁻¹, R^2 = 0.154, p < 0.001; Section 3.6.4). The post-2013 period has run 0.87°C above the pre-2013 baseline mean of 17.66°C, with six of the twelve summers between 2014 and 2025 exceeding this baseline by more than 1°C (2014, 2018, 2021, 2022, 2023 and 2025). The most extreme anomalies occurred in 2022 (+1.62°C) and 2023 (+1.92°C). Annual rainfall shows no significant trend over the full record (p = 0.498), but 2022 recorded only 747 mm — well below the long-term mean of 856 mm (Table 1) — representing the most severe dual-stress year in the monitoring period, combining an above-average temperature anomaly with substantially below-average rainfall.

The dipwell network records the hydrological consequence of this warming directly. Network-average summer minimum water table depth declined at a statistically significant rate of −0.012 m yr⁻¹ across the 2005–2025 monitoring period (R^2 = 0.213, p = 0.035), with the post-2013 mean summer minimum (−1.113 m) lying 0.111 m deeper than the pre-2013

mean (−1.002 m). As established in Section 4.8.1, this decline is consistent across all structural domains and is not attributable to progressive forest evapotranspiration — the Western Block and Forest clusters decline at near-identical rates despite very different canopy cover, and winter maxima show no significant trend in any cluster. The deterioration is therefore a seasonal climate signal, concentrated entirely in the summer drawdown phase and consistent with the observed warming trend in summer maximum temperatures.

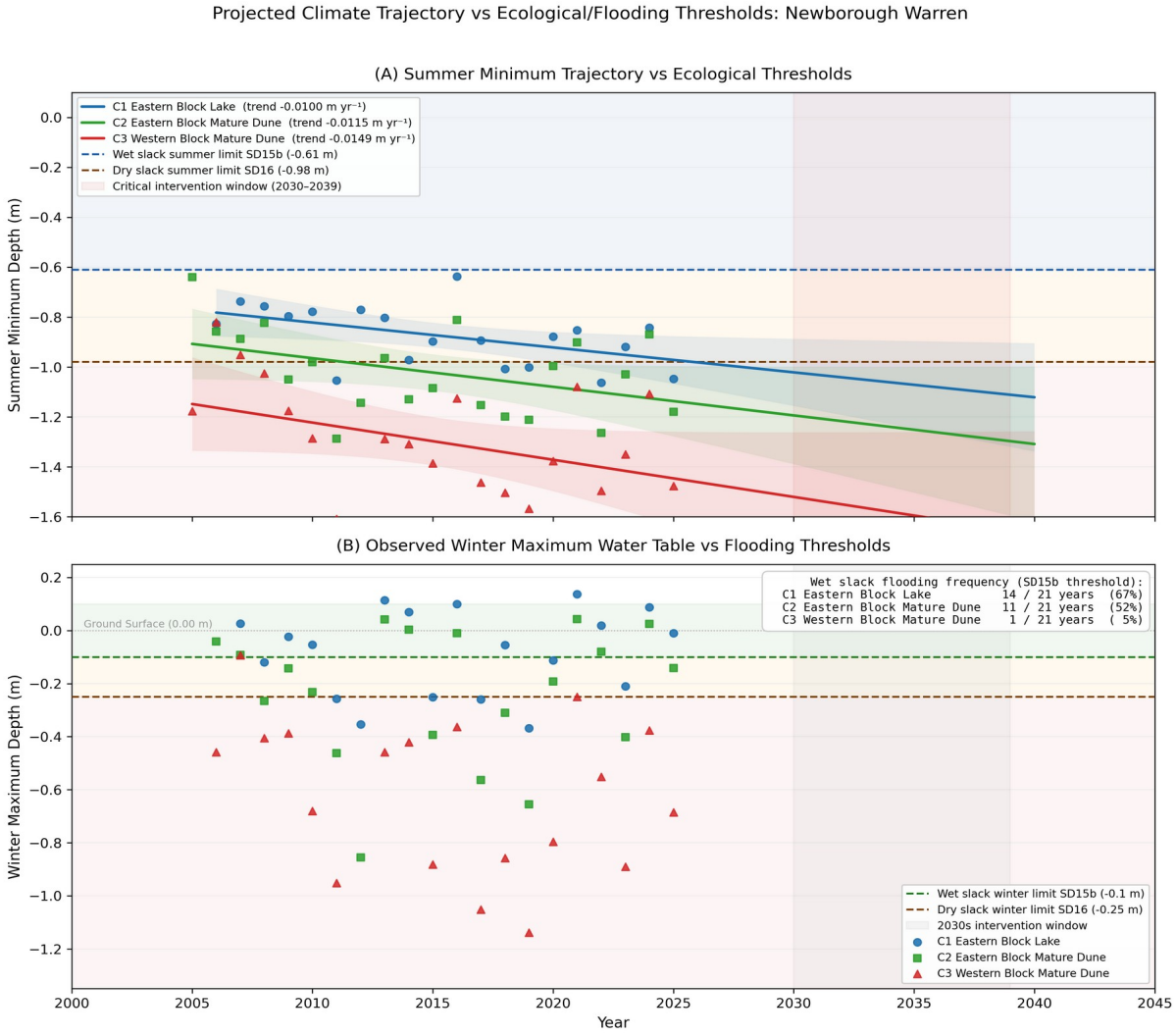


Figure 30. *Climate trajectory and threshold exceedance analysis for Newborough Warren, 2004–2025. Panel (a): observed annual summer minimum water table depth for C1 (Eastern Block Lake, blue), C2 (Eastern Block Mature Dune, green) and C3 (Western Block Mature Dune, red), with OLS linear trend lines and 95% confidence intervals extrapolated to 2040. Dashed horizontal lines indicate the wet slack summer viability limit (SD15b, −0.61 m) and dry slack viability limit (SD16, −0.98 m) from Curreli et al. (2013). The shaded critical intervention window (2030–2039) indicates the period within which C1 and C2 are projected to cross the dry slack threshold under their observed trends. Panel (b): observed annual winter maximum water table depth for the same three clusters, plotted against the wet slack winter flooding threshold (SD15b, −0.10 m) and dry slack winter flooding threshold (SD16, −0.25 m). No projection is shown for winter maxima; the exceedance statistics box reports the proportion of hydrological years in which each cluster exceeded the wet slack flooding threshold over the full monitoring period. (14_climate_trajectory_stacked.png)*

4.9. Spatial Groundwater Analysis

A steady-state 2D finite-difference model was developed to produce spatially explicit head surfaces and lateral flux fields across the 50 m grid covering the full dune system extent (94 × 56 cells). The model solves the Helmholtz form of the 2D steady-state groundwater flow equation (Section 3.8; Bear, 1972) with transmissivity T = K·b, where K = 6.0 m/day (Betson et al., 2002) and b is derived from the IDW thickness surface described in Section 3.8.Dirichlet boundary conditions (h = 0 m AOD) were applied at sea boundaries confirmed from the LiDAR DTM (Natural Resources Wales, 2023); Neumann no-flow conditions were applied at the bedrock ridge margin. The sparse linear system of approximately 4,200 equations was solved using the direct sparse solver scipy.sparse.linalg.spsolve (Virtanen et al., 2020). Aquifer thickness at the four borehole locations (3.65–12.8 m; Bristow, 2002; Betson et al., 2002) was used as hard constraints, with cluster-level priors applied at well locations and IDW interpolation (power = 1) used to produce the site-wide thickness surface.

The baseline steady-state solution produces a physically plausible head surface (minimum = −0.91 m AOD, maximum = 0.00 m AOD, mean = −0.63 m AOD), with the highest heads in the north-western forest zone adjacent to the ridge and a gradient descending toward the eastern and southern coastal margins consistent with the topographic configuration (Figure 31). Darcy lateral flux fields, computed as $Q = -K \cdot b \cdot \nabla h$ (Bear, 1972), confirm a predominantly south-eastward lateral flux consistent with the DEM-derived flow accumulation analysis (Conrad et al., 2015; Natural Resources Wales LiDAR DTM, 2023). The SSM water balance residual field (Figure 32) shows highest values in the Eastern Block — reflecting the structural sensitivity of C1 and C2's high β_2 and β_3 coefficients rather than the magnitude of lateral inflow directly. The Model B intercept audit (Section 4.4.2; Figure 15a) provides the more direct evidence of lateral inflow, with the highest persistent subsidies concentrated in the northern Forest margin (CEH14: α = +0.222 m/month; C4 cluster mean: +0.097 m/month). The independently computed Darcy flux field confirms a predominantly south-eastward lateral flux consistent with the topographic configuration, providing physical corroboration that the ridge-to-slack pathway identified by the intercept audit is hydraulically plausible.

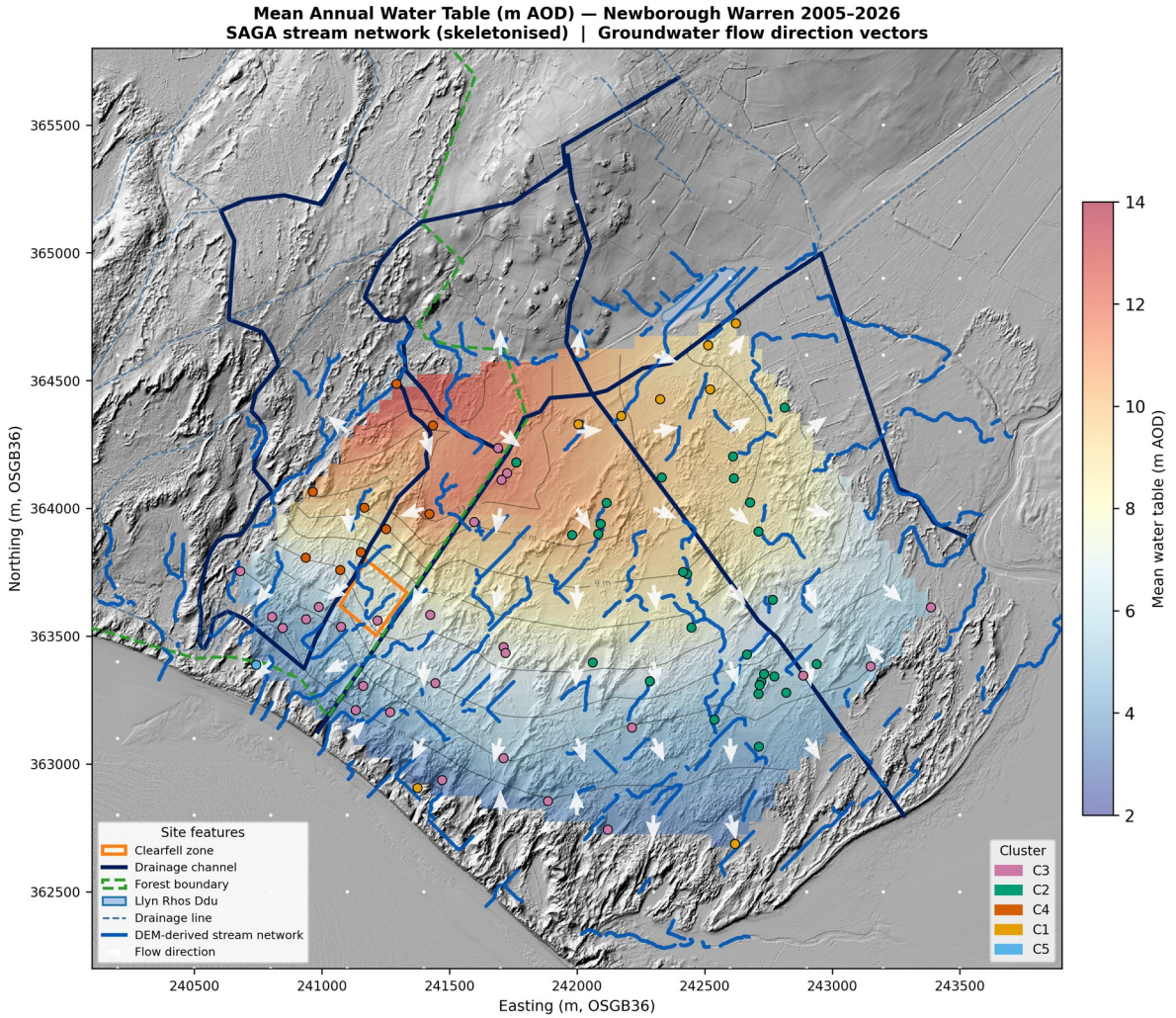


***Figure 31**. Mean annual water table elevation (m AOD) across the Newborough Warren dune aquifer, 2005–2026, derived by IDW interpolation of per-well mean annual heads to a 50 m grid. The highest heads occur in the north-western forest zone adjacent to the bedrock ridge, with a gradient descending toward the eastern and southern coastal margins. Normalised groundwater flow direction vectors (white arrows; direction only, magnitude not scaled) are derived from the head gradient field $Q = -K \cdot b \cdot \nabla h$, using $K = 6.0$ m/day (Betson et al., 2002) and the IDW aquifer thickness surface. The SAGA-derived stream network (blue lines; skeletonised from 9,847 stream cells filtered to 1–18 m AOD) provides independent topographic context. Well symbols are coloured by cluster. Background: LiDAR greyscale hillshade (LiDAR data: © NRW & OS; see Section 9 for full attribution). Generated by 20_spatial_figures.py.*

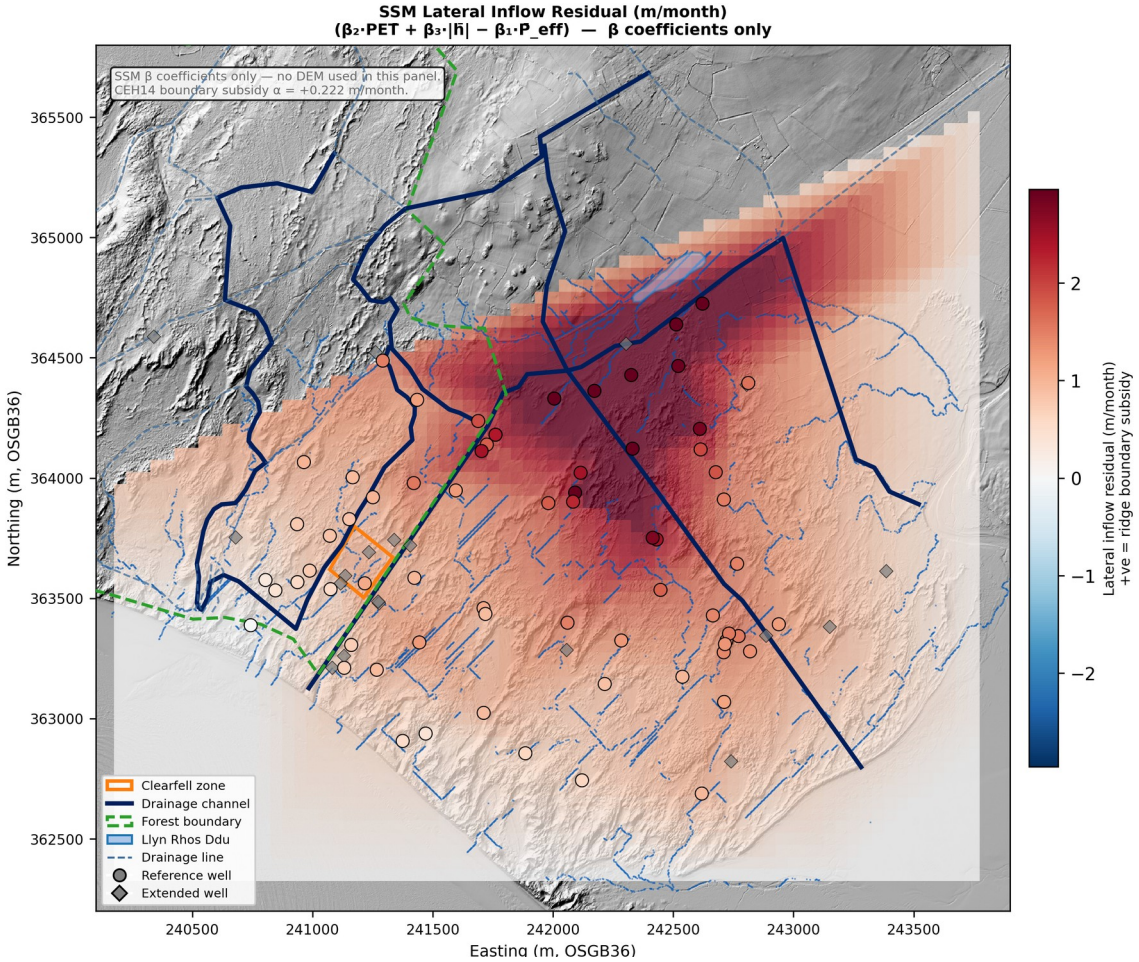


Figure 32. *SSM lateral inflow residual across the Newborough Warren dune aquifer, 2005–2026. The residual at each reference well is computed from β coefficients only as ($\beta_2 \cdot PE\bar{T} + \beta_3 \cdot |\bar{h}|$) − $\beta_1 \cdot \bar{P}_eff$ and IDW-interpolated to a 50 m grid with sea boundary anchors at zero. Positive values (red) indicate locations where the water balance requires external lateral inflow to close; negative values (blue) indicate locally drainage-dominated conditions. Normalised Darcy flow direction vectors (white arrows) derived from the mean head gradient field are shown for spatial context and are computed independently of the β coefficients. The concentration of positive residual in the north-western sector reflects the structural sensitivity of C1 and C2's high β_2 and β_3 coefficients rather than the magnitude of lateral inflow directly — for direct per-well estimates of persistent lateral flux refer to the Model B intercept audit (Section 4.4.2; Figure 15a). Well symbols coloured by residual value. Background: LiDAR greyscale hillshade. Generated by 20_spatial_figures.py (output: 20_residual_ssm.png).Note: the spatial pattern of residuals reflects the structural sensitivity of each cluster's β coefficients and mean head position as much as the magnitude of actual lateral inflow. For direct per-well estimates of persistent lateral flux, refer to the Model B intercept audit (Figure 15a; Section 4.4.2).*

Scenario difference maps were computed as Δh = h(scenario) − h(baseline) for five scenarios: full clearfell, forest thinning, broadleaf conversion, climate-dry (ΔP −10%, ΔPET +10%), and climate-wet (ΔP +10%, ΔPET −5%). Climate scenarios used the PDE solver applied to modified P and PET fields across all wells simultaneously. Forest management scenarios used an analytical spatial shift derived from the BACI-observed displacement (Section 4.6.1) applied to the C4 zone, with distance-decay propagation to adjacent C3 wells (exponential decay, e-folding distance 500 m).

Climate scenario difference maps are spatially coherent and physically plausible: the climate-dry scenario produces a mean Δh of −0.159 m across the active domain, and the climate-wet scenario produces +0.117 m, with larger responses in the western interior consistent with the lower β_3 drainage feedback in that zone. Forest management scenario difference maps produced near-zero site-wide differences when computed via the PDE solver (mean Δh = −0.003 to −0.008 m for clearfell and thinning). This is not a numerical failure but reflects three structural limitations of the spatial model in its current configuration: first, the β coefficient changes for forest management scenarios affect only the nine C4 wells, which when IDW-interpolated to 4,235 grid cells are heavily diluted by the surrounding clusters; second, the weak drainage feedback in C4 (β_3 = 0.078, the lowest in the network) means a moderate source term change produces only a small equilibrium head change; and third, the ridge boundary subsidy — the dominant control on C4 absolute head — is not parameterised in the PDE source terms. For these reasons, the PDE solver was found to be appropriate for climate scenarios but not for forest management scenarios under the current dataset. Forest management scenario differences were therefore derived analytically from the BACI-observed displacement rather than from the spatial model. The limitations of both approaches and the field measurements required to support a fully calibrated spatial model are discussed in Section 5.6.

5. Discussion

5.1. Linking Groundwater Behaviour to Geology and Land Cover

The hierarchical clustering solution (Section 3.2; Section 4.2, Figures 5–7) successfully identifies six spatially coherent and physically interpretable hydrogeological units that collectively explain the principal axes of groundwater variability at Newborough Warren.

The fundamental binary divide between the Eastern Block (C1, C2) and Western Block (C3) reflects a well-documented contrast in sub-surface stratigraphy. Shallow till and estuarine deposits in the east constrain storage capacity and drive rapid, flashy water table responses. Conversely, deep sand accumulation in the west supports a capacious, heavily buffered aquifer (Stratford et al., 2007; Grootjans et al., 2004). The LCSC values detailed in Table 2 quantify this physical contrast precisely: the Eastern Block exhibits LCSC values of 63.9% (C1) and 59.7% (C2), whereas the Western Block achieves 83.3% (C3) and the Forest cluster 104.5% (C4). This translates a qualitative geological description into a management-relevant metric expressing the exact precipitation depth required to achieve a unit rise in the water table — 1 mm of rain raises the Eastern Block mature dune (C2) water table by 1.674 mm, the Western Block (C3) by 1.201 mm, and the Forest cluster (C4) by only 0.957 mm, the last reflecting recharge interception by the canopy rather than storage capacity alone.

Crucially, as mapped in Figure 11, the plantation-covered Forest cluster (C4) shares the identical deep sandy substrate of the Western Block, yet segregates into its own distinct hydrogeological unit — indicating a structural override of the underlying geology by land cover. Notably, the Forest cluster also occupies the highest mean water table elevation in the network (9.52 m AOD, compared with 7.85–7.86 m AOD in the Eastern Block and 6.09 m AOD in the Western Block), yet exhibits the weakest drainage decay coefficient in the network ($-\beta_3 = 0.078$). This apparent paradox — high topographic position but slow drainage — is not explained by hydraulic gradient but by canopy interception dampening the recharge pulse: the water table never rises sufficiently above its suppressed equilibrium to generate the strong head-dependent drainage response seen in the open dune clusters.

This physical reality is independently cross-validated by the state-space model parameters (Section 4.2.2, Table 2). The Western Block's lower recharge sensitivity ($\beta_1 = 1.201$ versus 1.565–1.674 for the Eastern Block clusters) reflects the deep unsaturated sand buffer that attenuates the rainfall-to-water-table transfer. The Eastern Block's higher β_1 and $-\beta_3$ values are consistent with a fill-and-spill mechanism in which shallow till promotes surface ponding and rapid lateral overflow rather than slow drainage through the aquifer matrix. The practical consequence of this structural contrast is directly visible in the winter flooding record: C1 exceeded the wet slack flooding threshold in 14 of 21 monitored years (67%) and C2 in 11 of 21 years (52%), while C3 achieved this only once in 21 years (5%) in 2007 — confirming that the buffered Western Block aquifer is structurally incapable of delivering reliable ecological flooding under the current climate regime (Section 4.8.2, Figure 30).

The implication for conservation management is that Eastern Block slacks respond sharply to individual rainfall events but retain little water between events — making them sensitive to both exceptional winter rainfall and prolonged summer drought in ways that the buffered Western Block is not.

The gradational nature of the aquifer between these structural units is confirmed by the Pearson affinity analysis (Section 3.3; Section 4.3). The high proportion of Fuzzy members (46 of 69) — of whom 23 also carry a Multi-Cluster Affinity flag — reflects the continuous nature of dune hydrogeology rather than an analytical weakness in the clustering algorithm (Rao and Srinivas, 2006). The spatial concentration of MCA flags along the C1/C2/C3 transitional zone and the C2/C3/C4 forest–dune boundary (Figure 11) identifies the precise areas where geological and vegetative controls overlap. Hydrogeologically, these constitute the most structurally sensitive nodes within the monitoring network — the specific locations most likely to shift cluster membership and exhibit rapid state changes in response to sustained conservation management or climatic intervention (Grootjans et al., 2004).

The persistence of Forest cluster affinity at the clearfell treatment wells (FE1, FE2, FE4 and LIS1 all $r > 0.97$ with C4; Section 4.3) admits at least three interpretations: residual suppression by the surrounding intact canopy extending beyond the felling boundary; incomplete hydrological recovery within the felled footprint itself; or, most fundamentally, that the Forest cluster signature at these wells reflects deep sandy substrate at high topographic elevation rather than canopy influence alone — in which case convergence toward open dune behaviour may not be achievable regardless of management intervention. Distinguishing between these mechanisms requires continued monitoring beyond the current record.

5.2. Boundary Conditions, Model Limitations and Forecasting

The boundary subsidy audit, which compares steady-state models with and without an intercept for every well (Model A and Model B respectively; Section 3.4.3; Section 4.4.2), reveals a spatially structured pattern of unmeasured lateral recharge with direct implications for the application of P_flood management tools. The concentration of positive intercepts along the northern forest margin and lake-buffer sector (Figure 15a) is consistent with the vegetation-mediated ridge recharge mechanism described in Section 5.1. Because this ridge-derived input is not captured by local rainfall forcing alone, Model A systematically underestimates water table levels at these wells.

The isolation of the specialist boundary clusters — C5 (Coastal) and C6 (Lake) — demonstrates that continuous tidal and lacustrine conditions impose qualitatively different hydraulic regimes at the site margins. Consequently, LCSC values and P_flood thresholds at these locations must be interpreted with caution, as standard unconfined aquifer assumptions break down at these boundaries. At the northern wells, critical rainfall thresholds derived from cluster-average equations overestimate the rainfall required to flood the slacks, creating a conservative bias where flooding may occur at lower totals than predicted. At the coastal margin (C5), suppressed recharge sensitivity and near-zero storage decay ($-\beta_3 = 0.017$, ns; Table 2) quantitatively confirm that tidal exchange rather than gravity drainage governs baseline water table recession. At the lake boundary (C6), the near-zero and non-significant atmospheric draw parameter captures the lake acting as a recharge window during periods of high evaporative demand rather than an evaporative sink. Site managers should treat the northern and coastal boundary wells as bracketing the range of model uncertainty rather than over-relying on a single cluster-average threshold.

The physical distinction between the SSM and the traditional linear model (TLM) is most directly revealed by long-range iterative simulation (Section 3.4.2; Section 4.4.1). In one-step diagnostic mode both models perform almost identically, with median R^2 of 0.81 and 0.82 respectively (Table 4), because each prediction is corrected with the observed water level from the previous month. The divergence becomes stark under iterative simulation: the TLM median Nash-Sutcliffe Efficiency (NSE) of −0.01 indicates that on average the TLM performs no better than simply predicting the long-term mean at every time step — a failure of basic physical coherence for a model intended for management planning. In contrast, the SSM's median NSE of +0.62 represents genuinely useful predictive skill sustained over a 100-month horizon without observational correction (Table 4; Figure 13). The greatest modelling gains are concentrated in the north-eastern and central Eastern Block, precisely where unmeasured boundary fluxes create systematic drift in a mass-balanced model. The explicit $-\beta_3$ term in the SSM captures storage decay memory, preventing the cumulative drift that otherwise causes TLM collapse. The Forest cluster (C4) is the notable exception: the TLM already achieves a median iterative NSE of +0.281 across all nine Forest wells and the SSM improvement is only +0.166, confirming that canopy interception smooths the recharge signal into a near-linear climate response that the TLM can handle adequately without a storage decay term.

The demonstration that winter rainfall amount is a poor predictor of peak flood conditions in the Western Block (P_winter $p = 0.818$; Table 8), whilst antecedent summer minimum depth is dominant ($p < 0.001$), is the most consequential single finding for conservation management. It means that managers cannot reliably forecast slack flooding from seasonal rainfall forecasts alone — what matters is how deep the aquifer drains each summer. The ecological stakes of this finding are clarified by the eco-hydrological thresholds established by Curreli et al. (2013), whose four-year monitoring programme showed that wet slack vegetation (SD15b) requires mean annual maximum water levels above approximately −0.10 m relative to ground surface, and dry slack communities (SD16) persist at winter maxima around −0.25 m. Summer minimum depth sets the precondition for achieving these winter maxima: a water table that has drained deeply through summer must receive proportionally greater winter recharge to recover to flooding height. The P_flood equations quantify this precondition directly (Section 3.6.3; Table 10): at a typical Western Block well with a summer minimum of 1.2 m below surface, the critical rainfall required to achieve surface flooding is approximately 882 mm — more than the entire mean annual rainfall of 890 mm. Surface flooding from such a summer minimum is therefore practically impossible under current conditions.

Under a warming climate, summer minima will deepen further and the following winter's floods will become both later and shallower regardless of winter rainfall totals. Network-average summer minima have declined at −0.012 m yr^{-1} since 2005 (Section 4.8.1, Figure 22), the post-2013 period is running 0.87°C above the pre-2013 temperature baseline, and the 2022 dual-stress event illustrates precisely the compounding conditions most threatening to wet slack communities (Section 4.8.3). Recent ecological field surveys provide direct observational validation for this rapidly closing intervention window. Investigating the population dynamics of the protected liverwort *Petalophyllum ralfsii* within six-year-old restoration scrapes at Newborough Warren, Callaghan et al. (2021) found that despite active topographical intervention, populations remained restricted due to limited water availability, and recorded a complete absence of winter flooding following the severe 2018 summer

drought. Furthermore, their finding that *P. ralfsii* survival is currently facilitated by localised soil compaction from pony trampling — which artificially prolongs surface water retention — indicates that the underlying aquifer has already dropped beneath the critical capillary zone required by early-successional slack communities. Curreli et al. (2013) projected that conditions supporting wet slack vegetation may become non-viable by the 2030s under central climate estimates. The present study's monitoring record is consistent with this trajectory, and the implication is that the intervention window may be measured in one to two decades rather than the longer timeframes typically assumed in dune restoration planning (Figure 30).

The Eastern Block, by contrast, is better characterised as near-perennial: the shallow till-floored aquifer regularly reaches or approaches the surface in wet winters and flooding is driven by within-season rainfall accumulation rather than antecedent drought depth. The Forest transfer function result ($R^2 = 0.894$, summer minimum dominant, winter rainfall non-significant $p = 0.557$; Table 8) reinforces the canopy interception interpretation: under the intact plantation, summer drought conditions were attenuated by the canopy sponge mechanism, producing a more compressed range of summer minima and consequently more predictable winter peaks. Post-felling, this buffering is lost and the Forest water table is exposed to the full severity of summer PET — a further argument for the increased seasonal amplitude documented in the BACI analysis (Section 4.6; Figure 22).

5.3. Recharge, Loss and the Boundary Subsidy: A Cluster-Level Water Balance

The cluster-level water balance decomposition (Section 3.7.2; Section 4.2.3; Table 3a, 3b) makes explicit a structural feature that is implicit throughout the state-space model results: the monthly water balance is negative in every cluster under mean climate conditions. Atmospheric draw ($\beta_2 \cdot PET$) accounts for 62–65% of total water table losses across the four principal clusters, with gravity drainage ($\beta_3 \cdot |\bar{h}|$) contributing the remaining 35–38%. Despite this broadly consistent loss structure, recharge into the water table declines systematically from the Eastern Block (C1: 0.112 m/month; C2: 0.120 m/month) through the Western Block (C3: 0.086 m/month) to the Forest (C4: 0.069 m/month), reflecting increasing storage capacity and — in C4 — the additional suppression of throughfall by canopy interception, which is captured implicitly in the lower β_1 rather than as a separate model term.

The net monthly deficit ranges from 0.227 m/month in C3 to 0.337 m/month in C1 and must be sustained by boundary fluxes that differ in character across the site. It should be noted that in open-boundary systems, lumped parameter models of this form are known to absorb unmeasured boundary fluxes into the rainfall coefficient when no intercept term is fitted (Young, 2011; Knotters and van Walsum, 1997), which means the recharge term ($\beta_1 \cdot P$) should not be read as a direct measure of local precipitation reaching the water table. The dual-model intercept audit (Section 3.4.3; Section 4.4.2) addresses this directly: the fitted intercept α of Model B provides a well-level estimate of the steady-state lateral flux operating independently of meteorological forcing (von Asmuth et al., 2002), and the spatially structured pattern of positive α values along the northern forest margin (Figure 15a) confirms that the boundary contribution is real and not an artefact of model misspecification.

The intercept audit reveals an important spatial distinction. For the Forest cluster and its immediate margin, ridge-derived lateral recharge is a primary sustaining mechanism — confirmed by the positive model intercepts concentrated in CEH14 and adjacent C4 wells (Figure 15a; C4 mean α = +0.097 m/month, the highest of any cluster). Expressed volumetrically (Table 3b), the water balance residual — the flux required to close the monthly deficit under mean conditions — ranges from 314 mm/year in C4 to 351 mm/year in C2. It is important to note that this residual reflects the structural sensitivity of each cluster's β coefficients and mean head rather than the magnitude of lateral inflow directly: C1 and C2 show large residuals primarily because of their high β_2 and β_3 values, not because they receive proportionally more ridge recharge. Local recharge from rainfall contributes only 99–173 mm/year to the water balance depending on cluster; the remainder is sustained by boundary fluxes that differ in character — persistent ridge-derived lateral inflow concentrated in C4, and a combination of topographic position and high recharge sensitivity sustaining the open dune clusters.

Under a warming climate, the dominance of atmospheric draw in the water balance means that increasing summer evaporative demand will disproportionately deepen summer minima regardless of land cover — a finding that qualifies any management strategy focused solely on recharge recovery from canopy removal, and which sets the context for the clearfell analysis in Section 5.5.

The WTF analysis (Section 4.2.4; Table 3c; Figure 9) yields a further finding of hydrogeological interest. Once corrected for canopy interception, the Forest cluster specific yield estimate (median 0.227) is indistinguishable from those of the open dune clusters (0.223–0.259), implying a broadly uniform aquifer storage coefficient across the site. This is physically consistent with the uniform aeolian dune sand substrate and confirms that the Forest cluster's distinctive β-coefficient signature — lower β_1, lower β_2, higher LCSC — is entirely attributable to canopy boundary conditions operating at the surface rather than to any difference in subsurface storage architecture. The plantation modifies how rainfall reaches the aquifer and how evaporative demand is expressed at the water table, but it does not alter the fundamental storage properties of the sand. This distinction matters for management: the hydrological functions of the plantation that this study identifies as beneficial are surface-mediated and therefore reversible through silvicultural intervention, but the aquifer's capacity to store and transmit water is a fixed geological property that management cannot change.

5.4. The Plantation as a Multi-Functional Hydrological Structure

The preceding analyses have treated the Corsican pine plantation primarily as a source of recharge suppression — a canopy sponge (C4, LCSC = 104.5%; Table 2) that intercepts approximately 24% of rainfall before it reaches the aquifer (Freeman, 2008; Gash, 1979). This framing is correct but incomplete. The state-space coefficients, boundary subsidy mapping, and BACI results together suggest that the plantation simultaneously performs several hydrological functions whose loss through clearance has consequences that extend beyond the recovery of intercepted recharge. These functions are examined in turn below, and their implications for management are considered.

The emergence of the Forest cluster (C4) as a structurally distinct unit (Section 4.2.2; Table 2), despite its immediate geographical proximity to C3, is a highly significant finding. The fact that plantation land cover can override geological substrate as the primary control on groundwater behaviour confirms that vegetation-driven recharge suppression operates at a scale comparable to the lithological controls separating the eastern and western blocks (Clarke and Sanitwong Na Ayutthaya, 2010).

The LCSC differential between the mature dune (C3: 83.3%) and the plantation (C4: 104.5%) — a margin of 21 percentage points — reflects the combined effect of canopy interception and root-zone transpiration, both of which reduce the rainfall flux reaching the water table. Direct canopy interception measurements at the same site reported gross interception of approximately 24% of precipitation at 15% canopy cover (Freeman, 2008), and recent numerical modelling of comparable shallow sandy outwash aquifers under sparse pine canopy found similar losses of approximately 27% (Gumuła-Kawęcka et al., 2021). The broader conifer plantation literature reports interception losses of 25–45% of annual rainfall (Rutter et al., 1971; Gash, 1979), though these figures derive predominantly from dense Sitka spruce stands in UK upland catchments where losses converge on 35–40% relative to moorland (Calder and Newson, 1979) — a very different canopy structure and climate exposure to the partially thinned Corsican pine at Newborough. Freeman's site-specific figure and the Gumuła-Kawęcka estimate are therefore the more appropriate comparators for this exposed coastal plantation. The LCSC differential of 21 percentage points is broadly consistent with these direct interception measurements, confirming that canopy interception is the dominant mechanism suppressing recharge sensitivity in the Forest cluster.

The canopy influence extends beyond simple interception loss — it also governs the timing and character of recharge delivery to the water table. Recent numerical modelling of shallow sandy aquifers under pine plantation has demonstrated that canopy density, expressed as Leaf Area Index (LAI), is the dominant control on this process (Gumuła-Kawęcka et al., 2021). Under a sparse to moderately thinned canopy (LAI ≈ 2.10), flux reaching the water table is attenuated and delivered gradually over longer time periods, preventing winter rainfall from passing rapidly through the highly transmissive dune sand and instead producing a prolonged, smoothed lateral discharge. Beneath a dense commercial canopy (LAI ≈ 5.47), this buffering mechanism breaks down: chronic interception starves the aquifer and the limited water that does penetrate arrives as flashy, isolated peak fluxes rather than sustained baseflow.

At the ridge margin of Newborough Warren, the plantation canopy — largely intact in the northern block but variably thinned across its extent — likely operates in the intermediate regime, acting as a temporal release valve that smooths and prolongs the lateral delivery of ridge-derived recharge into the down-gradient dune system. This provides a physical mechanism consistent with the persistent positive model intercepts concentrated in the northern forest and lake-buffer sector (Section 4.4.2; Figure 15a): the boundary subsidy signal is not simply a product of topographic head gradient but may also reflect vegetation-mediated attenuation of recharge timing, which would sustain a more prolonged lateral flux into the eastern dune field than topographic gradient alone would predict.

5.4.1 Forest Drawdown and the Analogy with Dune Scraping

Dune scraping lowers the elevation of the slack floor toward the water table, reducing the unsaturated zone between the surface and the capillary fringe. Where sufficient material is removed, the water table intersects or approaches the new ground surface, creating the shallow, seasonally waterlogged conditions characteristic of humid dune slacks. The net effect is a rise in relative water table position at the scraped site, confirmed by the net BACI benefit of +0.143 m at CEH36 during the pure scraping era (Section 3.5.4; Section 4.5; Figure 18), a finding independently corroborated by synthetic control analysis (−0.116 m gap; Section 4.5).

The ecological significance of this localised head change is substantial and can be precisely quantified against the eco-hydrological gradient established by Curreli et al. (2013), who monitored permanent quadrats across Newborough Warren and three comparable UK west coast dune systems. Their multivariate analysis of 245 quadrats identified maximum winter water level as the single strongest hydrological predictor of community composition (6.3% variance explained), with mean annual minimum water levels spanning only 37 cm between the wettest slack community (SD15b, mean summer minimum −0.61 m) and the driest (SD16, mean summer minimum −0.98 m), and approximately 46 cm between these communities in mean annual maximum winter water levels. Because the entire ecological gradient of dune slack vegetation is compressed into this exceptionally narrow hydrological window, small shifts in hydraulic head dictate wholesale shifts in plant communities. The +0.143 m recovery achieved through localised scraping therefore represents an engineered shift across approximately one third of both the summer minimum and winter maximum gradients required to transition a dry slack back toward a wet slack state.

The plantation produces a directionally similar depression of hydraulic head through an entirely different mechanism. Canopy interception routes approximately 24% of incoming rainfall directly to evapotranspiration (Freeman, 2008), reducing the recharge flux that would otherwise drive the water table upward. Root-zone transpiration draws additionally on the capillary fringe during summer, lowering the water table further during the period of peak atmospheric demand. The combined effect is that the water table beneath the plantation requires substantially more rainfall to achieve a given rise than the open dune clusters — reflected in the Forest cluster's LCSC of 104.5% relative to the Western Block (83.3%; Table 2), a differential of 21 percentage points broadly consistent with Freeman's site-specific interception estimate of 24%. It should be noted, however, that canopy removal does not raise the water table to open dune levels: as the clearfell BACI demonstrates, removal deepens the water table further through drainage acceleration and elimination of the sacrificial evaporative layer (Section 4.6; Table 7). This is independently corroborated at the Ainsdale dune system on the Sefton Coast, where Clarke and Sanitwong Na Ayutthaya (2010) found that pine trees evaporated 214 mm/year more than open dune vegetation, resulting in water tables 0.5–1.0 m lower beneath the plantation than beneath open dunes.

The analogy between the two mechanisms is not exact: scraping acts immediately and locally on a defined area, while the plantation sets the absolute level of the water table beneath it through a sustained recharge deficit. However, as demonstrated in Section 4.8.1 (Figure 30), the long-term declining trend in summer minima is consistent across all structural domains including those most distant from the forest boundary, with all clusters declining at similar rates despite very different canopy cover, and with winter maxima

showing no significant trend in any cluster. The forest therefore sets the absolute depth of the water table beneath it but is not driving the network-wide trajectory — that signal is wholly climatic, concentrated in the summer season and consistent with increasing evaporative demand rather than declining winter recharge. Both scraping and plantation management nonetheless represent interventions that can be calibrated against the Curreli et al. (2013) community thresholds, and the spatial positioning of the plantation boundary and canopy density constitute hydrological design variables in the same sense that the location and depth of a scraping event do. This framing has not previously been applied to dune plantation management at Newborough Warren.

5.4.2 The β_2 Evapotranspiration Coefficient and Protection of Summer Minima

A result that initially appears counter-intuitive in the cluster-level mechanistic equations (Table 2; Section 4.2.2) is that the Forest cluster has the lowest atmospheric draw coefficient in the network (β_2 = 3.414), lower than both the Eastern Block (C1: 4.978, C2: 4.376) and the Western Block (β_2 = 3.592). Given the large transpiring leaf area of a mature conifer plantation, one might expect the forest to show the strongest atmospheric draw at the water table. The opposite is observed.

The explanation lies in where evapotranspiration is occurring. The pine canopy and its bramble and scrub understorey together intercept a substantial fraction of incoming rainfall and satisfy a significant portion of atmospheric demand at the surface, before that demand can propagate downward to the water table. The canopy functions as a sacrificial evaporative layer: by consuming water at the surface, it partially shields the water table from the full PET forcing that open, unvegetated, or short-vegetation surfaces transmit directly to the shallow aquifer. The water table beneath the plantation is therefore less responsive to short-term variations in atmospheric evaporative demand than water tables in the open dune system — captured precisely by the lower β_2. This interpretation is directly supported by the post-clearfell SSM coefficient analysis (Section 4.6.5; Table 7): following removal of the canopy, β_2 increased at all 13 paired wells, with zone mean gains of +0.759 (core impact), +0.674 (edge) and +0.577 (regional control), confirming that the water table became directly exposed to atmospheric evaporative demand once the sacrificial layer was removed. The Forest cluster's anomalously low β_2 is therefore not a modelling artefact but a physically meaningful signature of canopy-mediated decoupling between the water table and the atmosphere.

A complementary modelling experiment provides independent corroboration of this interpretation (Section 3.7.1; Section 4.2.2). An extended form of the state-space model was tested in which the β_2 coefficient was made depth-dependent, declining exponentially with water table depth according to $\beta_2(d) = \beta_2 \cdot \exp(-\lambda d)$, where d is the depth of the water table below ground surface and λ is a decay parameter controlling the rate at which evapotranspiration influence diminishes as the capillary fringe retreats from the root zone. A grid search over λ for each cluster, selecting the value that maximised iterative simulation NSE, produced modest improvements for the Western Block open dune cluster (C3: ΔNSE = +0.037 at λ = 0.45 m^{-1}) and a negligible gain for the Eastern Block lake-buffer cluster (C1: ΔNSE = +0.009 at λ = 0.45 m^{-1}); the consistency of the optimal decay length across open dune clusters ($1/\lambda \approx 2.2$ m) is broadly consistent with estimated capillary fringe depths in fine dune sand, though the modest magnitude of these improvements warrants caution before

treating depth-dependency as a confirmed process. For the Forest cluster, however, the optimal λ was zero — the standard fixed-β_2 formulation was not improved upon by any degree of depth-dependent decay. Since the Forest water table occupies the deepest position in the network, the depth-dependency mechanism should, if anything, be most pronounced there. Its absence confirms that the Forest's anomalously low β_2 cannot be attributed to capillary disconnection at depth, and instead reflects genuine canopy interception operating at the surface — precisely the sacrificial evaporative layer mechanism established above.

Because summer minimum water table depth (h_min) is the dominant control on winter flooding severity in the Western Block ($R^2 = 0.774$, $p < 0.001$ for h_min; P_winter non-significant, $p = 0.818$; Table 8; Section 4.7), the β_2 finding connects the plantation's canopy structure directly to the ecological outcome of most concern: a lower β_2 under managed canopy means shallower summer minima, and shallower summer minima mean a substantially lower P_flood requirement the following winter. Under a warming climate with increasing summer PET, the buffering function represented by a low β_2 acquires increasing management significance. Complete canopy removal optimises winter recharge at the cost of deepening summer minima; maintaining canopy at intermediate density may preserve the β_2 buffer whilst partially recovering the recharge penalty. Integrating Freeman's (2008) local baseline with recent numerical models (Gumuła-Kawęcka et al., 2021) suggests this functional balance is achieved under a sparse canopy regime (15–20% cover, LAI ≈ 2.10) — sufficient to maintain the sacrificial evaporative layer and attenuate extreme winter recharge spikes without chronically starving the aquifer.

5.4.3 The Plantation as a Ridge Recharge Buffer and Passive Hydraulic Barrier

The boundary subsidy mapping (Section 4.4.2; Figure 15) identifies the north-western forest margin as the zone of highest positive model intercepts in the network, with CEH14 showing $\alpha = +0.222$ m/month at the forest margin below the rock ridge crest, and Forest cluster wells CEH34 ($\alpha = +0.142$), CEH13 ($\alpha = +0.112$) and CEH2 ($\alpha = +0.112$) also carrying substantial positive intercepts consistent with their position in the principal lateral recharge pathway descending from the ridge. The C4 cluster mean intercept (+0.097 m/month) is the highest of any cluster, confirming that the Forest zone — not the open dune clusters — is the primary recipient of persistent ridge-derived lateral inflow. The NSE effect map (Figure 15b) confirms that removing the intercept term degrades forecasting stability across this north-western boundary zone. The plantation's interception and transpiration losses act on this laterally flowing groundwater in the same way they act on direct vertical recharge: a fraction of the water moving off the ridge is consumed by evapotranspiration rather than transmitted onward into the open dunes, moderating the rate at which ridge-derived recharge pulses reach the broader system.

A second related function concerns the clearfell zone's position within the broader groundwater system. Geographically, the clearfell zone sits directly astride a primary lateral drainage pathway — as mapped by DEM-derived flow accumulation analysis (SAGA v2, Conrad et al., 2015; Natural Resources Wales LiDAR DTM, 2023) — that channels groundwater from the up-gradient forest and adjacent Western Block toward the south-western coastal margin. The 79.3% reduction in climate sensitivity of the impact zone documented in the ANCOVA-BACI analysis (Section 4.6.3; Figure 22) confirms that the hydrological behaviour of the impact zone changed after felling, though as established in

Section 4.6.4 this cannot be attributed specifically to the clearfell given the scraping confound and the wet 2015–16 baseline. The accompanying network-wide β_3 increases are attributed to SSM fitting artefacts of the transition between equilibria rather than genuine drainage acceleration (Section 4.6.5)."

The inter-annual variability of C4 summer minima provides additional evidence for the episodic character of ridge recharge into the Forest zone, and illustrates what has been lost through canopy removal. Pre-felling summer minimum depths show a standard deviation of 0.45 m — substantially higher than any other cluster (C1: 0.13 m; C2: 0.15 m; C3: 0.23 m) — despite summer PET varying by only 3.8% between years. This excess variability cannot be explained by local climate forcing alone and is consistent with intermittent ridge saturation events delivering episodic lateral pulses during years of exceptional antecedent rainfall. Monthly totals of 190 mm at RAF Valley in October 2005 and 165 mm in October 2008 are consistent with ridge saturation events capable of driving non-linear lateral discharge into the adjacent dune aquifer; the ridge well CEH12 (ground elevation 34.9 m AOD) recorded water levels of −1.3 to −1.5 m from pipe top during 2006–2007, consistent with a charged ridge aquifer following these events. In good years the pre-felling C4 distribution approached the SD16 threshold (0.98 m) — a condition entirely absent post-felling. Post-felling, C4 summer minimum variability narrows (SD = 0.31 m) but the mean deepens by 0.145 m annually and 0.218 m in summer specifically — the good years have been removed without the bad years improving proportionally. This interpretation is appropriately hedged: the CEH12 record spans only 2006–2010 with substantial gaps and cannot support a formally parameterised ridge recharge term; the mechanism remains a plausible inference supported by circumstantial evidence rather than a demonstrated finding.

A practical risk associated with the spatial mapping of summer minimum depths (Figure 26) warrants explicit discussion before drawing management implications. The relatively shallow summer minima in the forest-adjacent zone — notably around NW2 and CEH2, which sit at the northern forest margin where the ridge recharge signal is strongest — may give the impression to a site manager that clearance in this area would expose a zone of near-viable dry dune slack habitat. This interpretation would be incorrect, and the hydrological consequences of acting on it would be the opposite of the intended outcome.

The shallow summer minima at NW2 and CEH2 are not a consequence of reduced canopy cover but a reflection of their structural position at the ridge-to-dune recharge boundary. These wells receive the highest persistent lateral subsidy in the network — CEH14, immediately adjacent, carries α = +0.222 m/month, and the C4 cluster mean of α = +0.097 m/month is the highest of any cluster. It is this ridge-derived inflow, not any property of the canopy, that sustains their relatively favourable water table position. Clearance of the plantation in this zone would eliminate the canopy's sacrificial evaporative shielding, increasing β_2 at the water table and deepening rather than raising summer minima.". The BACI analysis (Section 4.6.3) confirms a climate-corrected upper bound displacement of −0.093 m in the core impact zone directly down-gradient of this area; seasonal analysis shows summer deepening proportionally larger than winter improvement regardless of baseline choice (Section 5.5).

The summer minimum depth map should therefore be read as identifying where the aquifer is most structurally resilient — not as a target for intervention. The forest-adjacent margin around NW2 and CEH2 represents the zone in the network where ridge-derived lateral

inflow most actively sustains water table levels against the background climate drying trend. Protecting that recharge pathway, and the canopy conditions that moderate atmospheric draw on the water table, is a higher conservation priority than exposing it through clearance.

5.4.4 Implications, Broadleaf Conversion and the NW10 Natural Experiment

The two functions described above — head depression analogous to scraping, and β_2-mediated protection of summer minima — operate at different spatial scales and timescales but converge on a common implication: the hydrological consequences of canopy removal are not limited to the recovery of intercepted recharge and cannot be assessed from recharge gains alone.

This does not constitute an argument for retaining the plantation in its current extent or configuration. The chronic recharge suppression associated with a dense, mature canopy (LCSC = 104.5%; Table 2) has been cited as a constraint on aquifer recovery (Stratford et al., 2007; Clarke and Sanitwong Na Ayutthaya, 2010), and the case for targeted clearance paired with complementary interventions has been made on that basis. However, as the water balance decomposition (Section 5.3; Table 3a, 3b) and clearfell BACI (Section 4.6; Table 7) together demonstrate, recharge suppression is not the binding constraint — the system-wide deficit is structural and independent of canopy cover, and clearance has not demonstrably improved the net water balance — the β_2 increase eliminates the sacrificial evaporative layer and the monitoring data do not support the recovery of summer water tables that the intervention was intended to achieve (Section 4.6.4; Section 5.5).

Broadleaf conversion — observational evidence from the 1993 block. The monitoring network contains a direct observational test of broadleaf conversion effects. A block of mixed broadleaf woodland was established following clearfell in 1993 and restocked in 1996 on the north-western margin of the plantation, with dipwell NW10 situated within the block and monitored continuously from 2007. CEH2 and CEH34, both classified as C4 Forest cluster under intact Corsican pine canopy, serve as the principal comparators; NW11 and NW13 provide a paired pine-warren boundary transect approximately 90 m apart at virtually identical elevation.

The state-space β coefficients at NW10 are instructive. Its recharge sensitivity $\beta_1 = 1.532$ substantially exceeds the C4 cluster median of 0.882 and the adjacent Corsican pine well CEH2 ($\beta_1 = 0.939$), confirming that the broadleaf canopy intercepts less rainfall than dense pine and allows more recharge to reach the water table per millimetre of rainfall. This is consistent with the lower interception fraction expected from a deciduous canopy (~20%) relative to Corsican pine (~24%; Freeman, 2008), though the magnitude of the contrast is larger than interception alone would predict, suggesting differences in canopy structure and throughfall pathways also contribute. Critically, however, the atmospheric draw coefficient $\beta_2 = 4.119$ at NW10 is essentially identical to CEH2 ($\beta_2 = 4.222$) — a difference of less than 3%. The feared increase in summer evapotranspiration draw under broadleaf is not observed in the 18-year SSM record. This is consistent with the seasonal offset of deciduous leaf-off during October–March partially compensating for higher growing-season ET demand, such that the year-round integrated atmospheric draw at the water table is no greater than under adjacent pine.

Despite the similar β_2 values, the summer minimum water table at NW10 is consistently deeper than all pine comparators over the 18-year shared record, as detailed below. After

correcting for upstand height using the LiDAR DEM (Pipe_Top_Elev − DEM_Ground_Elev), NW10 records a mean September depth of −1.289 m below ground surface, compared with −0.887 m at CEH2 (Corsican pine, DEM = 12.03 m) and −1.185 m at CEH34 (Corsican pine, DEM = 9.80 m). NW10 is deeper than CEH2 in all 18 years of common record, and deeper than CEH34 in 14 of 15 years. The CEH34 comparison is particularly informative because CEH34 sits 2.0 m lower on the ground than NW10 yet consistently records a shallower summer minimum — a result that cannot be attributed to topographic position or boundary subsidy, since CEH34 also carries a higher boundary subsidy (α = +0.142 m/month) than NW10 (α = +0.072 m/month).

A spatial transect extending east from the forest block to the open warren reinforces this picture. The paired wells NW11 (pine, forest side of boundary, DEM = 12.10 m) and NW13 (open warren, DEM = 12.09 m) are separated by approximately 90 m at virtually identical elevation. The pine side (NW11, mean summer minimum −1.071 m) is consistently shallower than the adjacent warren side (NW13, −1.149 m) by a mean of 0.078 m, independently corroborating the β_2-mediated shielding function of the pine canopy described in Section 5.4.2. NW10 under broadleaf (−1.289 m) sits deeper than NW13 (open warren, −1.149 m) by 0.140 m — the broadleaf interior is drier in summer than even the adjacent open warren at equivalent elevation.

When each well is normalised to its mean over the shared record to remove the absolute depth differences driven by elevation and position, NW10 tracks the interannual climate signal almost identically to the adjacent pine and warren wells throughout the monitoring period. The mean anomaly of NW10 relative to the pine interior composite (CEH2 + CEH34) over the bramble-dominated phase 2010–2021 is +0.002 m — indistinguishable from zero. This confirms that during the bramble-dominated phase, when dense understorey maintained high ground-level ET demand, the broadleaf stand produced no detectable divergence from adjacent pine behaviour once the shared climate forcing is removed. From approximately 2022 the maturing tree canopy has begun to suppress understorey growth. An emerging trend of −34.1 mm/yr in the NW10 anomaly over 2019–2025 ($p = 0.062$) is consistent with the expected direction of canopy closure deepening summer minima, but does not meet conventional significance thresholds, is sensitive to a single influential value in 2019, and is treated as a monitoring hypothesis rather than a confirmed result.

Implications for management. The observational record at NW10 suggests that broadleaf conversion at this site may produce deeper summer minima than adjacent Corsican pine at comparable elevation and topographic position, across 18 years of common record. The mechanism is not an amplified atmospheric ET draw — β_2 at NW10 and CEH2 are essentially identical — but rather reflects the higher recharge sensitivity under broadleaf (β_1 = 1.532 vs 0.939): more rainfall reaching the water table during the recharge season also means a larger seasonal drawdown as that stored water drains through the transmissive dune sand during summer. The reduced boundary subsidy at NW10 relative to adjacent pine wells, despite its closer proximity to the ridge, further suggests that the broadleaf canopy is consuming a greater fraction of the lateral ridge-derived recharge flux before it can sustain the water table through the summer period. Whether this balance will shift as the canopy fully closes — and tree transpiration progressively displaces the bramble ET demand it is currently replacing — cannot yet be determined from the available record, and represents a priority question for continued monitoring through the 2026–2030 period.

The question of thinning as an alternative to either full clearfell or broadleaf conversion is not directly addressed by the monitoring data. No thinning experiment has been conducted within the study period and no thinned-stand wells exist in the network at comparable positions to NW10 or CEH2. The theoretical argument that an intermediate canopy density could preserve the β_2 buffer whilst partially recovering the recharge penalty (Section 5.4.2) is mechanistically plausible and consistent with the Dothistrōma blight trajectory, but it remains a prediction derived from the β coefficient framework rather than an observed outcome. Monitoring of β_2 responses adjacent to blight-affected stands of varying severity, as suggested in Section 5.4.5, represents the most direct available route to empirical validation of this prediction.

Any strategic retention or planting of canopy near open slacks must observe a strict spatial exclusion zone. Buffer belts must be positioned up-gradient and maintained at a minimum distance of 30 m from the perimeter of any target dune slack. This threshold is informed by the physical root architecture of Pinus species in nutrient-poor, well-drained soils, which drives extensive horizontal foraging with lateral root networks frequently extending 15 to 25 m from the stem (Stone and Kalisz, 1991), and by evidence from comparable systems that the transpirational cone of depression generated by plantation margins can propagate laterally beyond the physical root extent into adjacent dune slack water tables (Clarke and Sanitwong Na Ayutthaya, 2010; Grootjans et al., 2004). However, the clearfell BACI results of this study suggest that the net lateral hydrological influence of the Newborough plantation on adjacent water tables may be more complex than a simple transpirational cone of depression — the monitoring data indicate that canopy removal has not raised adjacent water tables, a finding that warrants further investigation before the exclusion zone distance is refined for this site.

5.4.5 *Silvicultural Management of Canopy Density as a Calibratable Hydrological Intervention*

The two hydrological functions established in Sections 5.4.1–5.4.2 — recharge suppression (LCSC = 104.5%; Table 2) and sacrificial evaporative shielding (β_2 = 3.414, the lowest in the network; Table 2) — are not fixed properties of the plantation as it currently exists but parameters that demonstrably respond to silvicultural intervention. The post-clearfell β_2 zone mean increases of +0.689 (core impact), +0.544 (edge) and +0.605 (regional control; Section 4.6.5; Table 7) confirm that canopy structure directly controls the magnitude of summer evaporative shielding at the water table.

This has a direct parallel with dune scraping. Scraping is a physical intervention that shifts the water table position relative to the ground surface by a quantifiable amount — the net BACI benefit of +0.143 m at CEH36 (Section 4.5; Figure 18) representing approximately one third of the hydrological gradient separating dry slack from wet slack communities in the Curreli et al. (2013) framework. Silvicultural management of canopy density operates on the same water balance through a different mechanism: by modulating β_2, it controls how much of the seasonal PET signal reaches the water table rather than being absorbed at the canopy surface. Both interventions are therefore calibratable against the same eco-hydrological thresholds, and both have quantifiable consequences for summer minima — the variable this study identifies as the dominant control on winter flooding severity ($R^2 = 0.774$, $p < 0.001$; Table 8; Section 4.7).

The critical constraint established by this study is that the relationship between canopy and water balance is not monotonic. Complete removal worsens summer minima by eliminating the sacrificial evaporative shielding function, as the clearfell BACI and β_2 analysis demonstrate (Section 4.6; Figures 17–19). Broadleaf conversion trades year-round interception capacity for a canopy that is absent during the winter recharge season, amplifying both hydrological stresses simultaneously. These findings point toward a different question: not whether there should be woodland in the back-dune position the plantation currently occupies, but what kind.

On prograding Atlantic shorelines, stabilised back dunes naturally succeed to woodland as part of the dune formation process — woodland is not the terminus of dune succession but a component of its natural cycling, providing the hydrological stability that underpins slack communities in the dunes in front of it. The geomorphological position of the Newborough plantation is therefore not anomalous; what is anomalous is the species and management regime. Freeman's (2008) site-specific measurement of 24–25% interception at Newborough — confirmed by Gumuła-Kawęcka et al. (2021) for comparable coastal sandy aquifers under sparse pine — sits at the low end of the 25–45% range typically reported for UK conifer plantations, precisely because of the exposed coastal setting and partial thinning. The hydrological cost of the plantation is not an inherent property of pine woodland on this site; it is a product of the closed commercial stand density imposed by forestry management.

The hydrological functions established in Sections 5.4.1–5.4.3 are properties of canopy cover rather than of any particular species — including broadleaf species, which shed their leaves in autumn precisely when winter recharge begins, removing the sacrificial evaporative layer at the moment it matters most whilst retaining full summer transpiration demand. Stand density is therefore the primary silvicultural management variable, not species composition per se. A more open canopy at reduced stand density would recover the majority of the recharge deficit whilst retaining the shielding, buffering and drainage-attenuation functions quantified above. The β-coefficient framework developed here provides a direct means of calibrating this balance: canopy density can be modulated toward a target β_2 reduction whilst maintaining sufficient crown cover to attenuate extreme winter recharge pulses and seaward drainage acceleration.

It is notable in this context that progressive *Dothistroma* needle blight infection is already driving natural canopy thinning across parts of the Newborough plantation. To the extent that this reduces stand LAI toward the 15–20% cover range identified here as functionally optimal, the disease trajectory is inadvertently moving the plantation in the hydrologically beneficial direction. Monitoring the β_2 response of wells adjacent to blight-affected stands would provide a direct field test of the canopy shielding mechanism quantified in this study, and would constitute a valuable natural experiment requiring no additional silvicultural intervention.

5.4.6 Canopy Salt Interception and Post-Felling Geochemical Consequences

A further geochemical consequence of canopy removal has received no attention in the dune restoration literature but follows directly from the LAI-deposition relationship established in forest hydrology. Pine plantation canopies intercept sea salt aerosols with substantially greater efficiency than open dune vegetation — throughfall studies in coastal pine forest (*Pinus radiata*) have demonstrated chloride deposition rates up to 89% higher beneath the

canopy than in adjacent open fields, an enhancement directly attributable to the high leaf area index of the needle canopy (Deng et al., 2013). Independent soil solution studies in coastal *Pinus radiata* plantations have confirmed large inputs of sea salts by enhanced canopy capture, contributing substantially larger solute concentrations and elemental fluxes in pine soils relative to adjacent pasture (Giddens et al., 1997). At a site exposed to frequent onshore Atlantic airflow such as Newborough Warren, this implies that the forest soils have been accumulating a chronically elevated Cl^- and Na^+ load relative to the open dune system throughout the plantation's 75-year history. Forest removal simultaneously terminates this elevated deposition input and, by increasing aquifer responsiveness following canopy removal — accelerates the downward flushing of accumulated ions through the highly transmissive dune sand. The geochemical consequence is a progressive leaching of base cations from soils that previously received sustained buffering from both canopy-concentrated deposition and the lateral groundwater flux. Given that Curreli et al. (2013) and Sival and Grootjans (1996) identify the maintenance of carbonate-rich groundwater buffering as critical for the persistence of basiphilous slack communities, this flushing dynamic represents an additional pathway by which large-scale clearance may accelerate ecological deterioration — operating on soil chemistry rather than water table depth, and potentially with a longer lag time before its effects become apparent in the vegetation. This hypothesis warrants targeted investigation through paired throughfall and soil porewater chemistry monitoring across the forest–dune boundary.

5.5. The Wrong Side of the Water Balance: Why Clearfell Cannot Restore the Western Block

Against the structural reality established in Section 5.3 — that every cluster runs a persistent monthly deficit sustained by ridge-derived lateral recharge (Table 3a, 3b) — plantation clearfell has an intuitively appealing rationale: removing the canopy eliminates approximately 24% interception loss, raising β_1 toward values more consistent with the open dune clusters and increasing the volume of rainfall reaching the water table. The results confirm that this recharge recovery effect is real: β_1 increased at 12 of 13 paired wells following the December 2017 clearfell (Section 4.6.6; Table 7), and the increase is directionally consistent, spatially coherent and physically expected from the removal of 142 mm/yr of canopy interception. Atmospheric draw (β_2) increased at all 13 paired wells, confirming that the sacrificial evaporative shielding function of the canopy has been eliminated (Section 5.4.2). These coefficient changes are the most robust findings from the intervention analysis and are not in dispute.

Before the clearfell signal can be interpreted, the hydrological legacy of the April 2015 dune scraping event must be acknowledged. CEH36 was scraped approximately 400 m seaward of the felling footprint in April 2015, 32 months before the clearfell, lowering the ground surface in the adjacent open dune slack system. Because CEH36 forms part of the BACI monitoring network, its pre-clearfell water table record reflects post-scraping conditions rather than undisturbed pre-intervention hydrology. The ANCOVA-BACI model (Section 4.6.3; Figure 22) estimated the scraping step change at −0.137 m ($p < 0.001$) — a larger individual displacement than the subsequent clearfell step of −0.093 m [−0.131, −0.055] ($p < 0.001$). The combined climate-corrected cost of −0.230 m therefore reflects the cumulative legacy of two sequential interventions, with the scraping having already substantially altered

the pre-felling baseline before a single tree was removed. This finding has important implications for the design of future management programmes: where scraping and canopy removal are planned in combination, the hydrological sequencing of interventions matters, and the compounding of their individual effects may produce cumulative drawdown substantially exceeding what either intervention would deliver alone.

The hydrological legacy of the December 2017 clearfell must also be interpreted within a longer climatic context that fundamentally qualifies earlier attributions of water table depression to the forest. The 1990s — the decade in which dipwell monitoring began and during which the forest effect was most forcefully argued (Cottingham, 1994; Bristow and Bailey, 1998; Betson et al., 2002) — was the driest decade in the 95-year RAF Valley record, with a mean annual rainfall of 793 mm against a long-run average of 853 mm (Section 4.1.1; Table 1). The studies that attributed widespread dune slack drying to forest evapotranspiration were conducted without reference to this climatic anomaly, and the apparent depression of the 1989–1995 water table baseline almost certainly reflects a compound signal of forest influence and exceptional climatic dryness that could not be separated without the kind of climate-partitioned ANCOVA framework applied in this study (Section 3.5.5).

The ANCOVA-BACI clearfell step of −0.093 m is statistically significant within its own framework, but three independent robustness analyses — SSM residual normalisation, synthetic control, and cluster transition testing — all failed to detect a clearfell-specific signal once the scraping confound and wet 2015–16 baseline were accounted for (Section 4.6.4). The spatially uniform post-felling recovery across the transect from uncleared plantation interior to clearfell core (Section 4.6.4; Figure 23) — CEH2 −0.111 m, CEH34 −0.093 m, WMC3 −0.140 m — provides the most direct visual demonstration that the ANCOVA step reflects a common climate baseline effect rather than a localised clearfell response. The −0.093 m figure should therefore be treated as an **upper bound** on any clearfell-specific hydrological effect rather than a precise causal attribution. The available monitoring data do not permit detection of a clearfell-specific signal above the noise of climate variability and the scraping confound.

Notwithstanding the difficulty of attribution, the seasonal structure of the post-felling displacement warrants attention. Measured against the full pre-clearfell period as baseline, the mean post-felling water table deepening at the core impact wells (FE2, FE4, WMC3) is strongly asymmetric: winter (November–March) mean depth increased by +0.089 m relative to the pre-felling period, while summer (June–September) mean depth increased by +0.292 m — more than three times the winter signal. These absolute figures must be treated with caution: the pre-clearfell period includes the anomalously wet 2015–16 years, which inflates the apparent post-felling summer deepening when used as the reference level. Using the post-scraping pre-felling era as baseline instead, the picture is a modest winter improvement (−0.155 m, water table shallower) and a moderate summer deepening (+0.111 m). Under either baseline the seasonal asymmetry is directionally consistent: the post-felling effect is proportionally larger in summer than in winter.

This asymmetry is physically expected from the β_2 mechanism. The canopy removal eliminates the sacrificial evaporative shielding function — the process by which the plantation canopy intercepted atmospheric ET demand before it could propagate to the water table. Without that shielding, the full summer PET forcing acts directly on the water table,

deepening summer minima. The recharge increase from reduced interception (β_1) operates year-round and is largest in winter when rainfall is highest, but provides proportionally less benefit in summer when PET dominates. The net result is a larger summer deepening than winter improvement, regardless of baseline choice. The summer deepening is concentrated precisely in the months that determine the following winter's flooding probability (Section 4.7.2; Table 8). Even if the causal attribution of the absolute magnitude remains uncertain, the β_2 physics provide a mechanistic basis for expecting that any clearfell effect would act most damagingly in the ecologically critical season.

Storage decay coefficients (β_3) increased at 13 of 14 paired wells but, as established in Section 4.6.5, these increases are attributed to SSM fitting artefacts of the transition between equilibria rather than genuine changes in aquifer drainage dynamics. The 79.3% reduction in climate sensitivity of the impact zone — derived independently from the ANCOVA interaction term — confirms that the hydrological behaviour of the impact zone changed after felling, but does not resolve whether that change reflects the clearfell, the scraping, the climate baseline shift, or a combination of all three

The critical management implication is not altered by these analytical caveats. Winter flooding in the Western Block is driven almost entirely by antecedent summer minimum depth ($R^2 = 0.684$, $p < 0.001$; Table 8), with winter rainfall amount non-significant ($p = 0.818$). Whatever the correct causal attribution of the post-felling displacement, the monitoring record shows unambiguously that the clearfell has not raised summer water tables at the core impact wells, and the β_2 physics provide a clear mechanistic reason to expect summer-dominant deepening from any canopy removal. Clearfell has therefore been acting on the wrong side of the seasonal cycle for the ecological objective it was intended to serve. The critical rainfall threshold equations reinforce this (Table 10; Section 4.7.5): at Forest cluster summer minima of approximately 1.8 m below ground, the P_flood equation yields a required winter rainfall that is physically unachievable under the current climate, confirming that the Forest water table has moved beyond the effective range of local rainfall-driven restoration regardless of canopy management.

Critically, it is the clearfell zone itself — approximately 8.4 ha, just over 1% of the plantation — whose water balance has shifted. Higher recharge within that footprint is accompanied by higher atmospheric draw on the water table and no detectable improvement in the lateral recharge budget, without reducing the fundamental deficit that must be met by the boundary subsidy (Table 3a, 3b; Section 5.3). The wider plantation and the open dune clusters that determine conservation condition remain unchanged. The spatial model result (Section 5.6) is consistent with this: no forest management scenario produces any detectable effect on C1 and C2 summer minima, confirming through an independent route that forest management in the C4 zone does not propagate hydraulically to the open dune slack clusters.

Whether this result is specific to Newborough Warren or represents a general mechanism warrants consideration. The seasonal asymmetry — proportionally larger summer deepening than winter improvement — is consistent with the β_2 increase from pioneer ground vegetation operating independently of site-specific ridge geometry. At a dune system without a bedrock ridge, clearfell might produce a different net annual result, but the summer deepening mechanism driven by the loss of canopy shielding would be expected to operate regardless of site. The Newborough finding therefore likely reflects a general caution about

clearfell as a hydrological restoration tool in any coastal dune system where summer minimum depth is the primary control on winter flooding.

The more fundamental threat is not the forest per se but the progressive redistribution of rainfall toward wetter winters and drier summers under a changing climate — a signal clearly visible in the declining summer minimum trends identified in Section 4.8.1 (Figure 30) and one that no amount of canopy removal can address. Large-scale clearance alters the amplitude of the seasonal cycle but does not change its direction of travel. The management challenge at Newborough Warren is therefore not simply hydrological restoration but hydrological adaptation — maintaining flooding capacity in an increasingly seasonal climate while preventing the accelerating summer drought trajectory from crossing the ecological viability thresholds identified by Curreli et al. (2013) before the 2030s intervention window closes (Section 4.8.1; Figure 30).

The conclusion is not that the plantation should remain untouched. The clearfell has demonstrably increased aquifer responsiveness to recharge and supports other conservation objectives including habitat diversification and access to dune slack communities currently suppressed by canopy shade. However, as a tool for restoring the Western Block water table, the monitoring data provide no evidence that it has succeeded, and the β_2 physics provide a clear mechanistic basis for expecting that its dominant hydrological effect — to the extent it has one — is to deepen summer minima rather than raise them. The primary lever for restoration is the boundary subsidy — protecting and if possible enhancing the lateral recharge pathway from the bedrock ridge. No realistic management of internal land cover alone can compensate for a reduction in that input, and any intervention that accelerates drainage toward the coast works directly against the structural condition on which the entire site depends.

5.6. Spatial Groundwater Modelling:Model Findings and Limitations

A steady-state 2D finite-difference model was developed to translate the SSM behavioural characterisation into a spatially explicit groundwater framework (Section 4.9). The baseline head surface and Darcy flux fields are physically plausible and spatially consistent with the DEM-derived flow network, providing independent corroboration of the ridge-derived boundary subsidy identified by the CEH14 intercept audit. Climate scenario responses are spatially coherent: the climate-dry scenario produces a mean head change of −0.159 m across the active domain and the climate-wet scenario +0.117 m, with larger responses in the western interior consistent with the lower drainage feedback in that zone. These climate outputs are presented as indicative spatial summaries of the SSM equilibrium framework rather than calibrated predictions.

For forest management scenarios the spatial model yields a finding that is not visible from the BACI results alone. The observed post-felling displacement in the C4 Forest zone (−0.201 m, still declining as of 2026; Section 4.6.1), applied analytically across the C4 spatial domain with exponential distance-decay (e-folding distance 500 m), confirms that even the full clearfell effect remains confined to the forest zone. This is not ecologically consequential in C4 itself, whose summer minimum is already approximately −1.7 m below ground — far below the SD15b (0.61 m) and SD16 (0.98 m) thresholds. The clusters where threshold exceedance is ecologically critical — C1 and C2, whose summer minima sit within 0.2–0.4 m of those thresholds — are not directly coupled to the C4 zone through the β-

coefficient framework, because lateral connectivity is parameterised implicitly through the ridge boundary subsidy rather than through cluster-to-cluster hydraulic flux. The failure of the PDE solver to produce informative forest management difference maps is therefore not a modelling weakness — it is itself a finding, confirming through an independent route that forest management in the C4 zone does not propagate hydraulically to the open dune slack clusters that determine conservation condition. The dominant control on threshold exceedance in C1 and C2 is the climate signal acting across the full dune plain.

The spatial model is practically limited by two data gaps that preclude defensible scenario predictions beyond this qualitative finding. Hydraulic conductivity is constrained by a single tracer test (K = 6.0 m/day; Betson et al., 2002) and aquifer thickness in the western interior is inferred from β_1 proxy values and cluster priors rather than measured, with only four borehole logs available across 700 ha (Bristow, 2002). Slug tests at two to three representative wells per cluster and a ground-penetrating radar survey of aquifer thickness are identified as the priority field measurements required to enable a properly calibrated model. These data would also support a HYDRUS-1D vadose zone analysis (Šimůnek et al., 2008) at the cluster level, providing an independent test of the vegetation-driven recharge suppression signal identified in C4 (Gumuła-Kawęcka et al., 2021).

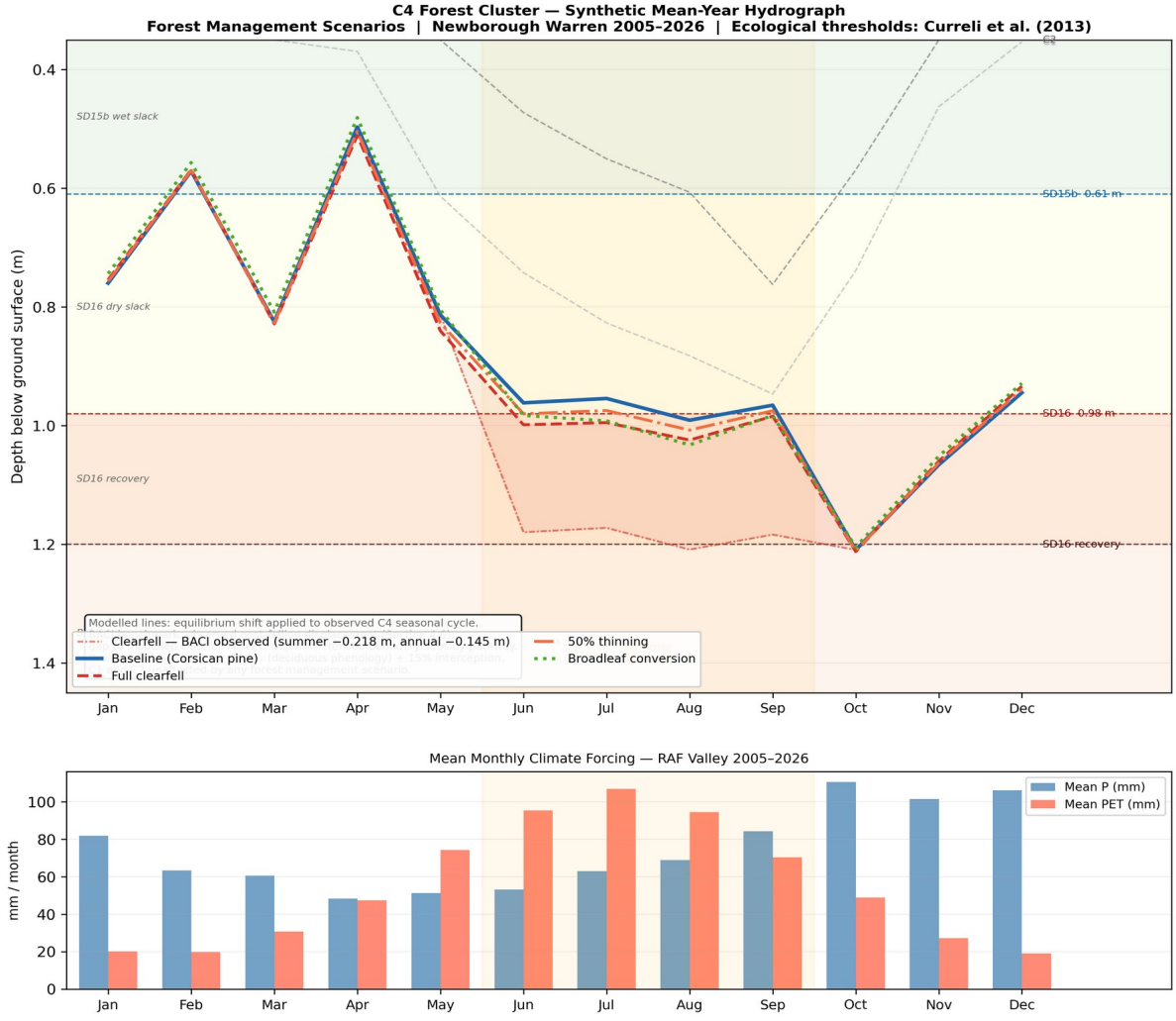


Figure 33. *Synthetic mean-year hydrograph for the C4 Forest cluster under four management scenarios: baseline Corsican pine, full clearfell, 50% thinning, and broadleaf conversion. Lines show equilibrium head shifts applied to the observed 2005–2026 C4 seasonal cycle, derived from scenario-specific β_1 (effective rainfall) and β_2 (ET draw) adjustments. The BACI-observed post-felling raw displacement at the core impact wells (summer +0.292 m deeper than the full pre-clearfell baseline; Section 4.6.3) is shown for context — the gap between the modelled clearfell line and the observed record reflects the baseline contamination from the 2015 scraping event and wet 2015–16 period, and should not be interpreted as evidence of a clearfell-specific effect of that magnitude (Section 4.6.4). C1 and C2 observed mean seasonal cycles are shown for ecological context; no management scenario produces any detectable effect on these clusters. Broadleaf conversion: seasonally-varying β_2 applied to reflect deciduous phenology (lower winter ET, higher summer ET). Lower panel: mean monthly rainfall and PET from RAF Valley 2005–2026. Ecological zone colouring follows Curreli et al. (2013). Generated from 03_master_data.csv and 03_cluster_averages_maod.csv. (21_forestry_01_hydrograph.png)*

The equilibrium shift approach provides a complementary perspective on forest management scenarios that is not available from the BACI analysis alone (Figure 33). Applying scenario-specific β coefficient adjustments to the observed C4 mean seasonal cycle confirms that all three management options — full clearfell, 50% thinning, and broadleaf conversion — produce deeper summer minima than the baseline Corsican pine under mean climate conditions. The summer-specific shifts are small (clearfell −0.033 m, thinning −0.017 m, broadleaf −0.042 m in August), reflecting the dominance of mean climate forcing over management signal in the equilibrium framework. The BACI-observed clearfell displacement (−0.218 m summer) is substantially larger than the modelled equilibrium shift, confirming that the β-coefficient framework does not capture the episodic ridge recharge pathway disrupted by canopy removal. The figure is presented as an analytical summary of the scenario parameter space rather than a predictive forecast.

5.7. Towards an Integrated Restoration Framework

Synthesising across all analytical strands — including the spatial groundwater analysis presented in Section 4.9 and discussed in Section 5.6 — it is clear that restoration can no longer rely on singular, isolated interventions. The following integrated management priorities emerge to establish a climate-resilient coastal aquifer:

Summer minima are the primary restoration lever. Given the dominance of antecedent summer depth in the Western Block peak flood transfer function ($R^2 = 0.684$ and $p < 0.001$; Table 8; Section 4.7), restoration success depends fundamentally on raising summer water table floors through topographic modification and reduced evaporative demand — not on maximising winter recharge. This reframing has direct implications for monitoring design: summer dipwell readings are more diagnostically informative than winter peak measurements for assessing restoration trajectory.

Topographical scraping is a valid maintenance tool, not a standalone solution. The +0.143 m net benefit at CEH36 during the pure scraping era (Section 4.5; Figure 18) confirms that topographic modification of slack floors can raise local water tables measurably. However, the progressive erosion of this benefit by coastal erosion and background climate-driven decline at CEH18 and CEH21 demonstrates that scraping cannot arrest system-wide deterioration alone. Priority targets for future scraping are C1/C2/C3 transitional wells in the mid-eastern dune sector, which combine structural sensitivity with

boundary subsidy access and are most likely to show persistent benefit. It should also be noted that the measured benefit at CEH36 reflects the slack floor being lowered relative to the ambient water table rather than the water table itself rising; the SSM forward residual shows the water table drifting progressively deeper through subsequent eras, confirming that the initial topographic benefit is being eroded by background hydrological decline rather than sustained by any scraping-induced change to aquifer dynamics. CEH18 and CEH21 showed no measurable scraping response in 2015 under any analytical method, consistent with the fact that neither well was scraped until 2023; their subsequent limited response to the 2023 intervention, particularly at CEH21, suggests that position within the coastal drainage pathway is a significant determinant of scraping efficacy, with wells closer to the coast subject to background deterioration that outpaces any localised topographic benefit.

Plantation clearance is a conditional and paired intervention. The β_1 and $-\beta_3$ responses confirm that felling is increasing aquifer responsiveness across a broader spatial footprint than previously recognised (Section 4.6.5; Table 7). However, large-scale clearance treats only the input side of the water balance. The net hydrological effect — increased seasonal amplitude rather than elevated summer water tables — addresses the wrong side of the seasonal cycle for the ecological target (Section 5.5). Extended clearance is therefore recommended only as part of a coupled strategy where scraping or biological buffering interventions are in place to raise summer minima and attenuate lateral drainage losses.

Silviculture should be considered as a potential climate valve, subject to empirical validation. The β-coefficient analysis identifies the plantation's sacrificial evaporative shielding function as a property of canopy density rather than species, suggesting that a more open canopy structure could recover the recharge deficit whilst partially retaining this function (Section 5.4.2). However, no thinning experiment exists within the monitoring record and the optimal canopy density prescription remains a prediction from the β framework rather than an observed outcome. The Dothistroma needle blight trajectory currently affecting parts of the plantation represents a natural experiment that could provide direct empirical evidence: monitoring β_2 responses at wells adjacent to blight-affected stands of varying severity would test the canopy shielding mechanism identified here without requiring additional silvicultural intervention. Any canopy retention must observe a strict 30-metre exclusion zone from the perimeter of target dune slacks.

The P_flood tools should be adopted as standard management practice. The cluster-specific threshold equations (Table 10; Section 4.7.2) enable managers to compute the minimum winter rainfall required to flood a given slack from any current dipwell reading, providing a quantitative basis for adaptive management under climate uncertainty. These tools operate entirely from routinely available climate data and require no specialist modelling capability to apply. The equations have been implemented in the Newborough Warren Hydrogeological Modelling Suite scenario viewer (Hollingham, 2026b; available at https://newbroman.github.io/Newborough-Hydrology_models/scenario_viewer.html), a companion browser-based tool designed for practitioner field use. The viewer operationalises the P_flood forecasting equations, provides indicative sensitivity analysis under projected climate forcing, and hosts the full spatial scenario outputs from the groundwater analysis. Scenario simulation of land cover change is included as an illustrative application of the SSM equilibrium framework; the limitations of this approach are discussed below.

State-space forecasting merits adoption over traditional regression as the standard approach. The median NSE improvement of +0.586 over traditional regression — with 36 of 69 traditional models performing worse than a climatological mean forecast, compared to only 1 of 69 state-space models (Table 4; Section 4.4.1) — demonstrates that the SSM's explicit parameterisation of storage decay provides substantially superior long-term forecast skill. The SSM framework merits serious consideration for future management planning models at Newborough Warren and comparable coastal dune systems.

Future modelling priorities. The analytical framework established here provides a robust foundation for a next generation of site management tools, but realising its full potential requires targeted field investment. The two measurements that would unlock a properly calibrated spatial groundwater model — slug tests at two to three representative wells per cluster and a ground-penetrating radar survey of aquifer thickness (Section 5.6) — would also directly serve the management priorities identified above. Cluster-specific hydraulic conductivity values would refine the P_flood threshold equations and reduce uncertainty in the volumetric water balance; a mapped thickness surface would enable explicit modelling of the ridge recharge pathway that currently operates as an unmeasured boundary term. Together these would allow the scenario difference maps to move from indicative to defensible, supporting evidence-based decisions on the scale and sequencing of scraping, canopy management, and climate adaptation interventions at Newborough Warren and comparable Atlantic dune systems.

6. Limitations

6.1. Temporal resolution and conservative volatility estimates

The reliance on monthly-averaged groundwater levels mathematically dampens the event-scale fluctuations inherent to highly conductive unconfined aquifers. Because daily peaks and troughs of individual storm events are smoothed over a 30-day window, the derived water balance parameters are optimised for seasonal trend analysis rather than event-based flood routing. This is a standard constraint in historic groundwater modelling (Taylor and Alley, 2001), and means the model captures the overall volumetric water balance while inherently underestimating daily transient spikes. The assessment of seasonal amplitude presented here is therefore conservative: the actual event-scale fluctuation — and any periodic severing of the capillary fringe from the root zone — is likely more pronounced than the monthly averages depict.

6.2. Topographic proxies, aeolian deflation, and bedrock causation

The strong spatial correlation between DEM-derived drainage pathways and the hydrogeological cluster boundaries confirms that surface topography acts as a reliable operational proxy for subsurface flow divides. However, this correlation does not imply direct causation: surface sand deposition patterns do not dictate groundwater divides. Both are co-determined by the underlying architecture of glacial till and bedrock ridges, where present. In coastal dune systems this structural relationship is mediated by aeolian deflation — wind erosion is physically limited by the capillary fringe, as cohesive damp sand resists aeolian transport (Bristow and Bailey, 2001; Hesp and Thom, 1990), and because the regional water table is governed by subsurface geology, that geology indirectly controls the baseline limits of surface erosion (Haitjema and Mitchell-Bruker, 2005). Surface topography at Newborough Warren is therefore a physical expression of subsurface hydrological structure, supporting the use of the DEM as a proxy for regional groundwater routing in the absence of high-resolution bedrock mapping.

6.3. The NW10 broadleaf comparison — a single uncontrolled well

The comparison of NW10 (broadleaf block, established 1993–1996) with adjacent Corsican pine wells represents the only direct observational evidence of species conversion effects within the monitoring network, but it is subject to several limitations that preclude a fully controlled comparison. NW10 is a single well in a single block; replication across multiple broadleaf wells and multiple blocks would be required to establish a statistically robust species effect independent of local topographic and substrate variation. The comparison wells (CEH2, CEH34) differ from NW10 in ground elevation, distance from the ridge, and boundary subsidy magnitude, and whilst the direction of the summer minimum contrast is consistent across all comparators the absolute magnitude is sensitive to these positional differences. Most significantly, the broadleaf stand was characterised by a dense bramble understorey throughout most of the monitoring period, which maintained high ground-level evapotranspiration demand and confounded the tree canopy signal; the normalised comparison confirms no detectable divergence from pine behaviour during this phase. The stand has only begun to approach a closed canopy configuration from approximately 2022, leaving fewer than four years of post-closure record available at the time of writing. The

conclusions drawn from NW10 should therefore be treated as indicative observations from a natural experiment rather than confirmed causal attributions, and will require validation against a longer post-closure record and, ideally, against additional broadleaf wells installed within the block.

6.4. Spatial groundwater model — hydraulic parameter uncertainty

The limitations of the steady-state finite-difference spatial model are discussed in Section 5.6. In summary, hydraulic conductivity is constrained by a single tracer test (K = 6.0 m/day; Betson et al., 2002) and aquifer thickness in the western interior is inferred from β_1 proxy values and cluster priors rather than measured. As a consequence, forest management scenario difference maps are not reliably distinguishable from interpolation artefacts, and the spatial model yields informative results only for climate scenarios where all wells are affected simultaneously. The priority field measurements required to address these constraints are identified in Section 5.6.

6.5. BACI experiment — network design limitations

The clearfell BACI experiment (Section 4.6) was conducted on a monitoring network designed for long-term water table surveillance rather than as a purpose-built experimental design. Several consequent limitations apply. The clearfell zone and scraping sites were not randomly assigned; both interventions occurred at locations selected on conservation management grounds, which introduces the possibility that pre-existing site characteristics influenced the selection and could confound the intervention signal. The monitoring network was not designed to capture the inland propagation of the drainage acceleration signal beyond the felling boundary, and the transition zone between the clearfell boundary and the open warren contains no dipwells; the extent to which clearfell-driven drainage acceleration has affected dune slack water tables deeper within the Western Block therefore cannot be quantified from the current dataset. Additionally, three of the 16 BACI wells were installed after the December 2017 clearfell and have no usable pre-felling record, limiting the paired before-after contrast to 13 wells for coefficient-level analysis.

7. Conclusions

This study provides the first network-wide, multi-method quantitative characterisation of groundwater behaviour at Newborough Warren, integrating over 70 dipwells, 21 years of monitoring and nine complementary analytical approaches. The principal conclusions are:

1. A fundamental structural divide governs site hydrology. The Eastern Block (C1: LCSC = 63.9%, C2: LCSC = 59.7%) and Western Block (C3: LCSC = 83.3%) are hydrologically distinct. The Eastern Block behaves as a shallow pan (low LCSC ≈ 60–64%, high β_1 ≈ 1.57–1.67): impermeable till and estuarine deposits leave little pore space, so rainfall spikes the water table sharply. The Western Block behaves as a deep sponge (LCSC = 83.3%, β_1 = 1.20): thick open sand absorbs rain slowly across a large unsaturated zone, holding the critical long-term groundwater reservoir. The Forest behaves as a canopy sponge (LCSC = 104.5%, β_1 = 0.957): the plantation intercepts approximately 24% of rainfall before it reaches the aquifer, producing a high LCSC through overhead loss rather than underground storage. This three-point spectrum is the conceptual backbone of the management implications.

2. State-space modelling transforms forecast skill. The median iterative NSE improvement of +0.586 over traditional regression confirms that aquifer hydrological memory ($-\beta_3$) is indispensable for accurate multi-step prediction, with 36 of 69 traditional regression models performing worse than a climatological mean forecast compared to only 1 of 69 state-space models. Adoption of SSM approaches should be recommended for all conservation planning forecasts at Newborough Warren and comparable coastal dune systems.

3. Plantation clearance produces measurable changes in SSM coefficients extending across the full experimental network, but the available monitoring data do not permit detection of a clearfell-specific hydrological signal above the noise of climate variability and the scraping confound. The ANCOVA-BACI model estimates a climate-corrected upper bound displacement of −0.093 m in the core impact zone (R^2 = 0.729; Section 4.6.3), but three independent robustness analyses — SSM residual normalisation, synthetic control, and cluster transition testing — found no statistically significant clearfell-specific step change once the April 2015 scraping confound and the anomalously wet 2015–16 baseline were accounted for (Section 4.6.4). Post-felling β_3 increases across all 16 BACI wells, including regional controls several hundred metres from the felling boundary, are attributed to SSM fitting artefacts of the transition between equilibria rather than genuine drainage acceleration (Section 4.6.5). The β_1 and β_2 increases are genuine physical responses to canopy removal — the β_2 increase in particular is ecologically consequential, deepening summer minima proportionally more than any winter improvement regardless of baseline choice (Section 5.5).

4. The forest sets the absolute depth of the water table beneath it but is not driving the network-wide declining trend. Summer minimum water table depths have declined at statistically significant or near-significant rates in all clusters (C1: −0.010 m yr^{-1}, C2: −0.012 m yr^{-1}, C3: −0.015 m yr^{-1}, C4: −0.017 m yr^{-1}) despite their very different canopy cover, and winter maxima show no significant trend in any cluster. The deterioration is entirely concentrated in the summer season, consistent with the observed warming trend in summer

maximum temperatures (+0.013°C yr^{-1}, $p < 0.001$) rather than any forest-induced progressive drawdown. The β-coefficient framework identifies two quantifiable hydrological functions of the plantation — recharge suppression and sacrificial evaporative shielding of summer minima — that are directly comparable in management effect to dune scraping. The monitoring data do not support the inference that clearance improves the net water balance; the β_2 evidence indicates summer minima are deepened rather than raised (Section 5.4; Section 5.5). Observational evidence from the 1993 broadleaf block (dipwell NW10, 18-year record) confirms that broadleaf conversion may produce deeper summer minima than adjacent Corsican pine at comparable elevation— though a single dipwell is insufficient for confident attribution given the positional confounders identified in Section 6.3, with the mechanism reflecting higher recharge-season throughfall producing a larger seasonal drawdown rather than amplified atmospheric draw (β_2 essentially identical between NW10 and adjacent pine wells). No temporal divergence from the pine baseline is detectable during the bramble-dominated phase of the monitoring record; an emerging trend consistent with canopy closure deepening summer minima has not yet reached statistical significance ($p = 0.062$) and is treated as a monitoring hypothesis for the 2026–2030 period. Stand density rather than species composition is the primary silvicultural management variable; the pathway toward improved dune slack hydrology runs through managed canopy reduction rather than wholesale removal or broadleaf conversion.

5. Dune scraping delivers a measurable but climate-limited benefit. The net BACI benefit of +0.143 m at CEH36 during the pure scraping era — independently corroborated by synthetic control analysis (−0.116 m; Section 4.5) — confirms that topographic modification of slack floors raises local water tables measurably.. However, background climate-driven decline offsets most benefits elsewhere, and scraping must be understood as a maintenance tool that slows deterioration rather than reverses it under current climate trajectories. It is most effective when combined with interventions that raise summer water table floors.

6. Winter rainfall amount is a poor predictor of slack flooding in the Western Block. Summer minimum depth is the dominant control on peak winter water table in the Western Block ($R^2 = 0.684$; h_min $p < 0.001$; winter rainfall non-significant, $p = 0.818$). Restoration strategy must therefore prioritise raising summer water table floors, not simply maximising winter recharge. This finding reframes the entire management problem: the dipwell readings most diagnostic of restoration trajectory are summer minima, not winter peaks, and the interventions most likely to deliver ecological benefit are those that reduce summer evaporative draw rather than those that maximise winter recharge inputs.

7. Operational P_flood threshold equations are now available for C1–C4. These tools, validated against 21 years of hydrological data, enable site managers to forecast minimum winter rainfall requirements from any current dipwell reading, providing a quantitative basis for adaptive management under climate uncertainty. The equations have been implemented in the companion Newborough Warren Hydrogeological Modelling Suite scenario viewer (Hollingham, 2026b), a browser-based tool enabling scenario simulation of forest clearance, dune scraping, and UKCP18 climate projections, and hosting the full spatial groundwater scenario outputs from this study.

8. The site water balance is structurally dependent on ridge-derived lateral recharge, not local rainfall. Water balance decomposition confirms a persistent monthly deficit in every cluster under mean climate conditions, with the water balance residual — the flux required to close the monthly deficit — ranging from 314–351 mm/year across clusters (Tables 3a, 3b). Local recharge from direct rainfall contributes only 99–173 mm/year depending on cluster. For the Forest cluster specifically, ridge-derived lateral recharge is the primary sustaining mechanism — confirmed by the positive model intercepts concentrated in the northern forest margin. Across the broader dune system, open dune clusters are sustained by a combination of high recharge sensitivity and topographic position. Any land management that interrupts ridge-to-site lateral flow — including vegetation clearance on ridge slopes — therefore poses a greater long-term hydrological risk than any management of within-site land cover.

9. Water table fluctuation analysis confirms broadly uniform aquifer storage properties across the site. Event-based WTF Sy estimates (Table 3c) range from 0.223 to 0.259 across the four principal clusters. After correction for 24% canopy interception (Freeman, 2008), the Forest cluster estimate (0.227) is indistinguishable from the open dune clusters, confirming that the plantation's distinctive hydrological signature is entirely a surface boundary effect rather than a property of the sand. The assumed Sy values used in the volumetric water balance (Table 3b) fall below the lower quartile of the WTF distributions in all clusters, confirming they represent conservative lower bounds. Individual well WTF analysis across the reference and extended networks (Figure 9, Table S4) confirms that this uniformity holds spatially — the interpolated Sy surface shows no abrupt discontinuities coinciding with cluster boundaries, and the extended network contour is consistent with the reference-only surface following Pearson cluster validation. Direct field measurement by slug test or pumping test at two to three representative wells per cluster is recommended as a priority for future work.

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10. Data Availability

10.1. . Data Availability and Software

The complete dipwell monitoring network, comprising 117 wells across Newborough Warren and the adjacent dune systems, is mapped at: https://www.google.com/maps/d/edit?mid=1hXLAauiMeaVsXhBR_IoUTziAtjk (M. Hollingham, unpublished).

The analytical pipeline underpinning this study is documented at https://newbroman.github.io/Newborough-Hydrology_models/ and comprises sixteen modules covering data preparation, state-space modelling, and intervention analysis and mapping. All analyses were implemented in Python using the SciPy ecosystem (*pandas*, *NumPy*, *statsmodels*, *scikit-learn*) and *seaborn* for visualization. Full diagnostic outputs for all 69 reference network wells and all 16 clearfell BACI wells are reproducible by running the published pipeline. Groundwater monitoring data supporting this study are also available.

Source Data Credits:

- **Climate Data:** Meteorological records for RAF Valley were obtained from the UK Met Office (MIDAS Open Government Licence).
- **Topographic Data:** Terrain analysis was performed using 1-metre resolution LiDAR composite datasets provided by Natural Resources Wales (NRW) and accessed via the Welsh Government's Lle Geo-Portal.
- **Groundwater Data:** The primary dipwell monitoring record was independently maintained by the author.

11. Supplementary Material

Supplementary File S1:

Interactive map of the full dipwell monitoring network (Google Maps, accessible at :https://www.google.com/maps/d/viewer?mid=1hXLAauiMeaVsXhBR_IoUTziAtjk

Supplementary Note S2: Aquifer Geometry, Hydraulic Parameterisation and the Basis of the Spatial Flow Model

Hollingham (2026) — Newborough Warren Coastal Sand Dune Aquifer, Isle of Anglesey

2.1 The Borehole Data Problem

Darcy's law and any spatially explicit groundwater flow model require three spatially resolved inputs: hydraulic conductivity (K), aquifer thickness (b, from which transmissivity T = K·b is derived), and hydraulic gradient (∇h). For the Newborough Warren dune system, only the third of these — the hydraulic gradient — is well constrained, through the 72-well dipwell network. Hydraulic conductivity and aquifer thickness are poorly constrained, because borehole logs providing lithological or geophysical evidence of the depth to the bedrock or glacial till substrate are absent across the great majority of the 700 ha dune system.

The available borehole evidence is confined to four locations documented in Betson et al. (2002), a report commissioned by the Countryside Council for Wales (Contract FC†73-05-18) covering hydrogeological investigations at the site. These boreholes provide minimum depth-to-basement constraints at four spatial points and constitute the only direct subsurface lithological evidence available for the site. They are summarised in Table S1.

Borehole	Easting (OSGB36)	Northing (OSGB36)	Saturated thickness (m)	Source
Water borehole	241,721	364,133	12.8 (min.)	Betson et al. (2002)
NH1	241,734	363,306	6.5	Betson et al. (2002)
NH2	242,024	363,107	6.5	Betson et al. (2002)
Borehole 3	242,837	363,288	3.65	Betson et al. (2002)

Table S1. Hard borehole constraints used as fixed nodes in the aquifer thickness IDW surface. Values represent confirmed minimum depth to the low-permeability substrate (glacial till or bedrock). The eastern borehole (Borehole 3) shows progressive thinning toward the Menai Strait.

The tracer test reported by Betson et al. (2002) provides the single direct estimate of hydraulic conductivity for the site, yielding K = 6.0 m/day at the central dune plain (range

3.0–9.0 m/day from the sensitivity analysis). This value is used unchanged as the site-wide K constant in the spatial model, with the range providing the basis for the exploratory uncertainty envelope noted in the figure annotations.

S2.2. Deriving the Aquifer Thickness Surface

S2.2.1. The β_1 Proxy

The four borehole constraints are insufficient to interpolate a site-wide thickness surface from direct evidence alone. The principal additional constraint used here is an indirect one, derived from the SSM β_1 (recharge sensitivity) coefficient.

The physical interpretation of β_1 is the fraction of a unit rainfall pulse that reaches the water table as a head rise, integrated over the monthly timestep. In an unconfined aquifer, this is governed by the depth of the unsaturated zone and the specific yield: a thick unsaturated zone above a large storage volume attenuates the rainfall-to-head transfer, producing a low β_1. A shallow, low-storage aquifer (thin unsaturated zone, low Sy) produces a high β_1 — rainfall arrives at the water table quickly and raises it sharply. Although β_1 cannot be inverted algebraically to yield aquifer thickness without independent knowledge of the recharge flux and specific yield, its spatial pattern is a reliable proxy for the relative contrast in aquifer thickness across the site.

The Eastern Block (C1: $\beta_1 = 1.565$; C2: $\beta_1 = 1.674$) and Western Block (C3: $\beta_1 = 1.201$) contrast is consistent with the geological interpretation of Stratford et al. (2007): the Eastern Block sits on shallow till and estuarine deposits, producing a thin, storage-limited aquifer with a flashy response; the Western Block occupies deep, clean aeolian sand, producing a capacious, buffered aquifer with an attenuated response. The Forest cluster (C4: $\beta_1 = 0.957$) is anomalously low even relative to C3, reflecting canopy interception rather than additional aquifer depth — confirmed by the depth-dependent PET analysis (script 15) which found $\lambda = 0$ optimal for C4, indicating that capillary disconnection at depth is not the primary control.

Cluster	Label	β_1 (median)	Thickness prior (m)	Basis
C1	Eastern lake-buffer	1.565	7.0	NH1, NH2 boreholes; shallow till substrate
C2	Eastern mature dune	1.674	7.0	NH1, NH2 boreholes; consistent with C1
C3	Western mature dune	1.201	16.0	β_1 proxy (deep sand); Betson et al. (2002) western section; DEM ridge geometry
C4	Forest	0.957	16.0	Same deep sand

				substrate as C3; confirmed by Pearson affinity persistence post-felling
C5	Coastal	1.047	4.0	Geomorphological thinning toward Menai Strait
C6	Lake	—	0.5	Near-zero: open water at Llyn Rhos Ddu

Table S2. Cluster thickness priors used in the IDW thickness surface. C3 and C4 share the same prior because the Forest plantation sits on the same deep aeolian sand body as the Western Block, confirmed by β_1 similarity and Pearson affinity analysis.

S2.2.2. Per-Well Overrides and Boundary Pinch-out

CEH14 (241,292 E, 364,488 N), classified as C4, sits at the crest of the bedrock ridge at 14.4 m AOD with a mean water table head of 13.3 m AOD. The water table is only approximately 1.1 m below ground surface here. Given the ridge geology (fractured bedrock with thin sand veneer), the saturated thickness is estimated at 5 m rather than the C4 prior of 16 m.

Wells CEH7 (243,386 E, 363,613 N) and CEH8 (243,150 E, 363,382 N) sit at the far eastern estuarine margin of the site, where the DEM confirms ground elevation of 1.2–5.2 m AOD. Both are treated as pinch-out nodes with thickness 1.0 m. Shore boundary nodes (0.5 m) are placed along the southern coast, eastern coast, and at Llyn Rhos Ddu.

S2.2.3 IDW Interpolation and Smoothing

All hard borehole nodes and per-well prior/override values are combined into a single point set and interpolated to the 50 m grid using inverse-distance weighting (IDW) with power = 1 (linear distance weighting). Power = 2 was tested first but produced bullseye artefacts around the isolated borehole nodes. Power = 1 gives a smoother surface that interpolates plausibly between the sparse hard constraints and the denser well-level priors. A physical minimum of 0.3 m is enforced. The resulting thickness surface ranges from 0.3 m at the shore margins to approximately 18 m in the western interior.

S2.3. Why C4 Retains Its Cluster Identity After Clearfell

A key question for the scenario analysis is whether the clearfell treatment wells (FE1, FE2, FE4, LIS1) converged toward C3 open-dune behaviour after the December 2017 felling. The Pearson affinity analysis (Section 4.3) provides a direct empirical test: all four impact wells maintained $r > 0.97$ affinity with C4 even in the post-felling period (2018–2026), with no significant shift toward C3 affinity.

This persistence is consistent with the interpretation that the Forest cluster signature at these wells reflects deep sandy substrate at high topographic elevation rather than canopy influence alone — in which case convergence toward open-dune behaviour may not be achievable regardless of management intervention. The third interpretation is consistent with the β_1 proxy evidence: C4 wells sit at the highest topographic positions in the dune system (mean head 9.52 m AOD), and their deep unsaturated zones and low recharge sensitivity reflect their geological position, not only their canopy cover. This conclusion is independently supported by the NW10 broadleaf comparison (Section 4.6.6): NW10, situated within the 1993 clearfell and 1996 broadleaf restocking block, has maintained C4 cluster affinity ($r = 0.986$) throughout the 18-year monitoring period despite its different canopy cover, confirming that substrate position rather than canopy type is the dominant control on cluster identity.

S2.4. Summary of Model Parameterisation

Parameter	Value	Source / basis
Hydraulic conductivity K	6.0 m/day (range 3–9)	Betson et al. (2002) tracer test, central dune plain
Aquifer thickness (hard constraints)	3.65–12.8 m	Betson et al. (2002) borehole logs (4 locations)
Aquifer thickness (C1/C2 prior)	7.0 m	NH1/NH2 borehole evidence; Eastern Block β_1 contrast
Aquifer thickness (C3/C4 prior)	16.0 m	β_1 proxy (deep sand); Betson et al. (2002) western section
CEH14 override	5.0 m	Ridge crest geology; WT only 1.1 m below ground
CEH7/CEH8 override	1.0 m	Estuarine margin pinch-out; DEM ground elevation 1–5 m AOD
Grid resolution	50 m	Balances spatial detail against borehole data density
IDW power	1 (linear)	Power = 2 produced bullseye artefacts at sparse borehole nodes
Sea boundary (south/east)	h = 0 m AOD	Dirichlet BC; Menai Strait / Caernarfon Bay at OD
Sea boundary (west cap)	N ≤ 363,400	DEM confirms ridge (>15 m AOD) above this Northing on western margin

Table S3. Summary of spatial model parameterisation. All values are directly traceable to borehole evidence, published tracer tests, or the SSM β_1 proxy analysis described in this note.

The principal uncertainty in the spatial model is the aquifer thickness surface in the western interior, where the C3 prior of 16 m is derived indirectly from the β_1 proxy and the Betson et al. (2002) western section rather than from direct borehole evidence. Slug tests or pumping tests at two to three representative C3 wells and a ground-penetrating radar survey of aquifer thickness are identified as the highest-priority data gaps for future field campaigns (Section 6.4).

Supplementary Note S3: Scenario Modelling Framework and Limitations

S3.1. Purpose and Scope of the Scenario Analysis

An exploratory scenario analysis was developed to investigate whether the site's own SSM β coefficients and the steady-state spatial model could yield defensible quantitative predictions of water table response to management interventions and climate change. This note documents the approach taken, why the spatial model proved informative for climate scenarios but not for forest management scenarios, and what field data would be required to support a fully calibrated spatial model in future work. The scenario outputs are presented in Section 4.9 and discussed in Section 5.6 of the main paper; this note provides the technical reasoning behind the methodological choices and their limitations.

S3.2. The Steady-State Spatial Model: Approach and Reasoning

The spatial groundwater model (script 19) solves the 2D steady-state groundwater flow equation on a 50 m grid covering the full dune system extent. The governing equation is a Helmholtz-type PDE (Bear, 1972; Freeze and Cherry, 1979):

$$\nabla\cdot(T\cdot\nabla h) - \beta_3\cdot h = -(\beta_1\cdot P_{eee} - \beta_2\cdot PET)$$

where $T = K\cdot b$ is the transmissivity (hydraulic conductivity $K = 6.0$ m/day after Betson et al., 2002, multiplied by the IDW-interpolated aquifer thickness b), and the right-hand side is the net recharge source term expressed directly in terms of the SSM β coefficients. The $\beta_3\cdot h$ term represents the internal drainage-to-baseflow feedback from the SSM. Dirichlet boundary conditions ($h = 0$ m AOD) are applied at the sea boundaries; Neumann no-flow conditions apply at the ridge. The system was solved using a sparse direct solver (scipy.sparse.linalg.spsolve; Virtanen et al., 2020) on a system of approximately 4,200 equations.

This approach is well-posed in a mathematical sense — the Helmholtz structure guarantees a unique solution with the given boundary conditions — and is physically consistent in that the β coefficients entering the source term are the same parameters estimated from the observed well records. The head surface produced by the solver is therefore the spatial equilibrium consistent with the observed site-wide β field, the estimated transmissivity distribution, and the sea boundary conditions.

S3.3. Why Scenario Difference Maps Are Not Informative for Forest Management

Despite the mathematical validity of the steady-state solver, the scenario difference maps — computed as $\Delta h = h(\text{scenario}) - h(\text{baseline})$ — consistently showed near-zero differences for the forest management scenarios (full clearfell, thinning, broadleaf conversion), with site-wide mean Δh values of order 0.001–0.010 m. This is not a coding error. It reflects a fundamental incompatibility between the spatial model and the physical structure of forest management at this site, for three reasons.

First, the forest management scenarios modify β coefficients only for the nine C4 Forest cluster wells. These per-well β values are then IDW-interpolated to the full 4,235-cell grid. Because the nine C4 wells constitute a small minority of the 72-well network, the modified β

values are heavily diluted by the surrounding C1–C3 clusters during interpolation. The source term $\beta_1 \cdot P_{eee} - \beta_2 \cdot PET$ therefore changes by only a small amount at each cell, and the resulting change in the steady-state head solution is proportionally small.

Second, the internal drainage term $\beta_3 \cdot h$ provides a strong stabilising feedback in the Helmholtz equation. C4 has the lowest β_3 in the network (0.096, compared with 0.135 for C3 and 0.214 for C2). This means that in the C4 zone, the drainage feedback is weak and the source term change required to produce a given head change is proportionally larger. The PDE solution is therefore relatively insensitive to moderate changes in the C4 source term.

Third, and most fundamentally, the steady-state model does not capture the ridge boundary subsidy. The observed mean head in C4 is 9.52 m AOD — substantially higher than C3 at 6.09 m AOD — not because of C4's β coefficients but because C4 wells are proximal to the bedrock ridge and receive substantial lateral recharge (the largest boundary subsidy in the network, α = +0.222 m/month at CEH14). The steady-state solver does not include this boundary subsidy term because it was not parameterised into the β coefficient framework. As a result, the PDE solution for C4 head values under both baseline and scenario conditions underestimates the true water table by a systematic amount related to the ridge contribution, and the difference between the two solutions is correspondingly small.

S3.4. Why Climate Scenarios Do Work in the Spatial Model

In contrast to the forest management scenarios, the climate scenarios (climate-dry: ΔP = −10%, ΔPET = +10%; climate-wet: ΔP = +10%, ΔPET =−5%) produce spatially coherent and physically plausible difference maps, with mean Δh values of −0.159 m (dry) and +0.117 m (wet). This is because climate perturbations modify P and PET for all 72 wells simultaneously, so the IDW dilution effect does not apply. Every cell in the grid receives a modified source term, and the PDE solution responds accordingly. The spatial pattern of the head change — larger in the western interior than on the coastal fringe, consistent with the transmissivity distribution — is also physically reasonable.

S3.5. The Correct Approach for Forest Scenario Differences

For forest management scenarios, the scenario analysis uses an alternative approach that is directly grounded in the empirical BACI record rather than in the spatial model. The observed post-felling displacement at the core impact zone is −0.201 m (raw BACI mean, 2018–2026), and is still declining as of early 2026. This value is applied as an analytical spatial shift to the C4 zone, with distance-decay propagation to adjacent C3 wells (exponential decay, e-folding distance 500 m). No empirically supported estimate of the thinning effect exists within the monitoring data; a proportional estimate (−50% of the clearfell displacement) is used as an illustrative scenario only.

The broadleaf conversion scenario shift (−0.075 m) is estimated from the observed behaviour of NW10 (Section 4.6.6), which shows consistently deeper summer minima than adjacent pine wells across 18 years. This is used as a conservative estimate of the broadleaf effect relative to the existing plantation; it should be treated as indicative rather than calibrated, given that NW10 is a single well and the monitoring period spans the bramble-dominated phase rather than a mature closed canopy.

These cluster hydrograph comparisons reveal a key finding that the steady-state spatial model cannot show: the forest management scenarios affect only C4, which is already so deep (summer minimum −1.7 m) that the −0.201 m BACI displacement is ecologically irrelevant to slack threshold exceedance. The ecologically critical clusters are C1 and C2, whose summer minima are within 0.2–0.4 m of the SD15b and SD16 thresholds respectively. These clusters are not affected by forest management in the SSM framework, because the C4 and C1/C2 zones are spatially separated and the lateral connectivity between them is parameterised implicitly through the boundary subsidy rather than through the β coefficients.

S3.6. Implications and Future Modelling Requirements

The failure of the steady-state spatial model to produce informative forest management difference maps is not a weakness of the study — it is itself a finding. It confirms, through an independent modelling route, that the SSM β coefficients do not predict a large direct hydraulic response to forest management in the C1/C2 open dune clusters. The dominant control on slack threshold exceedance in those clusters is the climate signal acting through the entire dune plain, not the forest zone. The BACI experiment provides the ground-truth evidence for what clearfell actually does to the forest zone water table (it worsens it in the short-to-medium term), and the NW10 natural experiment provides the only available evidence on broadleaf conversion effects.

A fully coupled transient groundwater model with explicit ridge recharge, cluster-to-cluster lateral flux, and time-varying boundary conditions would be required to test whether and over what timescale forest management interventions propagate to the open dune slack clusters. Such a model is beyond the scope of the present study but represents a clear direction for future work. The two priority field measurements that would enable it are: (1) slug tests at two to three representative wells per cluster to constrain the spatial distribution of K; and (2) a ground-penetrating radar survey of aquifer thickness to replace the β_1-proxy thickness priors with direct measurements. The β coefficient framework, transmissivity field, and boundary condition geometry derived in the present study would form the parameterisation basis of such a model.

Summary of scenario modelling approach

Spatial difference maps: 2D steady-state PDE solver (Helmholtz equation with SSM β source terms). Valid for climate scenarios (all wells affected simultaneously). Not informative for forest management scenarios due to IDW dilution of C4 β changes, weak C4 drainage feedback ($\beta_3 = 0.096$), and absence of ridge boundary subsidy in the PDE.

Forest management scenarios: BACI-observed displacement (−0.201 m for full clearfell, empirical) applied directly to the observed seasonal cycle. Thinning uses a proportional estimate (illustrative only — no empirical basis). Broadleaf uses NW10 observational evidence (indicative). All forest management scenarios affect C4 only; C1/C2 threshold exceedance is unaffected by any forest management scenario.

Climate scenarios: PDE-derived equilibrium shifts affecting all clusters simultaneously (−0.159 m dry, +0.117 m wet). These are the most defensible scenario outputs and the only ones where the spatial model adds genuine value beyond the cluster hydrograph approach.

S4, Specific Yield mapping

Table S4. Individual Well WTF Specific Yield Estimates — Newborough Warren 2005–2026: *Event-based median specific yield (Sy) derived from the water table fluctuation (WTF) method (Healy and Cook, 2002) for 67 reference wells and 19 extended network wells. n events = number of qualifying monthly rising-limb events (criteria: Δh > 5 mm, net recharge P − PET > 10 mm). Q25/Q75 = interquartile range; wide IQR reflects month-to-month variability in event estimates rather than measurement error. Forest cluster (C4) wells marked Interception corrected = Yes have net recharge adjusted for 24% canopy interception (Freeman, 2008). Extended well cluster assignments derived from Pearson affinity sitewide audit (Section 4.3). *Excluded from IDW interpolation surface: CEH12 (bedrock ridge — WTF response reflects fractured rock); CEH15 (forest slack floor — slack topography dominates water table dynamics). Wells shown in red. Generated by 18_wtf_spatial.py.*

Well	Cluster	Network	n events	Sy median	Sy Q25	Sy Q75	Interception corrected	Excluded from interpolation
CEH11	1	Reference	50	0.223	0.167	0.322		
CEH22	1	Reference	26	0.313	0.178	0.366		
CEH23	1	Reference	34	0.192	0.121	0.275		
CEH25	1	Reference	49	0.172	0.142	0.286		
CEH26	1	Reference	39	0.223	0.130	0.315		
CEH27	1	Reference	43	0.212	0.150	0.314		
CEH5	1	Reference	52	0.192	0.139	0.323		
CEH6	1	Reference	55	0.211	0.142	0.309		
P2	1	Extended	17	0.238	0.106	0.296		
CEH10	2	Reference	50	0.245	0.119	0.328		
CEH24	2	Reference	43	0.216	0.144	0.323		
CEH28	2	Reference	35	0.227	0.146	0.374		
CEH40	2	Reference	22	0.271	0.168	0.321		
CEH41	2	Reference	24	0.222	0.132	0.289		
D10	2	Reference	37	0.235	0.156	0.369		
D15	2	Reference	36	0.198	0.143	0.308		

Well	Cluster	Network	n events	Sy median	Sy Q25	Sy Q75	Interception corrected	Excluded from interpolation
D17	2	Reference	42	0.251	0.175	0.338		
D38	2	Reference	47	0.216	0.144	0.307		
D41	2	Reference	37	0.227	0.154	0.377		
D43	2	Reference	45	0.226	0.158	0.340		
D44	2	Reference	45	0.237	0.179	0.355		
D5	2	Reference	40	0.259	0.140	0.382		
D6	2	Reference	48	0.267	0.158	0.415		
D7	2	Reference	42	0.213	0.128	0.333		
D8	2	Reference	36	0.219	0.140	0.367		
D9	2	Reference	38	0.211	0.153	0.354		
L7	2	Reference	47	0.212	0.139	0.299		
NW 4B	2	Reference	42	0.212	0.147	0.320		
NW13	2	Reference	36	0.207	0.127	0.266		
NW3	2	Reference	47	0.254	0.160	0.359		
NW4	2	Reference	42	0.233	0.158	0.333		
T41A	2	Reference	43	0.201	0.136	0.338		
T41B	2	Reference	41	0.212	0.163	0.397		
T41C	2	Reference	40	0.253	0.167	0.365		
T41D	2	Reference	36	0.214	0.157	0.320		
WMC1	2	Reference	35	0.232	0.135	0.350		
WMC4	2	Reference	36	0.199	0.117	0.278		
CEH12	2	Extended	8	0.366	0.338	0.380		Yes*
CEH35	2	Extended	8	0.218	0.120	0.330		
CEH37	2	Extended	28	0.197	0.128	0.308		

Well	Cluster	Network	n events	Sy median	Sy Q25	Sy Q75	Interception corrected	Excluded from interpolation
CEH38	2	Extended	13	0.303	0.170	0.450		
CEH7	2	Extended	22	0.209	0.144	0.318		
CEH8	2	Extended	25	0.246	0.151	0.348		
PE	2	Extended	16	0.243	0.182	0.299		
PW	2	Extended	17	0.220	0.136	0.229		
CEH 1	3	Reference	49	0.232	0.141	0.306		
CEH16	3	Reference	31	0.250	0.161	0.342		
CEH17	3	Reference	25	0.282	0.161	0.389		
CEH18	3	Reference	38	0.260	0.150	0.354		
CEH19	3	Reference	24	0.360	0.180	0.391		
CEH21	3	Reference	24	0.307	0.166	0.378		
CEH31	3	Reference	26	0.275	0.205	0.392		
CEH36	3	Reference	28	0.290	0.162	0.361		
CEH39	3	Reference	27	0.263	0.152	0.378		
CEH4	3	Reference	39	0.243	0.166	0.339		
CEH42	3	Reference	27	0.254	0.142	0.359		
CEH9	3	Reference	40	0.275	0.166	0.389		
D25	3	Reference	35	0.264	0.189	0.355		
NW1	3	Reference	47	0.173	0.118	0.282		
NW11	3	Reference	47	0.203	0.129	0.270		
NW2	3	Reference	47	0.204	0.115	0.296		
NW5	3	Reference	48	0.254	0.144	0.337		
NW6	3	Reference	41	0.237	0.188	0.309		
NW7	3	Reference	42	0.265	0.180	0.358		
NW9	3	Reference	34	0.282	0.166	0.339		

Well	Cluster	Network	n events	Sy median	Sy Q25	Sy Q75	Interception corrected	Excluded from interpolation
WMC2	3	Reference	38	0.234	0.176	0.348		
WMC3	3	Reference	34	0.258	0.226	0.382		
FE3	3	Extended	19	0.263	0.165	0.332		
NW8	3	Extended	14	0.233	0.140	0.357		
NW8B	3	Extended	26	0.266	0.222	0.363		
P1	3	Extended	7	0.313	0.225	0.394		
CEH13	4	Reference	38	0.237	0.143	0.333	Yes	
CEH14	4	Reference	40	0.154	0.094	0.290	Yes	
CEH2	4	Reference	42	0.182	0.116	0.252	Yes	
CEH20	4	Reference	37	0.197	0.133	0.271	Yes	
CEH30	4	Reference	34	0.223	0.138	0.293	Yes	
CEH32	4	Reference	35	0.197	0.143	0.274	Yes	
CEH33	4	Reference	35	0.221	0.148	0.286	Yes	
CEH34	4	Reference	35	0.196	0.124	0.278	Yes	
NW10	4	Reference	42	0.210	0.111	0.284	Yes	
CEH15	4	Extended	16	0.141	0.101	0.364	Yes	Yes*
FE1	4	Extended	24	0.196	0.109	0.267	Yes	
FE2	4	Extended	24	0.217	0.128	0.285	Yes	
FE4	4	Extended	20	0.228	0.160	0.281	Yes	
LIS1	4	Extended	20	0.183	0.129	0.258	Yes	
NW12	4	Extended	9	0.211	0.161	0.293	Yes	